HW-Z1-ZCU106 Evaluation Board (XCZU7EV-2FFVC1156)

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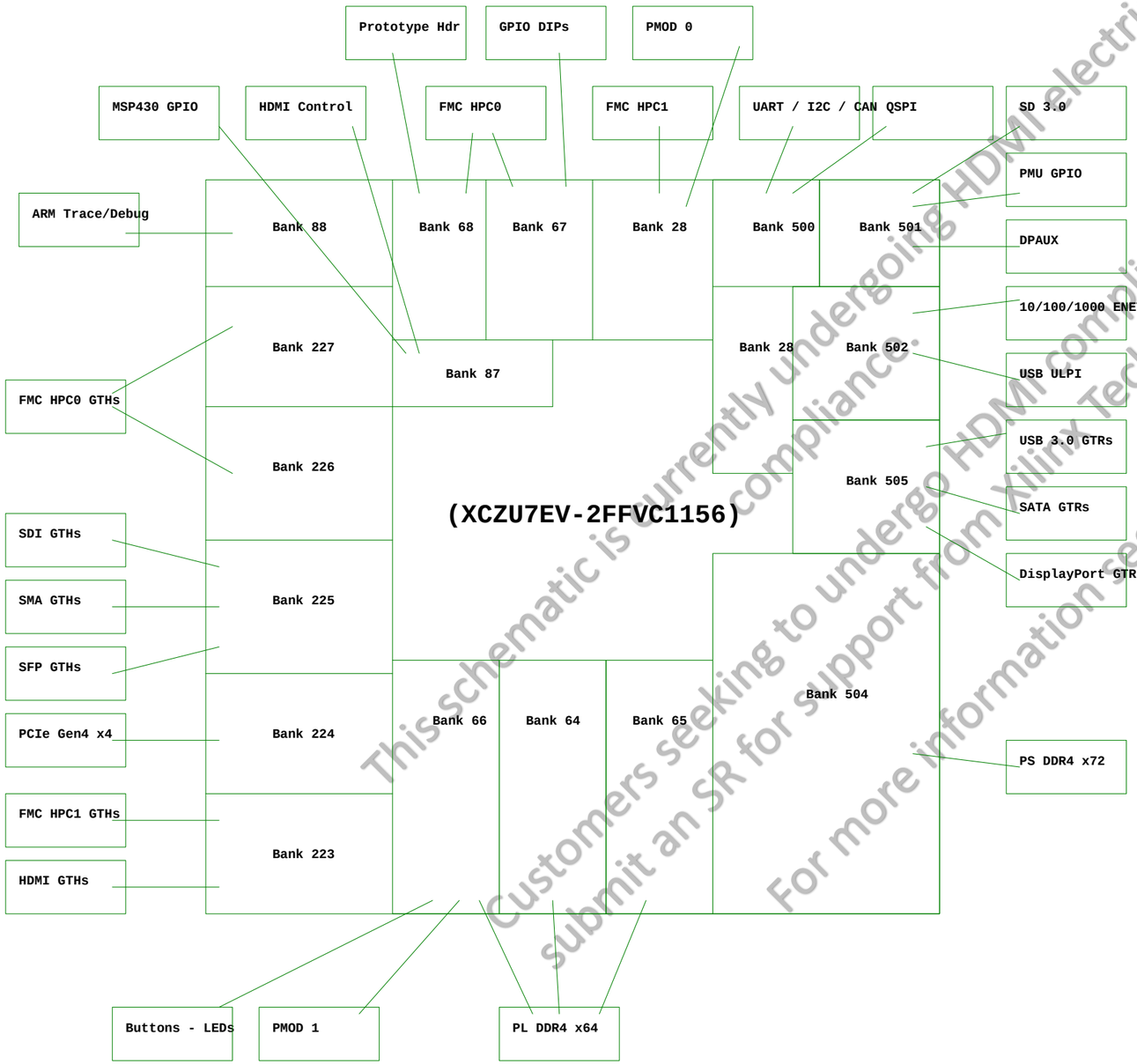
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submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

XILINX

TITLE: Title Page SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 1 OF 95	DRAWN BY: BF



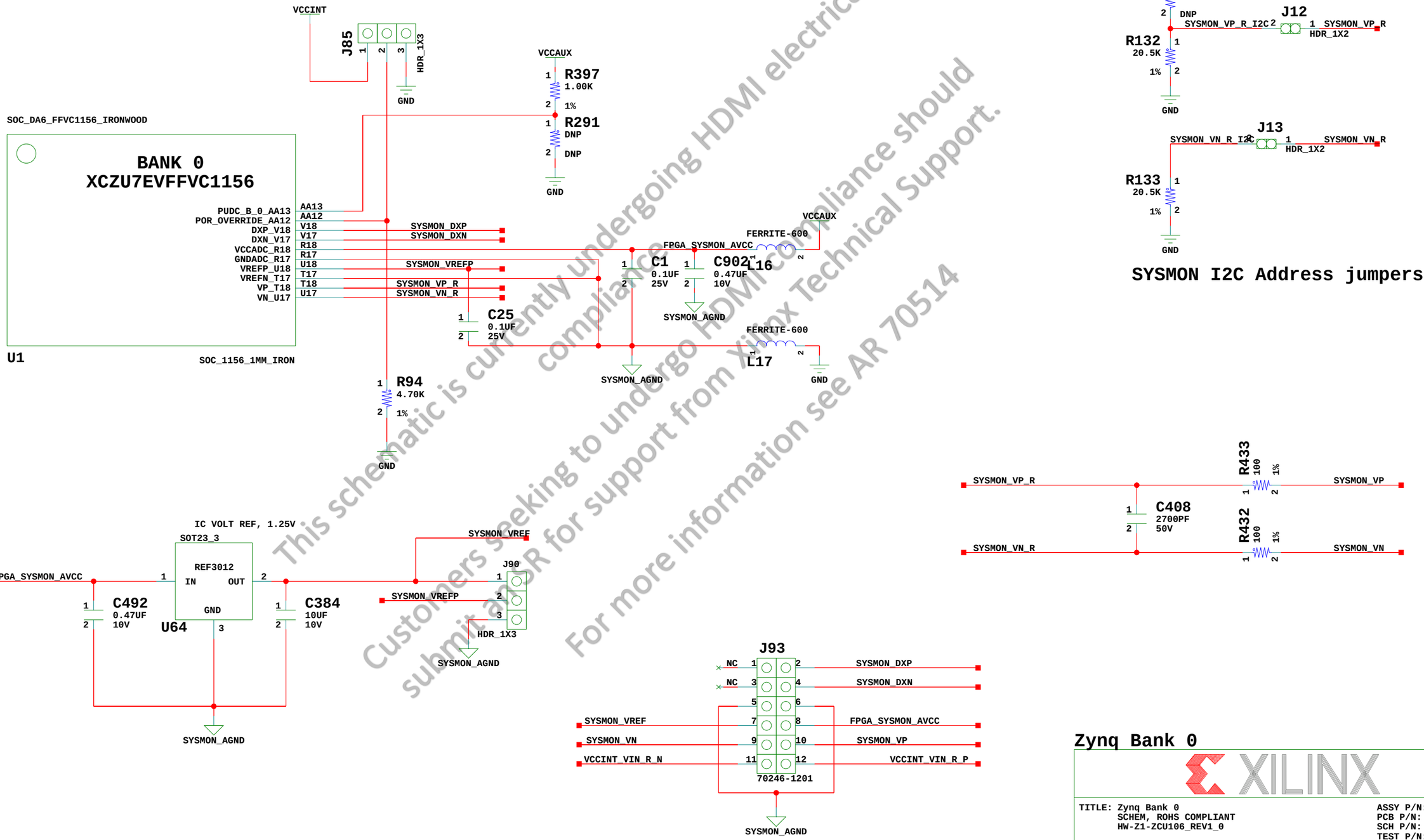
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For more information see AR 70514

XILINX

TITLE: Block Diagram
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 2 OF 95	DRAWN BY: BF



Zynq Bank 0

XILINX

TITLE: Zynq Bank 0 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 3 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

FMC_HPC1_VREF_A_M2C

R177 DNP / DNP / DNP

C542 DNP / DNP / DNP

GND

BANK 28
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 67
ZU7CG = BANK 28
ZU7EG = BANK 28

IO_T3U_N12_28_B23	B23	PMOD0_0_LS
IO_L24N_T3U_N11_28_B21	B21	SDI_SCLK_LS
IO_L24P_T3U_N10_28_B20	B20	SYSMON_SDA_LS
IO_L23N_T3U_N9_28_A23	A23	PMOD0_1_LS
IO_L23P_T3U_N8_28_A22	A22	SYSMON_SCL_LS
IO_L22N_T3U_N7_DBC_AD0N_28_B19	B19	FMC_HPC1_LA00_CC_N
IO_L22P_T3U_N6_DBC_AD0P_28_B18	B18	FMC_HPC1_LA00_CC_P
IO_L21N_T3L_N5_AD8N_28_A21	A21	FMC_HPC1_LA11_N
IO_L21P_T3L_N4_AD8P_28_A20	A20	FMC_HPC1_LA11_P
IO_L20N_T3L_N3_AD1N_28_C19	C19	FMC_HPC1_LA16_N
IO_L20P_T3L_N2_AD1P_28_C18	C18	FMC_HPC1_LA16_P
IO_L19N_T3L_N1_DBC_AD9N_28_A19	A19	FMC_HPC1_LA15_N
IO_L19P_T3L_N0_DBC_AD9P_28_A18	A18	FMC_HPC1_LA15_P
IO_T2U_N12_28_F25	F25	PMOD0_2_LS
IO_L18N_T2U_N11_AD2N_28_G26	G26	FMC_HPC1_LA05_N
IO_L18P_T2U_N10_AD2P_28_G25	G25	FMC_HPC1_LA05_P
IO_L17N_T2U_N9_AD10N_28_C23	C23	FMC_HPC1_LA07_N
IO_L17P_T2U_N8_AD10P_28_D22	D22	FMC_HPC1_LA07_P
IO_L16N_T2U_N7_QBC_AD3N_28_D24	D24	FMC_HPC1_LA01_CC_N
IO_L16P_T2U_N6_QBC_AD3P_28_E24	E24	FMC_HPC1_LA01_CC_P
IO_L15N_T2L_N5_AD11N_28_C22	C22	FMC_HPC1_LA13_N
IO_L15P_T2L_N4_AD11P_28_C21	C21	FMC_HPC1_LA13_P
IO_L14N_T2L_N3_GC_28_G24	G24	L14N_HDGC_65_N
IO_L14P_T2L_N2_GC_28_G23	G23	L14P_HDGC_65_P
IO_L13N_T2L_N1_GC_QBC_28_E23	E23	FMC_HPC1_CLK0_M2C_N
IO_L13P_T2L_N0_GC_QBC_28_F23	F23	FMC_HPC1_CLK0_M2C_P
IO_T1U_N12_28_E20	E20	PMOD0_3_LS
IO_L12N_T1U_N11_GC_28_F21	F21	HDMI_TX_LVDS_OUT_N
IO_L12P_T1U_N10_GC_28_G21	G21	HDMI_TX_LVDS_OUT_P
IO_L11N_T1U_N9_GC_28_E22	E22	FMC_HPC1_LA10_N
IO_L11P_T1U_N8_GC_28_F22	F22	FMC_HPC1_LA10_P
IO_L10N_T1U_N7_QBC_AD4N_28_F20	F20	FMC_HPC1_LA09_N
IO_L10P_T1U_N6_QBC_AD4P_28_G20	G20	FMC_HPC1_LA09_P
IO_L9N_T1L_N5_AD12N_28_D21	D21	FMC_HPC1_LA14_N
IO_L9P_T1L_N4_AD12P_28_D20	D20	FMC_HPC1_LA14_P
IO_L8N_T1L_N3_AD5N_28_H22	H22	FMC_HPC1_LA06_N
IO_L8P_T1L_N2_AD5P_28_H21	H21	FMC_HPC1_LA06_P
IO_L7N_T1L_N1_QBC_AD13N_28_D19	D19	FMC_HPC1_LA12_N
IO_L7P_T1L_N0_QBC_AD13P_28_E19	E19	FMC_HPC1_LA12_P
IO_T0U_N12_VRP_28_H23	H23	SDI_MISO_LS
IO_L6N_T0U_N11_AD6N_28_H24	H24	FMC_HPC1_LA04_N
IO_L6P_T0U_N10_AD6P_28_J24	J24	FMC_HPC1_LA04_P
IO_L5N_T0U_N9_AD14N_28_H26	H26	FMC_HPC1_LA08_N
IO_L5P_T0U_N8_AD14P_28_J25	J25	FMC_HPC1_LA08_P
IO_L4N_T0U_N7_DBC_AD7N_28_K23	K23	FMC_HPC1_LA02_N
IO_L4P_T0U_N6_DBC_AD7P_28_K22	K22	FMC_HPC1_LA02_P
IO_L3N_T0L_N5_AD15N_28_J22	J22	FMC_HPC1_LA03_N
IO_L3P_T0L_N4_AD15P_28_J21	J21	FMC_HPC1_LA03_P
IO_L2N_T0L_N3_28_K24	K24	PMOD0_4_LS
IO_L2P_T0L_N2_28_L23	L23	PMOD0_5_LS
IO_L1N_T0L_N1_DBC_28_L22	L22	PMOD0_6_LS
IO_L1P_T0L_N0_DBC_28_L21	L21	SDI_MOSI_LS
VREF_28_M23	M23	

VADJ_FMC

E21	VCCO_28_E21
F24	VCCO_28_F24
H20	VCCO_28_H20

U1

SOC_1156_1MM_IRON

This schematic is designed to support HDMI electrical compliance. Customers should submit their HDMI compliance report from Quantum Data Inc. See AR 70514 for more information.

Zynq Banks 28

XILINX

TITLE: Zynq Banks 28
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 4 OF 95	DRAWN BY: BF

Layout: Place resistor and capacitor for VREF

Underneath the FPGA via array
right next to the via

SOC_DA6_FFVC1156_IRONWOOD

BANK 64
XCZU7EVFFVC1156

IO_T3U_N12_64_AA17	AA17	MSP430_UCA1_TXD_LS
IO_L24N_T3U_N11_64_AD16	AD16	DDR4_DQ11
IO_L24P_T3U_N10_64_AD17	AD17	DDR4_DQ15
IO_L23N_T3U_N9_64_AB14	AB14	DDR4_DQ14
IO_L23P_T3U_N8_64_AA14	AA14	DDR4_DQ8
IO_L22N_T3U_N7_DBC_AD0N_64_AA15	AA15	DDR4_DQS1_C
IO_L22P_T3U_N6_DBC_AD0P_64_AA16	AA16	DDR4_DQS1_T
IO_L21N_T3L_N5_AD8N_64_AB15	AB15	DDR4_DQ10
IO_L21P_T3L_N4_AD8P_64_AB16	AB16	DDR4_DQ12
IO_L20N_T3L_N3_AD1N_64_AC16	AC16	DDR4_DQ9
IO_L20P_T3L_N2_AD1P_64_AC17	AC17	DDR4_DQ13
IO_L19N_T3L_N1_DBC_AD9N_64_AE15	AE15	GPIO_LED_3_LS
IO_L19P_T3L_N0_DBC_AD9P_64_AD15	AD15	DDR4_DM1
IO_T2U_N12_64_AH16	AH16	MSP430_UCA1_RXD_LS
IO_L18N_T2U_N11_AD2N_64_AG14	AG14	DDR4_DQ6
IO_L18P_T2U_N10_AD2P_64_AG15	AG15	DDR4_DQ2
IO_L17N_T2U_N9_AD10N_64_AF15	AF15	DDR4_DQ4
IO_L17P_T2U_N8_AD10P_64_AF16	AF16	DDR4_DQ0
IO_L16N_T2U_N7_QBC_AD3N_64_AJ14	AJ14	DDR4_DQS0_C
IO_L16P_T2U_N6_QBC_AD3P_64_AH14	AH14	DDR4_DQS0_T
IO_L15N_T2L_N5_AD11N_64_AF17	AF17	DDR4_DQ3
IO_L15P_T2L_N4_AD11P_64_AE17	AE17	DDR4_DQ7
IO_L14N_T2L_N3_GC_64_AG18	AG18	DDR4_DQ5
IO_L14P_T2L_N2_GC_64_AF18	AF18	DDR4_DQ1
IO_L13N_T2L_N1_GC_QBC_64_AH17	AH17	UART2_TXD_O_FPGA_RXD
IO_L13P_T2L_N0_GC_QBC_64_AH18	AH18	DDR4_DM0
IO_T1U_N12_64_AL17	AL17	UART2_RXD_I_FPGA_TXD
IO_L12N_T1U_N11_GC_64_AJ15	AJ15	DDR4_DQ20
IO_L12P_T1U_N10_GC_64_AJ16	AJ16	DDR4_DQ16
IO_L11N_T1U_N9_GC_64_AK17	AK17	DDR4_DQ19
IO_L11P_T1U_N8_GC_64_AJ17	AJ17	DDR4_DQ17
IO_L10N_T1U_N7_QBC_AD4N_64_AK14	AK14	DDR4_DQS2_C
IO_L10P_T1U_N6_QBC_AD4P_64_AK15	AK15	DDR4_DQS2_T
IO_L9N_T1L_N5_AD12N_64_AL18	AL18	DDR4_DQ23
IO_L9P_T1L_N4_AD12P_64_AK18	AK18	DDR4_DQ21
IO_L8N_T1L_N3_AD5N_64_AL15	AL15	DDR4_DQ18
IO_L8P_T1L_N2_AD5P_64_AL16	AL16	DDR4_DQ22
IO_L7N_T1L_N1_QBC_AD13N_64_AM15	AM15	UART2_RTS_O_B
IO_L7P_T1L_N0_QBC_AD13P_64_AM16	AM16	DDR4_DM2
IO_T0U_N12_VRP_64_AP14	AP14	VRP_64
IO_L6N_T0U_N11_AD6N_64_AN16	AN16	DDR4_DQ27
IO_L6P_T0U_N10_AD6P_64_AN17	AN17	DDR4_DQ30
IO_L5N_T0U_N9_AD14N_64_AP15	AP15	DDR4_DQ26
IO_L5P_T0U_N8_AD14P_64_AP16	AP16	DDR4_DQ25
IO_L4N_T0U_N7_DBC_AD7N_64_AN14	AN14	DDR4_DQS3_C
IO_L4P_T0U_N6_DBC_AD7P_64_AM14	AM14	DDR4_DQS3_T
IO_L3N_T0L_N5_AD15N_64_AN18	AN18	DDR4_DQ31
IO_L3P_T0L_N4_AD15P_64_AM18	AM18	DDR4_DQ29
IO_L2N_T0L_N3_64_AP13	AP13	DDR4_DQ24
IO_L2P_T0L_N2_64_AN13	AN13	DDR4_DQ28
IO_L1N_T0L_N1_DBC_64_AP17	AP17	UART2_CTS_I_B
IO_L1P_T0L_N0_DBC_64_AP18	AP18	DDR4_DM3
VREF_64_AG16	AG16	

VCC1V2
AE16 VCCO_64_AE16
AH15 VCCO_64_AH15
AJ18 VCCO_64_AJ18

R683 1.00K 1/16W 1%
GND

U1 SOC_1156_1MM_IRON

VRP_64
R622 240 1/10W 1%
GND

SOC_DA6_FFVC1156_IRONWOOD

BANK 65
XCZU7EVFFVC1156

IO_T3U_N12_65_AE22	AE22	SFP0_TX_DISABLE
IO_L24N_T3U_N11_PERSTN0_65_AA20	AA20	DDR4_DQ36
IO_L24P_T3U_N10_PERSTN1_I2C_SDA_65_AA19	AA19	DDR4_DQ38
IO_L23N_T3U_N9_65_AD19	AD19	DDR4_DQ33
IO_L23P_T3U_N8_I2C_SCLK_65_AC18	AC18	DDR4_DQ34
IO_L22N_T3U_N7_DBC_AD0N_65_AB18	AB18	DDR4_DQS4_C
IO_L22P_T3U_N6_DBC_AD0P_65_AA18	AA18	DDR4_DQS4_T
IO_L21N_T3L_N5_AD8N_65_AE20	AE20	DDR4_DQ37
IO_L21P_T3L_N4_AD8P_65_AD20	AD20	DDR4_DQ39
IO_L20N_T3L_N3_AD1N_65_AC19	AC19	DDR4_DQ35
IO_L20P_T3L_N2_AD1P_65_AB19	AB19	DDR4_DQ32
IO_L19N_T3L_N1_DBC_AD9N_65_AE19	AE19	PL_I2C0_SCL_LS
IO_L19P_T3L_N0_DBC_AD9P_65_AE18	AE18	DDR4_DM4
IO_T2U_N12_65_AF20	AF20	SFP1_TX_DISABLE
IO_L18N_T2U_N11_AD2N_65_AE24	AE24	DDR4_DQ44
IO_L18P_T2U_N10_AD2P_65_AE23	AE23	DDR4_DQ46
IO_L17N_T2U_N9_AD10N_65_AF22	AF22	DDR4_DQ40
IO_L17P_T2U_N8_AD10P_65_AF21	AF21	DDR4_DQ47
IO_L16N_T2U_N7_QBC_AD3N_65_AG23	AG23	DDR4_DQS5_C
IO_L16P_T2U_N6_QBC_AD3P_65_AF23	AF23	DDR4_DQS5_T
IO_L15N_T2L_N5_AD11N_65_AG20	AG20	DDR4_DQ45
IO_L15P_T2L_N4_AD11P_65_AG19	AG19	DDR4_DQ42
IO_L14N_T2L_N3_GC_65_AH21	AH21	DDR4_DQ41
IO_L14P_T2L_N2_GC_65_AG21	AG21	DDR4_DQ43
IO_L13N_T2L_N1_GC_QBC_65_AH23	AH23	PL_I2C0_SDA_LS
IO_L13P_T2L_N0_GC_QBC_65_AH22	AH22	DDR4_DM5
IO_T1U_N12_65_AH19	AH19	PL_I2C1_SCL_LS
IO_L12N_T1U_N11_GC_65_AJ22	AJ22	DDR4_DQ49
IO_L12P_T1U_N10_GC_65_AJ21	AJ21	DDR4_DQ51
IO_L11N_T1U_N9_GC_65_AK20	AK20	DDR4_DQ52
IO_L11P_T1U_N8_GC_65_AJ20	AJ20	DDR4_DQ55
IO_L10N_T1U_N7_QBC_AD4N_65_AK23	AK23	DDR4_DQS6_C
IO_L10P_T1U_N6_QBC_AD4P_65_AK22	AK22	DDR4_DQS6_T
IO_L9N_T1L_N5_AD12N_65_AK19	AK19	DDR4_DQ54
IO_L9P_T1L_N4_AD12P_65_AJ19	AJ19	DDR4_DQ53
IO_L8N_T1L_N3_AD5N_65_AL23	AL23	DDR4_DQ50
IO_L8P_T1L_N2_AD5P_65_AL22	AL22	DDR4_DQ48
IO_L7N_T1L_N1_QBC_AD13N_65_AL21	AL21	PL_I2C1_SDA_LS
IO_L7P_T1L_N0_QBC_AD13P_65_AL20	AL20	DDR4_DM6
IO_T0U_N12_VRP_65_AM20	AM20	VRP_65
IO_L6N_T0U_N11_AD6N_65_AN23	AN23	DDR4_DQ63
IO_L6P_T0U_N10_AD6P_65_AM23	AM23	DDR4_DQ61
IO_L5N_T0U_N9_AD14N_65_AP23	AP23	DDR4_DQ59
IO_L5P_T0U_N8_AD14P_65_AN22	AN22	DDR4_DQ57
IO_L4N_T0U_N7_DBC_AD7N_65_AN21	AN21	DDR4_DQS7_C
IO_L4P_T0U_N6_DBC_AD7P_SMBALERT_65_AM21	AM21	DDR4_DQS7_T
IO_L3N_T0L_N5_AD15N_65_AP22	AP22	DDR4_DQ56
IO_L3P_T0L_N4_AD15P_65_AP21	AP21	DDR4_DQ58
IO_L2N_T0L_N3_65_AN19	AN19	DDR4_DQ62
IO_L2P_T0L_N2_65_AM19	AM19	DDR4_DQ60
IO_L1N_T0L_N1_DBC_65_AP20	AP20	GPIO_SW_S
IO_L1P_T0L_N0_DBC_65_AP19	AP19	DDR4_DM7
VREF_65_AB20	AB20	

VCC1V2
AF19 VCCO_65_AF19
AG22 VCCO_65_AG22
AK21 VCCO_65_AK21

R681 1.00K 1/16W 1%
GND

U1 SOC_1156_1MM_IRON

VRP_65
R621 240 1/10W 1%
GND

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Support.
For more information or support.

Zynq Banks 64 65

XILINX

TITLE: Zynq Banks 64 65
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 5 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 66
XCZU7EVFFVC1156

IO_T3U_N12_66_AC14	AC14	GPIO_SW_E
IO_L24N_T3U_N11_66_AC13	AC13	DDR4_PAR
IO_L24P_T3U_N10_66_AB13	AB13	DDR4_CKE
IO_L23N_T3U_N9_66_AF12	AF12	DDR4_RESET_B
IO_L23P_T3U_N8_66_AE12	AE12	DDR4_A15_CAS_B
IO_L22N_T3U_N7_DBC_AD0N_66_AD12	AD12	DDR4_CS_B
IO_L22P_T3U_N6_DBC_AD0P_66_AC12	AC12	DDR4_A14_WE_B
IO_L21N_T3L_N5_AD8N_66_AF13	AF13	AES_OUT_N
IO_L21P_T3L_N4_AD8P_66_AE13	AE13	AES_OUT_P
IO_L20N_T3L_N3_AD1N_66_AE14	AE14	DDR4_BG0
IO_L20P_T3L_N2_AD1P_66_AD14	AD14	DDR4_ACT_B
IO_L19N_T3L_N1_DBC_AD9N_66_AF10	AF10	DDR4_ODT
IO_L19P_T3L_N0_DBC_AD9P_66_AF11	AF11	DDR4_A16_RAS_B
IO_T2U_N12_66_AH8	AH8	DDR4_A5
IO_L18N_T2U_N11_AD2N_66_AG8	AG8	DDR4_A7
IO_L18P_T2U_N10_AD2P_66_AF8	AF8	DDR4_A13
IO_L17N_T2U_N9_AD10N_66_AG10	AG10	DDR4_A9
IO_L17P_T2U_N8_AD10P_66_AG11	AG11	DDR4_A1
IO_L16N_T2U_N7_QBC_AD3N_66_AH9	AH9	DDR4_A8
IO_L16P_T2U_N6_QBC_AD3P_66_AG9	AG9	DDR4_A11
IO_L15N_T2L_N5_AD11N_66_AH13	AH13	DDR4_A10
IO_L15P_T2L_N4_AD11P_66_AG13	AG13	GPIO_SW_N
IO_L14N_T2L_N3_GC_66_AJ11	AJ11	DDR4_CK_C
IO_L14P_T2L_N2_GC_66_AH11	AH11	DDR4_CK_T
IO_L13N_T2L_N1_GC_QBC_66_AJ12	AJ12	USER_SI570_N
IO_L13P_T2L_N0_GC_QBC_66_AH12	AH12	USER_SI570_P
IO_T1U_N12_66_AM13	AM13	DDR4_A12
IO_L12N_T1U_N11_GC_66_AK9	AK9	DDR4_A0
IO_L12P_T1U_N10_GC_66_AJ9	AJ9	DDR4_A6
IO_L11N_T1U_N9_GC_66_AK10	AK10	DDR4_A4
IO_L11P_T1U_N8_GC_66_AJ10	AJ10	DDR4_A2
IO_L10N_T1U_N7_QBC_AD4N_66_AL8	AL8	DDR4_A3
IO_L10P_T1U_N6_QBC_AD4P_66_AK8	AK8	DDR4_BA0
IO_L9N_T1L_N5_AD12N_66_AL12	AL12	DDR4_BA1
IO_L9P_T1L_N4_AD12P_66_AK12	AK12	GPIO_SW_W
IO_L8N_T1L_N3_AD5N_66_AL10	AL10	GPIO_SW_C
IO_L8P_T1L_N2_AD5P_66_AL11	AL11	GPIO_LED_0_LS
IO_L7N_T1L_N1_QBC_AD13N_66_AL13	AL13	GPIO_LED_1_LS
IO_L7P_T1L_N0_QBC_AD13P_66_AK13	AK13	GPIO_LED_2_LS
IO_T0U_N12_VRP_66_AP8	AP8	VRP_66
IO_L6N_T0U_N11_AD6N_66_AM8	AM8	GPIO_LED_4_LS
IO_L6P_T0U_N10_AD6P_66_AM9	AM9	GPIO_LED_5_LS
IO_L5N_T0U_N9_AD14N_66_AM10	AM10	GPIO_LED_6_LS
IO_L5P_T0U_N8_AD14P_66_AM11	AM11	GPIO_LED_7_LS
IO_L4N_T0U_N7_DBC_AD7N_66_AN8	AN8	PMOD1_0_LS
IO_L4P_T0U_N6_DBC_AD7P_66_AN9	AN9	PMOD1_1_LS
IO_L3N_T0L_N5_AD15N_66_AP11	AP11	PMOD1_2_LS
IO_L3P_T0L_N4_AD15P_66_AN11	AN11	PMOD1_3_LS
IO_L2N_T0L_N3_66_AP9	AP9	PMOD1_4_LS
IO_L2P_T0L_N2_66_AP10	AP10	PMOD1_5_LS
IO_L1N_T0L_N1_DBC_66_AP12	AP12	PMOD1_6_LS
IO_L1P_T0L_N0_DBC_66_AN12	AN12	PMOD1_7_LS
VREF_66_AB12	AB12	

VCC1V2

AF9	VCCO_66_AF9	
AG12	VCCO_66_AG12	
AK11	VCCO_66_AK11	

U1

SOC_1156_1MM_IRON

USER_SI570_P

R303 1 — 100 1/10W 1% — 2

USER_SI570_N

R723 1 — 1.00K 1/16W 1% — 2

GND

VRP_66

R968 1 — 240 1/10W 1% — 2

GND

SOC_DA6_FFVC1156_IRONWOOD

BANK 67
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 68
ZU7CG = BANK 67
ZU7EG = BANK 67

IO_T3U_N12_67_J20	J20	SDI_CS_RCVR_LS
IO_L24N_T3U_N11_67_L16	L16	FMC_HPC0_LA04_N
IO_L24P_T3U_N10_67_L17	L17	FMC_HPC0_LA04_P
IO_L23N_T3U_N9_67_K18	K18	FMC_HPC0_LA03_N
IO_L23P_T3U_N8_67_K19	K19	FMC_HPC0_LA03_P
IO_L22N_T3U_N7_DBC_AD0N_67_K15	K15	FMC_HPC0_LA10_N
IO_L22P_T3U_N6_DBC_AD0P_67_L15	L15	FMC_HPC0_LA10_P
IO_L21N_T3L_N5_AD8N_67_J17	J17	FMC_HPC0_LA05_N
IO_L21P_T3L_N4_AD8P_67_K17	K17	FMC_HPC0_LA05_P
IO_L20N_T3L_N3_AD1N_67_J15	J15	FMC_HPC0_LA07_N
IO_L20P_T3L_N2_AD1P_67_J16	J16	FMC_HPC0_LA07_P
IO_L19N_T3L_N1_DBC_AD9N_67_K20	K20	FMC_HPC0_LA02_N
IO_L19P_T3L_N0_DBC_AD9P_67_L20	L20	FMC_HPC0_LA02_P
IO_T2U_N12_67_J19	J19	SDI_CS_DRVR_LS
IO_L18N_T2U_N11_AD2N_67_G16	G16	FMC_HPC0_LA09_N
IO_L18P_T2U_N10_AD2P_67_H16	H16	FMC_HPC0_LA09_P
IO_L17N_T2U_N9_AD10N_67_F18	F18	FMC_HPC0_LA12_N
IO_L17P_T2U_N8_AD10P_67_G18	G18	FMC_HPC0_LA12_P
IO_L16N_T2U_N7_QBC_AD3N_67_H17	H17	FMC_HPC0_LA01_CC_N
IO_L16P_T2U_N6_QBC_AD3P_67_H18	H18	FMC_HPC0_LA01_CC_P
IO_L15N_T2L_N5_AD11N_67_G19	G19	FMC_HPC0_LA06_N
IO_L15P_T2L_N4_AD11P_67_H19	H19	FMC_HPC0_LA06_P
IO_L14N_T2L_N3_GC_67_F15	F15	FMC_HPC0_LA13_N
IO_L14P_T2L_N2_GC_67_G15	G15	FMC_HPC0_LA13_P
IO_L13N_T2L_N1_GC_QBC_67_F16	F16	FMC_HPC0_LA00_CC_N
IO_L13P_T2L_N0_GC_QBC_67_F17	F17	FMC_HPC0_LA00_CC_P
IO_T1U_N12_67_E13	E13	SDI_XALARM_RX_LS
IO_L12N_T1U_N11_GC_67_E14	E14	FMC_HPC0_CLK0_M2C_N
IO_L12P_T1U_N10_GC_67_E15	E15	FMC_HPC0_CLK0_M2C_P
IO_L11N_T1U_N9_GC_67_D14	D14	CLK_74_25_N
IO_L11P_T1U_N8_GC_67_D15	D15	CLK_74_25_P
IO_L10N_T1U_N7_QBC_AD4N_67_F13	F13	HDMI_REC_CLOCK_C_N
IO_L10P_T1U_N6_QBC_AD4P_67_G14	G14	HDMI_REC_CLOCK_C_P
IO_L9N_T1L_N5_AD12N_67_E17	E17	FMC_HPC0_LA08_N
IO_L9P_T1L_N4_AD12P_67_E18	E18	FMC_HPC0_LA08_P
IO_L8N_T1L_N3_AD5N_67_C17	C17	FMC_HPC0_LA16_N
IO_L8P_T1L_N2_AD5P_67_D17	D17	FMC_HPC0_LA16_P
IO_L7N_T1L_N1_QBC_AD13N_67_C16	C16	FMC_HPC0_LA15_N
IO_L7P_T1L_N0_QBC_AD13P_67_D16	D16	FMC_HPC0_LA15_P
IO_T0U_N12_VRP_67_C14	C14	SDI_XALARM_TX_LS
IO_L6N_T0U_N11_AD6N_67_C12	C12	FMC_HPC0_LA14_N
IO_L6P_T0U_N10_AD6P_67_C13	C13	FMC_HPC0_LA14_P
IO_L5N_T0U_N9_AD14N_67_A12	A12	FMC_HPC0_LA11_N
IO_L5P_T0U_N8_AD14P_67_A13	A13	FMC_HPC0_LA11_P
IO_L4N_T0U_N7_DBC_AD7N_67_B13	B13	GPIO_DIP_SW7
IO_L4P_T0U_N6_DBC_AD7P_67_B14	B14	GPIO_DIP_SW6
IO_L3N_T0L_N5_AD15N_67_A14	A14	GPIO_DIP_SW5
IO_L3P_T0L_N4_AD15P_67_A15	A15	GPIO_DIP_SW4
IO_L2N_T0L_N3_67_B15	B15	GPIO_DIP_SW3
IO_L2P_T0L_N2_67_B16	B16	GPIO_DIP_SW2
IO_L1N_T0L_N1_DBC_67_A16	A16	GPIO_DIP_SW1
IO_L1P_T0L_N0_DBC_67_A17	A17	GPIO_DIP_SW0
VREF_67_L18	L18	

VADJ_FMC

D13	VCCO_67_D13	
E16	VCCO_67_E16	
H15	VCCO_67_H15	

U1

SOC_1156_1MM_IRON

FMC_HPC0_VREF_A_M2C

R967 1 — DNP DNP 2 — DNP

C1061 1 — DNP 2 — DNP

GND

CLK_74_25_P

R953 1 — 100 1/10W 1% — 2

CLK_74_25_N

This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical support. For more information see UG...

Zynq Banks 66 67

XILINX

TITLE: Zynq Banks 66 67 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 6 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 68
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 69
ZU7CG = BANK 68
ZU7EG = BANK 68

IO_T3U_N12_68_A9	A9	SDI_CS_RCLKR_LS
IO_L24N_T3U_N11_68_A11	A11	FMC_HPC0_LA23_N
IO_L24P_T3U_N10_68_B11	B11	FMC_HPC0_LA23_P
IO_L23N_T3U_N9_68_A7	A7	FMC_HPC0_LA27_N
IO_L23P_T3U_N8_68_A8	A8	FMC_HPC0_LA27_P
IO_L22N_T3U_N7_DBC_AD0N_68_A10	A10	FMC_HPC0_LA21_N
IO_L22P_T3U_N6_DBC_AD0P_68_B10	B10	FMC_HPC0_LA21_P
IO_L21N_T3L_N5_AD8N_68_A6	A6	FMC_HPC0_LA24_N
IO_L21P_T3L_N4_AD8P_68_B6	B6	FMC_HPC0_LA24_P
IO_L20N_T3L_N3_AD1N_68_B8	B8	FMC_HPC0_LA26_N
IO_L20P_T3L_N2_AD1P_68_B9	B9	FMC_HPC0_LA26_P
IO_L19N_T3L_N1_DBC_AD9N_68_C6	C6	FMC_HPC0_LA25_N
IO_L19P_T3L_N0_DBC_AD9P_68_C7	C7	FMC_HPC0_LA25_P
IO_T2U_N12_68_G13	G13	CPU_RESET
IO_L18N_T2U_N11_AD2N_68_C11	C11	FMC_HPC0_LA19_N
IO_L18P_T2U_N10_AD2P_68_D12	D12	FMC_HPC0_LA19_P
IO_L17N_T2U_N9_AD10N_68_E12	E12	FMC_HPC0_LA20_N
IO_L17P_T2U_N8_AD10P_68_F12	F12	FMC_HPC0_LA20_P
IO_L16N_T2U_N7_QBC_AD3N_68_D10	D10	FMC_HPC0_LA18_CC_N
IO_L16P_T2U_N6_QBC_AD3P_68_D11	D11	FMC_HPC0_LA18_CC_P
IO_L15N_T2L_N5_AD11N_68_H12	H12	FMC_HPC0_LA22_N
IO_L15P_T2L_N4_AD11P_68_H13	H13	FMC_HPC0_LA22_P
IO_L14N_T2L_N3_GC_68_E10	E10	FMC_HPC0_LA17_CC_N
IO_L14P_T2L_N2_GC_68_F11	F11	FMC_HPC0_LA17_CC_P
IO_L13N_T2L_N1_GC_QBC_68_G11	G11	SFP_REC_CLOCK_C_N
IO_L13P_T2L_N0_GC_QBC_68_H11	H11	SFP_REC_CLOCK_C_P
IO_T1U_N12_68_D7	D7	PMOD0_7_LS
IO_L12N_T1U_N11_GC_68_F10	F10	FMC_HPC0_CLK1_M2C_N
IO_L12P_T1U_N10_GC_68_G10	G10	FMC_HPC0_CLK1_M2C_P
IO_L11N_T1U_N9_GC_68_G9	G9	CLK_125_N
IO_L11P_T1U_N8_GC_68_H9	H9	CLK_125_P
IO_L10N_T1U_N7_QBC_AD4N_68_D9	D9	FMC_HPC0_LA30_N
IO_L10P_T1U_N6_QBC_AD4P_68_E9	E9	FMC_HPC0_LA30_P
IO_L9N_T1L_N5_AD12N_68_E8	E8	FMC_HPC0_LA32_N
IO_L9P_T1L_N4_AD12P_68_F8	F8	FMC_HPC0_LA32_P
IO_L8N_T1L_N3_AD5N_68_C8	C8	FMC_HPC0_LA33_N
IO_L8P_T1L_N2_AD5P_68_C9	C9	FMC_HPC0_LA33_P
IO_L7N_T1L_N1_QBC_AD13N_68_E7	E7	FMC_HPC0_LA31_N
IO_L7P_T1L_N0_QBC_AD13P_68_F7	F7	FMC_HPC0_LA31_P
IO_T0U_N12_VRP_68_H14	H14	VRP_68
IO_L6N_T0U_N11_AD6N_68_K13	K13	L6N_AD6N_64_N
IO_L6P_T0U_N10_AD6P_68_L14	L14	L6P_AD6P_64_P
IO_L5N_T0U_N9_AD14N_68_J14	J14	L5N_AD14N_64_N
IO_L5P_T0U_N8_AD14P_68_K14	K14	L5P_AD14P_64_P
IO_L4N_T0U_N7_DBC_AD7N_68_J11	J11	L4N_AD7N_64_N
IO_L4P_T0U_N6_DBC_AD7P_68_K12	K12	L4P_AD7P_64_P
IO_L3N_T0L_N5_AD15N_68_L11	L11	L3N_AD15N_64_N
IO_L3P_T0L_N4_AD15P_68_L12	L12	L3P_AD15P_64_P
IO_L2N_T0L_N3_68_J10	J10	FMC_HPC0_LA29_N
IO_L2P_T0L_N2_68_K10	K10	FMC_HPC0_LA29_P
IO_L1N_T0L_N1_DBC_68_L13	L13	FMC_HPC0_LA28_N
IO_L1P_T0L_N0_DBC_68_M13	M13	FMC_HPC0_LA28_P
VREF_68_J12	J12	

VADJ_FMC		
F9	VCCO_68_F9	
G12	VCCO_68_G12	
K11	VCCO_68_K11	

U1

SOC_1156_1MM_IRON

FMC_HPC0_VREF_A_M2C

R180: DNP / DNP / DNP

C543: DNP / DNP

GND

CLK_125_P

R954: 100 / 1/10W / 1%

CLK_125_N

VRP_68

R345: 240 / 1/10W / 1%

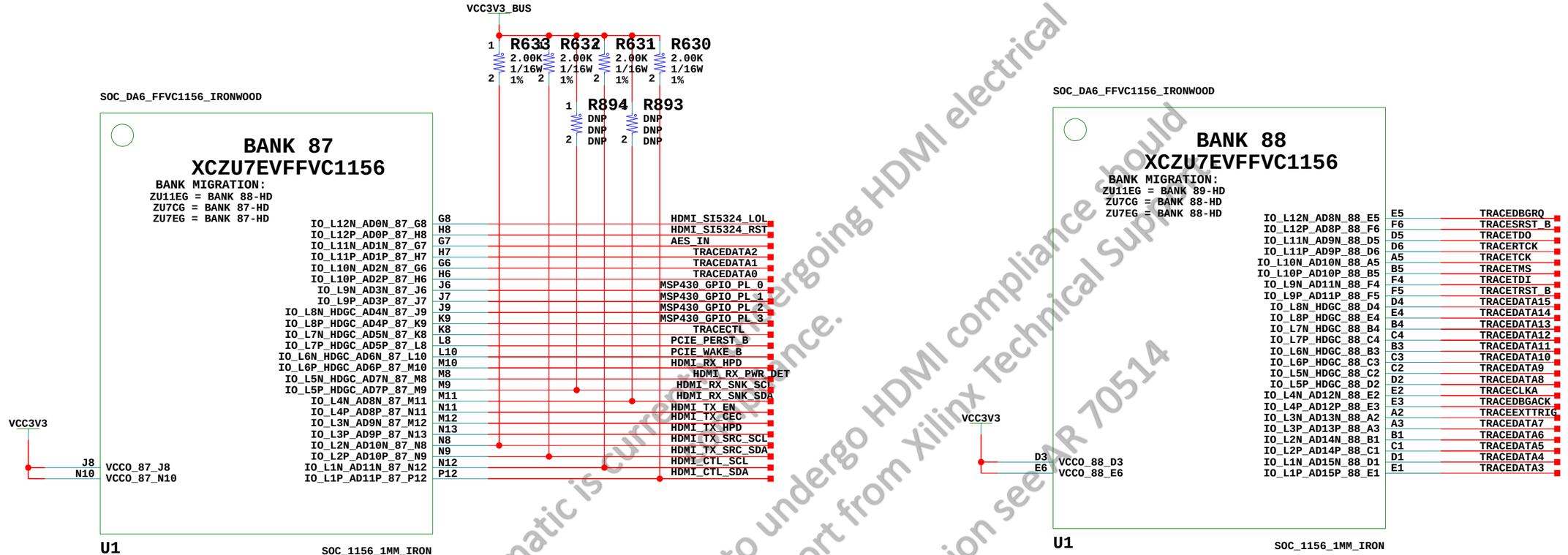
GND

Design currently undergoing HDMI electrical compliance.
Customer seeking to undergo HDMI compliance should contact an SR for support from Xilinx Technical Support.
For more information see AR 70514

Zynq Banks 68

XILINX

TITLE: Zynq Banks 68 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 7 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

Zynq Banks 87 88

XILINX

TITLE: Zynq Banks 87 88 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 8 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 223
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 224
ZU7CG = BANK 223
ZU7EG = BANK 223

MGTHTXP0_223_AN6	AN6	HDMI_TX0_P
MGTHTXN0_223_AN5	AN5	HDMI_TX0_N
MGTHRXP0_223_AP4	AP4	HDMI_RX0_C_P
MGTHRXN0_223_AP3	AP3	HDMI_RX0_C_N
MGTHTXP1_223_AM4	AM4	HDMI_TX1_P
MGTHTXN1_223_AM3	AM3	HDMI_TX1_N
MGTHRXP1_223_AN2	AN2	HDMI_RX1_C_P
MGTHRXN1_223_AN1	AN1	HDMI_RX1_C_N
MGTHTXP2_223_AL6	AL6	HDMI_TX2_P
MGTHTXN2_223_AL5	AL5	HDMI_TX2_N
MGTHRXP2_223_AL2	AL2	HDMI_RX2_C_P
MGTHRXN2_223_AL1	AL1	HDMI_RX2_C_N
MGTHTXP3_223_AJ6	AJ6	FMC_HPC1_DP0_C2M_P
MGTHTXN3_223_AJ5	AJ5	FMC_HPC1_DP0_C2M_N
MGTHRXP3_223_AK4	AK4	FMC_HPC1_DP0_M2C_P
MGTHRXN3_223_AK3	AK3	FMC_HPC1_DP0_M2C_N
MGTREFCLK0P_223_AD8	AD8	HDMI_SI5324_OUT_C_P
MGTREFCLK0N_223_AD7	AD7	HDMI_SI5324_OUT_C_N
MGTREFCLK1P_223_AC10	AC10	HDMI_RX_CLK_C_P
MGTREFCLK1N_223_AC9	AC9	HDMI_RX_CLK_C_N

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK 224
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 225
ZU7CG = BANK 224
ZU7EG = BANK 224

MGTHTXP0_224_AH4	AH4	PCIE_TX3_P
MGTHTXN0_224_AH3	AH3	PCIE_TX3_N
MGTHRXP0_224_AJ2	AJ2	PCIE_RX3_P
MGTHRXN0_224_AJ1	AJ1	PCIE_RX3_N
MGTHTXP1_224_AG6	AG6	PCIE_TX2_P
MGTHTXN1_224_AG5	AG5	PCIE_TX2_N
MGTHRXP1_224_AG2	AG2	PCIE_RX2_P
MGTHRXN1_224_AG1	AG1	PCIE_RX2_N
MGTHTXP2_224_AE6	AE6	PCIE_TX1_P
MGTHTXN2_224_AE5	AE5	PCIE_TX1_N
MGTHRXP2_224_AF4	AF4	PCIE_RX1_P
MGTHRXN2_224_AF3	AF3	PCIE_RX1_N
MGTHTXP3_224_AD4	AD4	PCIE_TX0_P
MGTHTXN3_224_AD3	AD3	PCIE_TX0_N
MGTHRXP3_224_AE2	AE2	PCIE_RX0_P
MGTHRXN3_224_AE1	AE1	PCIE_RX0_N
MGTREFCLK0P_224_AB8	AB8	PCIE_CLK_P
MGTREFCLK0N_224_AB7	AB7	PCIE_CLK_N
MGTREFCLK1P_224_AA10	AA10	USER_SMA_MGT_CLOCK_C_P
MGTREFCLK1N_224_AA9	AA9	USER_SMA_MGT_CLOCK_C_N
MGTRREF_R_AD10	AD10	MGTRREF_224
MGTAVTTRCAL_R_AD9	AD9	

U1

SOC_1156_1MM_IRON

MGTAVTT

R214 100 1% 1/10W

C385 0.01UF 25V
C386 0.01UF 25V

HDMI_RX_CLK_C_P → HDMI_RX_CLK_P
HDMI_RX_CLK_C_N → HDMI_RX_CLK_N

C987 0.1UF 25V
C988 0.1UF 25V

HDMI_RX0_C_P → HDMI_RX0_P
HDMI_RX0_C_N → HDMI_RX0_N

C989 0.1UF 25V
C990 0.1UF 25V

HDMI_RX1_C_P → HDMI_RX1_P
HDMI_RX1_C_N → HDMI_RX1_N

C991 0.1UF 25V
C992 0.1UF 25V

HDMI_RX2_C_P → HDMI_RX2_P
HDMI_RX2_C_N → HDMI_RX2_N

C1057 0.01UF 25V
C1058 0.01UF 25V

USER_SMA_MGT_CLOCK_C_P → USER_SMA_MGT_CLOCK_P
USER_SMA_MGT_CLOCK_C_N → USER_SMA_MGT_CLOCK_N

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see UG1144

Zynq Banks 223 224

XILINX

TITLE: Zynq Banks 223 224
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 9 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 225
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 226
ZU7CG = BANK 225
ZU7EG = BANK 225

MGTHTXP0_225_AC6	AC6	SDI_MGT_TX_P
MGTHTXN0_225_AC5	AC5	SDI_MGT_TX_N
MGTHRXP0_225_AC2	AC2	SDI_MGT_RX_P
MGTHRXN0_225_AC1	AC1	SDI_MGT_RX_N
MGTHTXP1_225_AA6	AA6	SMA_MGT_TX_P
MGTHTXN1_225_AA5	AA5	SMA_MGT_TX_N
MGTHRXP1_225_AB4	AB4	SMA_MGT_RX_C_P
MGTHRXN1_225_AB3	AB3	SMA_MGT_RX_C_N
MGTHTXP2_225_Y4	Y4	SFP0_TX_P
MGTHTXN2_225_Y3	Y3	SFP0_TX_N
MGTHRXP2_225_AA2	AA2	SFP0_RX_P
MGTHRXN2_225_AA1	AA1	SFP0_RX_N
MGTHTXP3_225_W6	W6	SFP1_TX_P
MGTHTXN3_225_W5	W5	SFP1_TX_N
MGTHRXP3_225_W2	W2	SFP1_RX_P
MGTHRXN3_225_W1	W1	SFP1_RX_N
MGTREFCLK0P_225_Y8	Y8	FMC_HPC1_GBTCLK0_M2C_C_P
MGTREFCLK0N_225_Y7	Y7	FMC_HPC1_GBTCLK0_M2C_C_N
MGTREFCLK1P_225_W10	W10	SFP_SI5328_OUT_C_P
MGTREFCLK1N_225_W9	W9	SFP_SI5328_OUT_C_N

U1 SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK 226
XCZU7EVFFVC1156

BANK MIGRATION:
ZU11EG = BANK 227
ZU7CG = BANK 226
ZU7EG = BANK 226

MGTHTXP0_226_U6	U6	FMC_HPC0_DP3_C2M_P
MGTHTXN0_226_U5	U5	FMC_HPC0_DP3_C2M_N
MGTHRXP0_226_V4	V4	FMC_HPC0_DP3_M2C_P
MGTHRXN0_226_V3	V3	FMC_HPC0_DP3_M2C_N
MGTHTXP1_226_T4	T4	FMC_HPC0_DP1_C2M_P
MGTHTXN1_226_T3	T3	FMC_HPC0_DP1_C2M_N
MGTHRXP1_226_U2	U2	FMC_HPC0_DP1_M2C_P
MGTHRXN1_226_U1	U1	FMC_HPC0_DP1_M2C_N
MGTHTXP2_226_R6	R6	FMC_HPC0_DP0_C2M_P
MGTHTXN2_226_R5	R5	FMC_HPC0_DP0_C2M_N
MGTHRXP2_226_R2	R2	FMC_HPC0_DP0_M2C_P
MGTHRXN2_226_R1	R1	FMC_HPC0_DP0_M2C_N
MGTHTXP3_226_N6	N6	FMC_HPC0_DP2_C2M_P
MGTHTXN3_226_N5	N5	FMC_HPC0_DP2_C2M_N
MGTHRXP3_226_P4	P4	FMC_HPC0_DP2_M2C_P
MGTHRXN3_226_P3	P3	FMC_HPC0_DP2_M2C_N
MGTREFCLK0P_226_V8	V8	FMC_HPC0_GBTCLK0_M2C_C_P
MGTREFCLK0N_226_V7	V7	FMC_HPC0_GBTCLK0_M2C_C_N
MGTREFCLK1P_226_U10	U10	USER_MGT_SI570_CLOCK1_C_P
MGTREFCLK1N_226_U9	U9	USER_MGT_SI570_CLOCK1_C_N

U1 SOC_1156_1MM_IRON

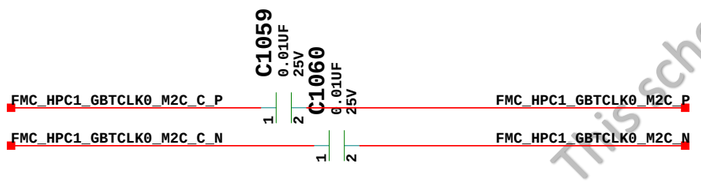
SOC_DA6_FFVC1156_IRONWOOD

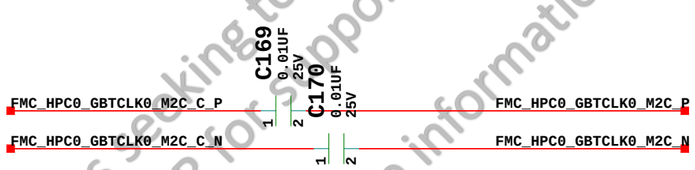
BANK 227
XCZU7EVFFVC1156

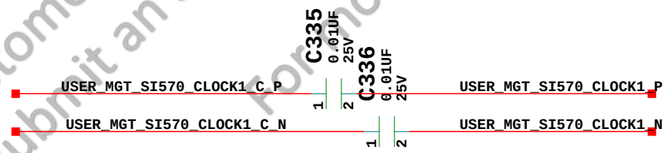
BANK MIGRATION:
ZU11EG = BANK 228
ZU7CG = BANK 227
ZU7EG = BANK 227

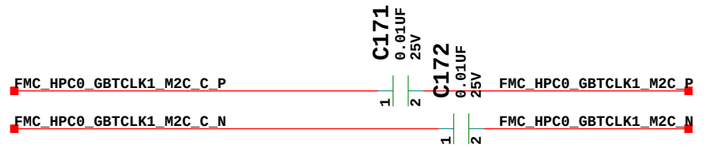
MGTHTXP0_227_M4	M4	FMC_HPC0_DP6_C2M_P
MGTHTXN0_227_M3	M3	FMC_HPC0_DP6_C2M_N
MGTHRXP0_227_N2	N2	FMC_HPC0_DP6_M2C_P
MGTHRXN0_227_N1	N1	FMC_HPC0_DP6_M2C_N
MGTHTXP1_227_L6	L6	FMC_HPC0_DP5_C2M_P
MGTHTXN1_227_L5	L5	FMC_HPC0_DP5_C2M_N
MGTHRXP1_227_L2	L2	FMC_HPC0_DP5_M2C_P
MGTHRXN1_227_L1	L1	FMC_HPC0_DP5_M2C_N
MGTHTXP2_227_K4	K4	FMC_HPC0_DP7_C2M_P
MGTHTXN2_227_K3	K3	FMC_HPC0_DP7_C2M_N
MGTHRXP2_227_J2	J2	FMC_HPC0_DP7_M2C_P
MGTHRXN2_227_J1	J1	FMC_HPC0_DP7_M2C_N
MGTHTXP3_227_H4	H4	FMC_HPC0_DP4_C2M_P
MGTHTXN3_227_H3	H3	FMC_HPC0_DP4_C2M_N
MGTHRXP3_227_G2	G2	FMC_HPC0_DP4_M2C_P
MGTHRXN3_227_G1	G1	FMC_HPC0_DP4_M2C_N
MGTREFCLK0P_227_T8	T8	FMC_HPC0_GBTCLK1_M2C_C_P
MGTREFCLK0N_227_T7	T7	FMC_HPC0_GBTCLK1_M2C_C_N
MGTREFCLK1P_227_R10	R10	USER_MGT_SI570_CLOCK2_C_P
MGTREFCLK1N_227_R9	R9	USER_MGT_SI570_CLOCK2_C_N

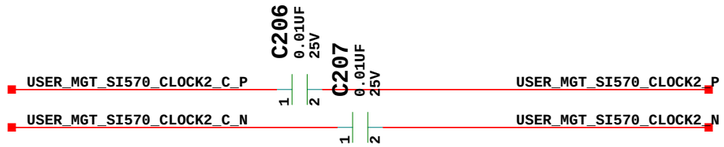
U1 SOC_1156_1MM_IRON











This schematic is currently undergoing technical electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

Zynq Banks 225 226 227

XILINX

TITLE: Zynq Banks 225 226 227 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 10 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 500
XCZU7EVFFVC1156

PS_MIO25_D29	D29	MIO25_CAN_RX
PS_MIO24_E28	E28	MIO24_CAN_TX
PS_MIO23_B29	B29	MIO23_LED
PS_MIO22_F28	F28	MIO22_BUTTON
PS_MIO21_C28	C28	MIO21_UART1_RXD
PS_MIO20_E29	E29	MIO20_UART1_TXD
PS_MIO19_B28	B28	MIO19_UART0_TXD
PS_MIO18_F27	F27	MIO18_UART0_RXD
PS_MIO17_C29	C29	MIO17_I2C1_SDA
PS_MIO16_A28	A28	MIO16_I2C1_SCL
PS_MIO15_E27	E27	MIO15_I2C0_SDA
PS_MIO14_A27	A27	MIO14_I2C0_SCL
PS_MIO13_D27	D27	MIO13_PS_GPIO2
PS_MIO12_C27	C27	MIO12_QSPI_UPR_CLK
PS_MIO11_B26	B26	MIO11_QSPI_UPR_DQ3
PS_MIO10_F26	F26	MIO10_QSPI_UPR_DQ2
PS_MIO9_C26	C26	MIO9_QSPI_UPR_DQ1
PS_MIO8_D26	D26	MIO8_QSPI_UPR_DQ0
PS_MIO7_B25	B25	MIO7_QSPI_UPR_CS_B
PS_MIO6_A26	A26	NC
PS_MIO5_D25	D25	MIO5_QSPI_LWR_CS_B
PS_MIO4_A25	A25	MIO4_QSPI_LWR_DQ0
PS_MIO3_E25	E25	MIO3_QSPI_LWR_DQ3
PS_MIO2_B24	B24	MIO2_QSPI_LWR_DQ2
PS_MIO1_C24	C24	MIO1_QSPI_LWR_DQ1
PS_MIO0_A24	A24	MIO0_QSPI_LWR_CLK

VCCOPS

C25	VCCO_PSIO0_500_C25
D28	VCCO_PSIO0_500_D28

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK 501
XCZU7EVFFVC1156

PS_MIO51_F34	F34	MIO51_SDIO_CLK_R
PS_MIO50_F33	F33	MIO50_SDIO_CMD_R
PS_MIO49_F32	F32	MIO49_SDIO_DAT3_R
PS_MIO48_F31	F31	MIO48_SDIO_DAT2_R
PS_MIO47_F30	F30	MIO47_SDIO_DAT1_R
PS_MIO46_E34	E34	MIO46_SDIO_DAT0_R
PS_MIO45_E33	E33	MIO45_SDIO_DETECT
PS_MIO44_E32	E32	MIO44_SDIO_PROTECT
PS_MIO43_E30	E30	NC
PS_MIO42_D34	D34	MIO42_SDIO_DIR_DAT1_3
PS_MIO41_D32	D32	MIO41_SDIO_DIR_DAT0
PS_MIO40_D31	D31	MIO40_SDIO_DIR_CMD
PS_MIO39_D30	D30	MIO39_SDIO_SEL
PS_MIO38_C34	C34	MIO38_PS_GPIO1
PS_MIO37_C33	C33	MIO37_PMU_GPO5
PS_MIO36_C32	C32	MIO36_PMU_GPO4
PS_MIO35_C31	C31	MIO35_PMU_GPO3
PS_MIO34_B34	B34	MIO34_PMU_GPO2
PS_MIO33_B33	B33	MIO33_PMU_GPO1
PS_MIO32_B31	B31	MIO32_PMU_GPO0
PS_MIO31_B30	B30	NC
PS_MIO30_A33	A33	MIO30_DP_AUX_IN
PS_MIO29_A32	A32	MIO29_DP_OE
PS_MIO28_A31	A31	MIO28_DP_HPD
PS_MIO27_A30	A30	MIO27_DP_AUX_OUT
PS_MIO26_A29	A29	MIO26_PMU_INPUT

R931 R934 R935 R936 R937 R938 — 30 Ω

MIO51_SDIO_CLK_R	MIO51_SDIO_CLK
MIO50_SDIO_CMD_R	MIO50_SDIO_CMD
MIO49_SDIO_DAT3_R	MIO49_SDIO_DAT3
MIO48_SDIO_DAT2_R	MIO48_SDIO_DAT2
MIO47_SDIO_DAT1_R	MIO47_SDIO_DAT1
MIO46_SDIO_DAT0_R	MIO46_SDIO_DAT0

VCCOPS

B32	VCCO_PSIO1_501_B32
E31	VCCO_PSIO1_501_E31

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK 502
XCZU7EVFFVC1156

PS_MIO77_L34	L34	MIO77_ENET_MDIO
PS_MIO76_L33	L33	MIO76_ENET_MDC
PS_MIO75_L30	L30	MIO75_ENET_RX_CTRL
PS_MIO74_L29	L29	MIO74_ENET_RX_D3
PS_MIO73_K34	K34	MIO73_ENET_RX_D2
PS_MIO72_K33	K33	MIO72_ENET_RX_D1
PS_MIO71_K32	K32	MIO71_ENET_RX_D0
PS_MIO70_K31	K31	MIO70_ENET_RX_CLK
PS_MIO69_K30	K30	MIO69_ENET_TX_CTRL
PS_MIO68_K29	K29	MIO68_ENET_TX_D3
PS_MIO67_K28	K28	MIO67_ENET_TX_D2
PS_MIO66_J34	J34	MIO66_ENET_TX_D1
PS_MIO65_J32	J32	MIO65_ENET_TX_D0
PS_MIO64_J31	J31	MIO64_ENET_TX_CLK
PS_MIO63_J30	J30	MIO63_USB_DATA7_R
PS_MIO62_J29	J29	MIO62_USB_DATA6_R
PS_MIO61_H34	H34	MIO61_USB_DATA5_R
PS_MIO60_H33	H33	MIO60_USB_DATA4_R
PS_MIO59_H32	H32	MIO59_USB_DATA3_R
PS_MIO58_H31	H31	MIO58_USB_STP_R
PS_MIO57_H29	H29	MIO57_USB_DATA1_R
PS_MIO56_G34	G34	MIO56_USB_DATA0_R
PS_MIO55_G33	G33	MIO55_USB_NXT
PS_MIO54_G31	G31	MIO54_USB_DATA2_R
PS_MIO53_G30	G30	MIO53_USB_DIR
PS_MIO52_G29	G29	MIO52_USB_CLK

R941 R942 R943 R944 R945 R946 R947 R949 R948 — 30 Ω

MIO63_USB_DATA7_R	MIO63_USB_DATA7
MIO62_USB_DATA6_R	MIO62_USB_DATA6
MIO61_USB_DATA5_R	MIO61_USB_DATA5
MIO60_USB_DATA4_R	MIO60_USB_DATA4
MIO59_USB_DATA3_R	MIO59_USB_DATA3
MIO58_USB_STP_R	MIO58_USB_STP
MIO57_USB_DATA1_R	MIO57_USB_DATA1
MIO56_USB_DATA0_R	MIO56_USB_DATA0
MIO54_USB_DATA2_R	MIO54_USB_DATA2

VCCOPS

H30	VCCO_PSIO2_502_H30
J33	VCCO_PSIO2_502_J33

U1

SOC_1156_1MM_IRON

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

Zynq Banks 500 501 502 - MIO

XILINX

TITLE: Zynq Banks 500 501 502 - MIO SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 11 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 503
XCZU7EVFFVC1156

U1

SOC_1156_1MM_IRON

PS_POR_B_M24
PS_ERROR_STATUS_R25
PS_ERROR_OUT_T25
PS_SRST_B_P25
PS_REF_CLK_R24
PS_JTAG_TCK_K27
PS_JTAG_TMS_H28
PS_JTAG_TDO_G28
PS_JTAG_TDI_J27
PS_INIT_B_P24
PS_PROG_B_T24
PS_DONE_N24
PS_PADI_M25
PS_PADO_L25
PS_MODE3_K25
PS_MODE2_K26
PS_MODE1_J26
PS_MODE0_H27

M24 PS_POR_B
R25 PS_ERR_STATUS
T25 PS_ERR_OUT
P25 PS_SRST_B
R24 PS_REF_CLK
K27 JTAG_TCK_LS
H28 JTAG_TMS_LS
G28 FPGA_TDO_HPC0_TDI
J27 JTAG_TDI_LS
P24 PS_INIT_B
T24 PS_PROG_B
N24 PS_DONE
M25 PS_PADI
L25 PS_PADO
K25
K26
J26
H27

VCCOPS3
G27 VCCO_PSIO3_503_G27
N25 VCCO_PSIO3_503_N25

R921 20 1/16W 1%
C1033 33PF 50V C0G, NP0
GND

VCCOPS3_BUS
R19 R18 R17 R16 4.70K 1/16W 1%

J77 HDR_1X4
PS_MODE3
PS_MODE2
PS_MODE1
PS_MODE0

SDA04H1SBD
SW6

R237 R238 R239 R240 499 1/10W 1%

PS_PADI
PS_PADO
R143 4.7M 1/10W 5%
32.768KHZ X1 X2 20PPM X5
C876 22PF 50V
C875 22PF 50V

1.8V VCCOPS3_BUS
UTIL_3V3

C183 0.1UF 25V
R15 4.70K 1/16W 1%
SN74AVC1T45
U43 SC70_6
PS_INIT_B
C86 0.1UF 25V
R323 261 1/10W 1%
R326 261 1/10W 1%
LED-GRN-RED
DS1

INIT_B = 0, RED LED ON
INIT_B = 1, GRN LED ON

C181 0.1UF 25V
R13 4.70K 1/16W 1%
SN74AVC1T45
U44 SC70_6
PS_DONE
C87 0.1UF 25V
R328 261 1/10W 1%
DS32 LED-GRN-SMT

DONE_0 = 0, DONE LED OFF
DONE_0 = 1, DONE LED ON

UTIL_3V3
VCCOPS3_BUS

R267 R266 R265 10.0K 1/10W 1%
R241 8.06K 1/10W 1%

PS_POR_B
Pushbutton TL3301EF100QG
SW4
PS_SRST_B
Pushbutton TL3301EF100QG
SW3

BAT54T1G D6
BAT54T1G D5

J22 HDR_1X2

R271 10.0K 1/10W 1%

MAX16025
U22 TQFN_16
VCC OUT1 OUT2 RST_B CDLY1 CDLY2 CRESET EPAD GND
IN1 IN2 EN1 EN2 MR_B TOL TH0 TH1
NC NC

PS_POR_SW_B
PS_SRST_SW_B
POR_RST_B

C188 0.1UF 25V

J20 HDR_1X2 PS_POR_B_HDR
J21 HDR_1X2 PS_SRST_B_HDR
OPEN DRAIN
PS_POR_B
PS_SRST_B

C185 C186 C187 0.1UF 25V

R273 R272 4.70K 1/16W 1%
R354 261 1/10W 1%
DS36 LED-RED-SMT

R355 R356 261 1/10W 1%
DS35 LED-RED-SMT
DS33 LED-GRN-SMT
Q13 Q14 NDS331N 460MW

PS_ERR_OUT
PS_ERR_STATUS
R164 R187 499 1/10W 1%

R317 R318 DNP

PS ERROR STATUS LEDs

R14 4.70K 1/16W 1%
Pushbutton TL3301EF100QG
SW5
PS_PROG_B

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70511

Zynq Banks 503 - Config

XILINX

TITLE: Zynq Banks 503 - Config SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 12 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 504
XCZU7EVFFVC1156

DDR4_SODIMM_A0	AN34	PS_DDR_A0_AN34	
DDR4_SODIMM_A1	AM34	PS_DDR_A1_AM34	
DDR4_SODIMM_A2	AM33	PS_DDR_A2_AM33	
DDR4_SODIMM_A3	AL34	PS_DDR_A3_AL34	
DDR4_SODIMM_A4	AL33	PS_DDR_A4_AL33	
DDR4_SODIMM_A5	AK33	PS_DDR_A5_AK33	
DDR4_SODIMM_A6	AK30	PS_DDR_A6_AK30	
DDR4_SODIMM_A7	AJ30	PS_DDR_A7_AJ30	
DDR4_SODIMM_A8	AJ31	PS_DDR_A8_AJ31	
DDR4_SODIMM_A9	AH31	PS_DDR_A9_AH31	
DDR4_SODIMM_A10	AG31	PS_DDR_A10_AG31	
DDR4_SODIMM_A11	AF31	PS_DDR_A11_AF31	
DDR4_SODIMM_A12	AG30	PS_DDR_A12_AG30	
DDR4_SODIMM_A13	AF30	PS_DDR_A13_AF30	
DDR4_SODIMM_WE_B	AG29	PS_DDR_A14_AG29	
DDR4_SODIMM_CAS_B	AG28	PS_DDR_A15_AG28	
DDR4_SODIMM_RAS_B	AF28	PS_DDR_A16_AF28	
NC	AF26	PS_DDR_A17_AF26	
DDR4_SODIMM_BA0	AE27	PS_DDR_BA0_AE27	
DDR4_SODIMM_BA1	AE28	PS_DDR_BA1_AE28	
DDR4_SODIMM_ACT_B	AE25	PS_DDR_ACT_N_AE25	
DDR4_SODIMM_ALERT_B	AB26	PS_DDR_ALERT_N_AB26	
DDR4_SODIMM_PARITY	AA26	PS_DDR_PARITY_AA26	
ZYNQ_DDR4_SODIMM_RESET_B	AD26	PS_DDR_RAM_RST_N_AD26	
DDR4_SODIMM_BG0	AD27	PS_DDR_BG0_AD27	
DDR4_SODIMM_BG1	AF27	PS_DDR_BG1_AF27	
DDR4_SODIMM_CS0_B	AP33	PS_DDR_CS_N0_AP33	
DDR4_SODIMM_CS1_B	AK32	PS_DDR_CS_N1_AK32	
DDR4_SODIMM_CK0_T	AL31	PS_DDR_CK0_AL31	
DDR4_SODIMM_CK0_C	AN32	PS_DDR_CK_N0_AN32	
DDR4_SODIMM_CK1_T	AL30	PS_DDR_CK1_AL30	
DDR4_SODIMM_CK1_C	AL32	PS_DDR_CK_N1_AL32	
DDR4_SODIMM_CKE0	AN33	PS_DDR_CKE0_AN33	
DDR4_SODIMM_CKE1	AH32	PS_DDR_CKE1_AH32	
DDR4_SODIMM_ODT0	AP32	PS_DDR_ODT0_AP32	
DDR4_SODIMM_ODT1	AJ32	PS_DDR_ODT1_AJ32	
DDR4_SODIMM_DQS0_T	AN26	PS_DDR_DQS_P0_AN26	
DDR4_SODIMM_DQS0_C	AN27	PS_DDR_DQS_N0_AN27	
DDR4_SODIMM_DQS1_T	AN29	PS_DDR_DQS_P1_AN29	
DDR4_SODIMM_DQS1_C	AP30	PS_DDR_DQS_N1_AP30	
DDR4_SODIMM_DQS2_T	AH26	PS_DDR_DQS_P2_AH26	
DDR4_SODIMM_DQS2_C	AJ26	PS_DDR_DQS_N2_AJ26	
DDR4_SODIMM_DQS3_T	AK28	PS_DDR_DQS_P3_AK28	
DDR4_SODIMM_DQS3_C	AK29	PS_DDR_DQS_N3_AK29	
DDR4_SODIMM_DQS4_T	AD30	PS_DDR_DQS_P4_AD30	
DDR4_SODIMM_DQS4_C	AD31	PS_DDR_DQS_N4_AD31	
DDR4_SODIMM_DQS5_T	Y27	PS_DDR_DQS_P5_Y27	
DDR4_SODIMM_DQS5_C	Y28	PS_DDR_DQS_N5_Y28	
DDR4_SODIMM_DQS6_T	AB33	PS_DDR_DQS_P6_AB33	
DDR4_SODIMM_DQS6_C	AB34	PS_DDR_DQS_N6_AB34	
DDR4_SODIMM_DQS7_T	W31	PS_DDR_DQS_P7_W31	
DDR4_SODIMM_DQS7_C	W32	PS_DDR_DQS_N7_W32	
DDR4_SODIMM_DQS8_T	AG33	PS_DDR_DQS_P8_AG33	
DDR4_SODIMM_DQS8_C	AG34	PS_DDR_DQS_N8_AG34	
DDR4_SODIMM_DM0_B	AN24	PS_DDR_DM0_AN24	
DDR4_SODIMM_DM1_B	AM29	PS_DDR_DM1_AM29	
DDR4_SODIMM_DM2_B	AH24	PS_DDR_DM2_AH24	
DDR4_SODIMM_DM3_B	AJ29	PS_DDR_DM3_AJ29	
DDR4_SODIMM_DM4_B	AD29	PS_DDR_DM4_AD29	
DDR4_SODIMM_DM5_B	Y29	PS_DDR_DM5_Y29	
DDR4_SODIMM_DM6_B	AC32	PS_DDR_DM6_AC32	
DDR4_SODIMM_DM7_B	Y32	PS_DDR_DM7_Y32	
DDR4_SODIMM_DM8_B	AF34	PS_DDR_DM8_AF34	

VCCO_PSDDR_504

AL29	VCCO_PSDDR_504_AL29
AJ33	VCCO_PSDDR_504_AJ33
AJ28	VCCO_PSDDR_504_AJ28
AG32	VCCO_PSDDR_504_AG32
AG27	VCCO_PSDDR_504_AG27
AE31	VCCO_PSDDR_504_AE31
AE26	VCCO_PSDDR_504_AE26

U1

PS_DDR_DQ0_AP27	AP27	DDR4_SODIMM_DQ0	
PS_DDR_DQ1_AP25	AP25	DDR4_SODIMM_DQ1	
PS_DDR_DQ2_AP26	AP26	DDR4_SODIMM_DQ2	
PS_DDR_DQ3_AM26	AM26	DDR4_SODIMM_DQ3	
PS_DDR_DQ4_AP24	AP24	DDR4_SODIMM_DQ4	
PS_DDR_DQ5_AL25	AL25	DDR4_SODIMM_DQ5	
PS_DDR_DQ6_AM25	AM25	DDR4_SODIMM_DQ6	
PS_DDR_DQ7_AM24	AM24	DDR4_SODIMM_DQ7	
PS_DDR_DQ8_AM28	AM28	DDR4_SODIMM_DQ8	
PS_DDR_DQ9_AN28	AN28	DDR4_SODIMM_DQ9	
PS_DDR_DQ10_AP29	AP29	DDR4_SODIMM_DQ10	
PS_DDR_DQ11_AP28	AP28	DDR4_SODIMM_DQ11	
PS_DDR_DQ12_AM31	AM31	DDR4_SODIMM_DQ12	
PS_DDR_DQ13_AP31	AP31	DDR4_SODIMM_DQ13	
PS_DDR_DQ14_AN31	AN31	DDR4_SODIMM_DQ14	
PS_DDR_DQ15_AM30	AM30	DDR4_SODIMM_DQ15	
PS_DDR_DQ16_AF25	AF25	DDR4_SODIMM_DQ16	
PS_DDR_DQ17_AG25	AG25	DDR4_SODIMM_DQ17	
PS_DDR_DQ18_AG26	AG26	DDR4_SODIMM_DQ18	
PS_DDR_DQ19_AJ25	AJ25	DDR4_SODIMM_DQ19	
PS_DDR_DQ20_AG24	AG24	DDR4_SODIMM_DQ20	
PS_DDR_DQ21_AK25	AK25	DDR4_SODIMM_DQ21	
PS_DDR_DQ22_AJ24	AJ24	DDR4_SODIMM_DQ22	
PS_DDR_DQ23_AK24	AK24	DDR4_SODIMM_DQ23	
PS_DDR_DQ24_AH28	AH28	DDR4_SODIMM_DQ24	
PS_DDR_DQ25_AH27	AH27	DDR4_SODIMM_DQ25	
PS_DDR_DQ26_AJ27	AJ27	DDR4_SODIMM_DQ26	
PS_DDR_DQ27_AK27	AK27	DDR4_SODIMM_DQ27	
PS_DDR_DQ28_AL26	AL26	DDR4_SODIMM_DQ28	
PS_DDR_DQ29_AL27	AL27	DDR4_SODIMM_DQ29	
PS_DDR_DQ30_AH29	AH29	DDR4_SODIMM_DQ30	
PS_DDR_DQ31_AL28	AL28	DDR4_SODIMM_DQ31	
PS_DDR_DQ32_AB29	AB29	DDR4_SODIMM_DQ32	
PS_DDR_DQ33_AB30	AB30	DDR4_SODIMM_DQ33	
PS_DDR_DQ34_AC29	AC29	DDR4_SODIMM_DQ34	
PS_DDR_DQ35_AD32	AD32	DDR4_SODIMM_DQ35	
PS_DDR_DQ36_AC31	AC31	DDR4_SODIMM_DQ36	
PS_DDR_DQ37_AE30	AE30	DDR4_SODIMM_DQ37	
PS_DDR_DQ38_AC28	AC28	DDR4_SODIMM_DQ38	
PS_DDR_DQ39_AE29	AE29	DDR4_SODIMM_DQ39	
PS_DDR_DQ40_AC27	AC27	DDR4_SODIMM_DQ40	
PS_DDR_DQ41_AA27	AA27	DDR4_SODIMM_DQ41	
PS_DDR_DQ42_AA28	AA28	DDR4_SODIMM_DQ42	
PS_DDR_DQ43_AB28	AB28	DDR4_SODIMM_DQ43	
PS_DDR_DQ44_W27	W27	DDR4_SODIMM_DQ44	
PS_DDR_DQ45_W29	W29	DDR4_SODIMM_DQ45	
PS_DDR_DQ46_W28	W28	DDR4_SODIMM_DQ46	
PS_DDR_DQ47_V27	V27	DDR4_SODIMM_DQ47	
PS_DDR_DQ48_AA32	AA32	DDR4_SODIMM_DQ48	
PS_DDR_DQ49_AA33	AA33	DDR4_SODIMM_DQ49	
PS_DDR_DQ50_AA34	AA34	DDR4_SODIMM_DQ50	
PS_DDR_DQ51_AE34	AE34	DDR4_SODIMM_DQ51	
PS_DDR_DQ52_AD34	AD34	DDR4_SODIMM_DQ52	
PS_DDR_DQ53_AB31	AB31	DDR4_SODIMM_DQ53	
PS_DDR_DQ54_AC34	AC34	DDR4_SODIMM_DQ54	
PS_DDR_DQ55_AC33	AC33	DDR4_SODIMM_DQ55	
PS_DDR_DQ56_AA30	AA30	DDR4_SODIMM_DQ56	
PS_DDR_DQ57_Y30	Y30	DDR4_SODIMM_DQ57	
PS_DDR_DQ58_AA31	AA31	DDR4_SODIMM_DQ58	
PS_DDR_DQ59_W30	W30	DDR4_SODIMM_DQ59	
PS_DDR_DQ60_Y33	Y33	DDR4_SODIMM_DQ60	
PS_DDR_DQ61_W33	W33	DDR4_SODIMM_DQ61	
PS_DDR_DQ62_W34	W34	DDR4_SODIMM_DQ62	
PS_DDR_DQ63_Y34	Y34	DDR4_SODIMM_DQ63	
PS_DDR_DQ64_AF32	AF32	DDR4_SODIMM_CB0	
PS_DDR_DQ65_AE32	AE32	DDR4_SODIMM_CB1	
PS_DDR_DQ66_AH33	AH33	DDR4_SODIMM_CB2	
PS_DDR_DQ67_AE33	AE33	DDR4_SODIMM_CB3	
PS_DDR_DQ68_AF33	AF33	DDR4_SODIMM_CB4	
PS_DDR_DQ69_AH34	AH34	DDR4_SODIMM_CB5	
PS_DDR_DQ70_AJ34	AJ34	DDR4_SODIMM_CB6	
PS_DDR_DQ71_AK34	AK34	DDR4_SODIMM_CB7	
PS_DDR_ZQ_AC26	AC26	UDIMM_PS_ZQ	

SOC_1156_1MM_IRON

R351
240
1/10W
1%
1
2
GND

This schematic is undergoing HDMI electrical compliance. Contact Xilinx support for more information. Copyright Xilinx 2018

Zynq Banks 504 - Memory

XILINX

TITLE: Zynq Banks 504 - Memory SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 13 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK 505
XCZU7EVFFVC1156

PS_MGTRTXP0_505_U29	U29	GT0_DP_TX_P
PS_MGTRTXN0_505_U30	U30	GT0_DP_TX_N
PS_MGTRXP0_505_U33	U33	NC
PS_MGTRXN0_505_U34	U34	NC
PS_MGTRTXP1_505_R29	R29	GT1_DP_TX_P
PS_MGTRTXN1_505_R30	R30	GT1_DP_TX_N
PS_MGTRXP1_505_T31	T31	NC
PS_MGTRXN1_505_T32	T32	NC
PS_MGTRTXP2_505_P31	P31	GT2_USB0_TX_P
PS_MGTRTXN2_505_P32	P32	GT2_USB0_TX_N
PS_MGTRXP2_505_R33	R33	GT2_USB0_RX_P
PS_MGTRXN2_505_R34	R34	GT2_USB0_RX_N
PS_MGTRTXP3_505_N29	N29	GT3_SATA1_TX_P
PS_MGTRTXN3_505_N30	N30	GT3_SATA1_TX_N
PS_MGTRXP3_505_N33	N33	GT3_SATA1_RX_P
PS_MGTRXN3_505_N34	N34	GT3_SATA1_RX_N
PS_MGTREFCLK0P_505_T27	T27	NC
PS_MGTREFCLK0N_505_T28	T28	NC
PS_MGTREFCLK1P_505_P27	P27	GTR_REF_CLK_SATA_C_P
PS_MGTREFCLK1N_505_P28	P28	GTR_REF_CLK_SATA_C_N
PS_MGTREFCLK2P_505_M27	M27	GTR_REF_CLK_USB3_C_P
PS_MGTREFCLK2N_505_M28	M28	GTR_REF_CLK_USB3_C_N
PS_MGTREFCLK3P_505_M31	M31	GTR_REF_CLK_DP_C_P
PS_MGTREFCLK3N_505_M32	M32	GTR_REF_CLK_DP_C_N
PS_MGTRREF_505_U31	U31	

U1

SOC_1156_1MM_IRON

R249
500
1/20W
0.1%

GND

GTR_REF_CLK_SATA_C_P	C191 0.01UF 25V	C192 0.01UF 25V	GTR_REF_CLK_SATA_P
GTR_REF_CLK_SATA_C_N	C193 0.01UF 25V	C194 0.01UF 25V	GTR_REF_CLK_SATA_N
GTR_REF_CLK_USB3_C_P	C195 0.01UF 25V	C196 0.01UF 25V	GTR_REF_CLK_USB3_P
GTR_REF_CLK_USB3_C_N			GTR_REF_CLK_USB3_N
GTR_REF_CLK_DP_C_P			GTR_REF_CLK_DP_P
GTR_REF_CLK_DP_C_N			GTR_REF_CLK_DP_N

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 75514

Zynq Banks 505 - GTR

XILINX

TITLE: Zynq Banks 505 - GTR
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 14 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK MGTAVCC_R
XCZU7EVFFVC1156

MGTAVCC_R_W8 W8
MGTAVCC_R_U8 U8
MGTAVCC_R_T10 T10
MGTAVCC_R_R8 R8
MGTAVCC_R_AC8 AC8
MGTAVCC_R_AB10 AB10
MGTAVCC_R_AA8 AA8

MGTAVCC

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK MGTAVTT_R
XCZU7EVFFVC1156

MGTAVTT_R_Y6 Y6
MGTAVTT_R_V6 V6
MGTAVTT_R_T6 T6
MGTAVTT_R_P6 P6
MGTAVTT_R_M6 M6
MGTAVTT_R_AP6 AP6
MGTAVTT_R_AM6 AM6
MGTAVTT_R_AK6 AK6
MGTAVTT_R_AH6 AH6
MGTAVTT_R_AF6 AF6
MGTAVTT_R_AD6 AD6
MGTAVTT_R_AB6 AB6

MGTAVTT

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK MGTVCCAUX_R
XCZU7EVFFVC1156

MGTVCCAUX_R_Y10 Y10
MGTVCCAUX_R_V10 V10

MGTVCCAUX

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCAUX
XCZU7EVFFVC1156

VCCAUX_V23 V23
VCCAUX_U23 U23
VCCAUX_R23 R23
VCCAUX_P23 P23

VCCAUX

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCAUX_IO
XCZU7EVFFVC1156

VCCAUX_IO_U22 U22
VCCAUX_IO_T22 T22
VCCAUX_IO_R22 R22
VCCAUX_IO_N22 N22

VCCAUX

U1

SOC_1156_1MM_IRON

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

Zynq Power 1

XILINX

TITLE: Zynq Power 1
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16
VER: 1.0
SHEET SIZE: B
REV: 01
SHEET 15 OF 95
DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCINT
XCZU7EVFFVC1156

VCCINT_Y19	Y19	VCCINT
VCCINT_Y17	Y17	
VCCINT_Y15	Y15	
VCCINT_W20	W20	
VCCINT_W18	W18	
VCCINT_W16	W16	
VCCINT_W14	W14	
VCCINT_V21	V21	
VCCINT_V19	V19	
VCCINT_V15	V15	
VCCINT_U20	U20	
VCCINT_U16	U16	
VCCINT_U14	U14	
VCCINT_T21	T21	
VCCINT_T19	T19	
VCCINT_T15	T15	
VCCINT_R20	R20	
VCCINT_R16	R16	
VCCINT_R14	R14	
VCCINT_P21	P21	
VCCINT_P19	P19	
VCCINT_P17	P17	
VCCINT_P15	P15	
VCCINT_N20	N20	
VCCINT_N18	N18	
VCCINT_N16	N16	
VCCINT_N14	N14	
VCCINT_M21	M21	
VCCINT_M19	M19	
VCCINT_M17	M17	
VCCINT_M15	M15	

U1 SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCBRAM
XCZU7EVFFVC1156

VCCBRAM_W12	W12	VCCBRAM
VCCBRAM_V12	V12	
VCCBRAM_T12	T12	
VCCBRAM_R12	R12	

U1 SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCINT_IO
XCZU7EVFFVC1156

VCCINT_IO_Y13	Y13	VCCBRAM
VCCINT_IO_V13	V13	
VCCINT_IO_U13	U13	
VCCINT_IO_T13	T13	
VCCINT_IO_P13	P13	

U1 SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK VCCINT_VCU
XCZU7EVFFVC1156

VCCINT_VCU_AD24	AD24	VCCINT_VCU
VCCINT_VCU_AD23	AD23	
VCCINT_VCU_AD22	AD22	
VCCINT_VCU_AD21	AD21	

U1 SOC_1156_1MM_IRON

This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx technical Support. For more information see AR 70514

Zynq Power 2

XILINX

TITLE: Zynq Power 2
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 16 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK PS_MGTRAVCC
XCZU7EVFFVC1156

PS_MGTRAVCC_R27
PS_MGTRAVCC_N27

R27
N27

MGTRAVCC

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK PS_MGTRAVTT
XCZU7EVFFVC1156

PS_MGTRAVTT_T29
PS_MGTRAVTT_P29
PS_MGTRAVTT_N31

T29
P29
N31

MGTRAVTT

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK PS_POWER
XCZU7EVFFVC1156

VCC_PSPLL_Y24
VCC_PSPLL_W24
VCC_PSPLL_V24
VCC_PSINTLP_Y22
VCC_PSINTLP_Y21
VCC_PSINTLP_W22
VCC_PSINTLP_W21
VCC_PSINTLP_V22
VCC_PSINTLP_AA22
VCC_PSINTFP_DDR_AC25
VCC_PSINTFP_DDR_AC24
VCC_PSINTFP_DDR_AB25
VCC_PSINTFP_AC22
VCC_PSINTFP_AC21
VCC_PSINTFP_AB24
VCC_PSINTFP_AB23
VCC_PSINTFP_AB22
VCC_PSINTFP_AB21
VCC_PSINTFP_AA23
VCC_PSDDR_PLL_V25
VCC_PSDDR_PLL_U25
VCC_PSBATT_Y23
VCC_PSAUX_Y25
VCC_PSAUX_W26
VCC_PSAUX_W25
VCC_PSAUX_V26
VCC_PSADC_AA25
GND_PSADC_AA24

Y24
W24
V24
Y22
Y21
W22
W21
V22
AA22
AC25
AC24
AB25
AC22
AC21
AB24
AB23
AB22
AB21
AA23
V25
U25
Y23
Y25
W26
W25
V26
AA25
AA24

VCCPSPLL
VCCPSINTLP
VCCPSINTFP
VCCPSDDRPLL
VCCPSAUX
VCC_PSBATT

U1

SOC_1156_1MM_IRON

VCCPSAUX
FERRITE-600
PS_SYSMON_AVCC
C168 0.1UF 25V X5R
C917 0.47UF 10V X5R
PS_SYSMON_AGND
FERRITE-600
L19
PS_SYSMON_AGND
GND

UTIL_1V8
NC
D40
40V
BAS40-04
200MW
R22
4.70K
1/16W
1%
VCC_PSBATT
TS518FE_F135E
B1
BAT_TS518_TS621_DUAL
GND

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to understand HDMI compliance should submit an SR for support from Xilinx Support.
For more information see AR#70514

Zynq Power 3

XILINX

TITLE: Zynq Power 3 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 17 OF 95	DRAWN BY: BF

SOC_DA6_FFVC1156_IRONWOOD

BANK GND1
XCZU7EVFFVC1156

J18	GND_J18	GND_AK2	AK2
J13	GND_J13	GND_AK16	AK16
H5	GND_H5	GND_AK1	AK1
H25	GND_H25	GND_AJ8	AJ8
H2	GND_H2	GND_AJ7	AJ7
H10	GND_H10	GND_AJ4	AJ4
H1	GND_H1	GND_AJ3	AJ3
G5	GND_G5	GND_AJ23	AJ23
G4	GND_G4	GND_AJ13	AJ13
G32	GND_G32	GND_AH7	AH7
G3	GND_G3	GND_AH5	AH5
G22	GND_G22	GND_AH30	AH30
G17	GND_G17	GND_AH25	AH25
F3	GND_F3	GND_AH20	AH20
F29	GND_F29	GND_AH2	AH2
F2	GND_F2	GND_AH10	AH10
F19	GND_F19	GND_AH1	AH1
F14	GND_F14	GND_AG7	AG7
F1	GND_F1	GND_AG4	AG4
E26	GND_E26	GND_AG3	AG3
E11	GND_E11	GND_AG17	AG17
D8	GND_D8	GND_AF7	AF7
D33	GND_D33	GND_AF5	AF5
D23	GND_D23	GND_AF29	AF29
D18	GND_D18	GND_AF24	AF24
C5	GND_C5	GND_AF2	AF2
C30	GND_C30	GND_AF14	AF14
C20	GND_C20	GND_AF1	AF1
C15	GND_C15	GND_AE9	AE9
C10	GND_C10	GND_AE8	AE8
B7	GND_B7	GND_AE7	AE7
B27	GND_B27	GND_AE4	AE4
B22	GND_B22	GND_AE3	AE3
B2	GND_B2	GND_AE21	AE21
B17	GND_B17	GND_AE11	AE11
B12	GND_B12	GND_AE10	AE10
AP7	GND_AP7	GND_AD5	AD5
AP5	GND_AP5	GND_AD33	AD33
AP34	GND_AP34	GND_AD28	AD28
AP2	GND_AP2	GND_AD25	AD25
AP1	GND_AP1	GND_AD2	AD2
AN7	GND_AN7	GND_AD18	AD18
AN4	GND_AN4	GND_AD13	AD13
AN30	GND_AN30	GND_AD11	AD11
AN3	GND_AN3	GND_AD1	AD1
AN25	GND_AN25	GND_AC7	AC7
AN20	GND_AN20	GND_AC4	AC4
AN15	GND_AN15	GND_AC30	AC30
AN10	GND_AN10	GND_AC3	AC3
AM7	GND_AM7	GND_AC23	AC23
AM5	GND_AM5	GND_AC20	AC20
AM32	GND_AM32	GND_AC15	AC15
AM27	GND_AM27	GND_AC11	AC11
AM22	GND_AM22	GND_AB9	AB9
AM2	GND_AM2	GND_AB5	AB5
AM17	GND_AM17	GND_AB32	AB32
AM12	GND_AM12	GND_AB27	AB27
AM1	GND_AM1	GND_AB2	AB2
AL9	GND_AL9	GND_AB17	AB17
AL7	GND_AL7	GND_AB11	AB11
AL4	GND_AL4	GND_AB1	AB1
AL3	GND_AL3	GND_AA7	AA7
AL24	GND_AL24	GND_AA4	AA4
AL19	GND_AL19	GND_AA3	AA3
AL14	GND_AL14	GND_AA29	AA29
AK7	GND_AK7	GND_AA21	AA21
AK5	GND_AK5	GND_AA11	AA11
AK31	GND_AK31	GND_A4	A4
AK26	GND_AK26	GND_A34	A34
		GND_A1	A1

GND GND

U1

SOC_1156_1MM_IRON

SOC_DA6_FFVC1156_IRONWOOD

BANK GND2
XCZU7EVFFVC1156

Y9	GND_Y9	GND_R21	R21
Y5	GND_Y5	GND_R19	R19
Y31	GND_Y31	GND_R15	R15
Y26	GND_Y26	GND_R13	R13
Y20	GND_Y20	GND_R11	R11
Y2	GND_Y2	GND_P9	P9
Y18	GND_Y18	GND_P8	P8
Y16	GND_Y16	GND_P7	P7
Y14	GND_Y14	GND_P5	P5
Y12	GND_Y12	GND_P34	P34
Y11	GND_Y11	GND_P33	P33
Y1	GND_Y1	GND_P30	P30
W7	GND_W7	GND_P26	P26
W4	GND_W4	GND_P22	P22
W3	GND_W3	GND_P20	P20
W23	GND_W23	GND_P2	P2
W19	GND_W19	GND_P18	P18
W17	GND_W17	GND_P16	P16
W15	GND_W15	GND_P14	P14
W13	GND_W13	GND_P11	P11
W11	GND_W11	GND_P10	P10
V9	GND_V9	GND_P1	P1
V5	GND_V5	GND_N7	N7
V34	GND_V34	GND_N4	N4
V33	GND_V33	GND_N32	N32
V32	GND_V32	GND_N3	N3
V31	GND_V31	GND_N28	N28
V30	GND_V30	GND_N26	N26
V29	GND_V29	GND_N23	N23
V28	GND_V28	GND_N21	N21
V20	GND_V20	GND_N19	N19
V2	GND_V2	GND_N17	N17
V16	GND_V16	GND_N15	N15
V14	GND_V14	GND_M7	M7
V11	GND_V11	GND_M5	M5
V1	GND_V1	GND_M34	M34
U7	GND_U7	GND_M33	M33
U4	GND_U4	GND_M30	M30
U32	GND_U32	GND_M29	M29
U3	GND_U3	GND_M26	M26
U28	GND_U28	GND_M22	M22
U27	GND_U27	GND_M20	M20
U26	GND_U26	GND_M2	M2
U24	GND_U24	GND_M18	M18
U21	GND_U21	GND_M16	M16
U19	GND_U19	GND_M14	M14
U15	GND_U15	GND_M1	M1
U12	GND_U12	GND_L9	L9
U11	GND_U11	GND_L7	L7
T9	GND_T9	GND_L4	L4
T5	GND_T5	GND_L32	L32
T34	GND_T34	GND_L31	L31
T33	GND_T33	GND_L3	L3
T30	GND_T30	GND_L28	L28
T26	GND_T26	GND_L27	L27
T23	GND_T23	GND_L26	L26
T20	GND_T20	GND_L24	L24
T2	GND_T2	GND_L19	L19
T16	GND_T16	GND_K7	K7
T14	GND_T14	GND_K6	K6
T11	GND_T11	GND_K5	K5
T1	GND_T1	GND_K21	K21
R7	GND_R7	GND_K2	K2
R4	GND_R4	GND_K16	K16
R32	GND_R32	GND_K1	K1
R31	GND_R31	GND_J5	J5
R3	GND_R3	GND_J4	J4
R28	GND_R28	GND_J3	J3
R26	GND_R26	GND_J28	J28
		GND_J23	J23

GND GND

U1

SOC_1156_1MM_IRON

This schematic is currently undergoing DDR4 electrical compliance testing. Customers seeking to do DDR4 compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70575.

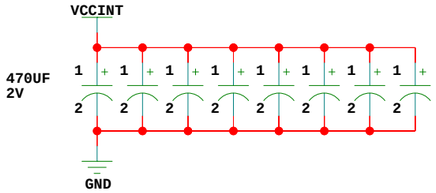
Zynq GND

XILINX

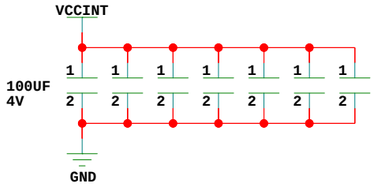
TITLE: Zynq GND SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 18 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to use the HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

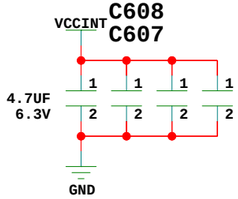
C696 C700
C697 C701
C698 C702
C699 C703



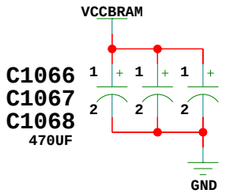
C424 C428
C425 C431
C426 C432
C427

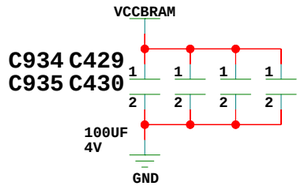


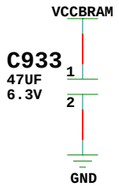
C602
C606
C608
C607



VCCINT

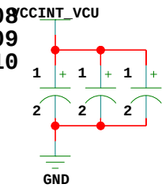


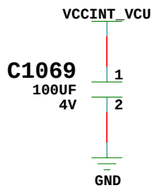




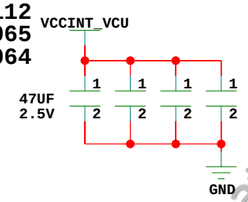
VCCBRAM

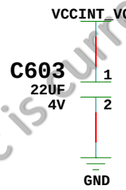
C1108
C1109
C1110

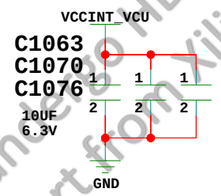




C1111
C1112
C1065
C1064

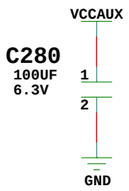




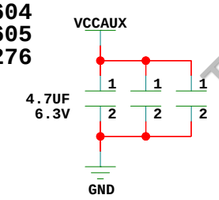


VCCINT_VCU

Place these 0201/0402 dual pad capacitors directly under U1 near pins AD24/AD23/AD22/AD21

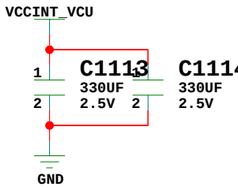


C604
C605
C276



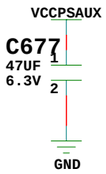
VCCAUX / VCCAUX_IO

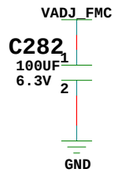
VCCINT_VCU

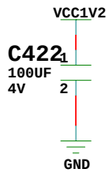


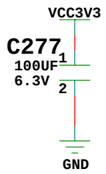
Place these 1210 capacitors close to U1

VCCO BANKS 0 28 64 65 66 67 68 87 88





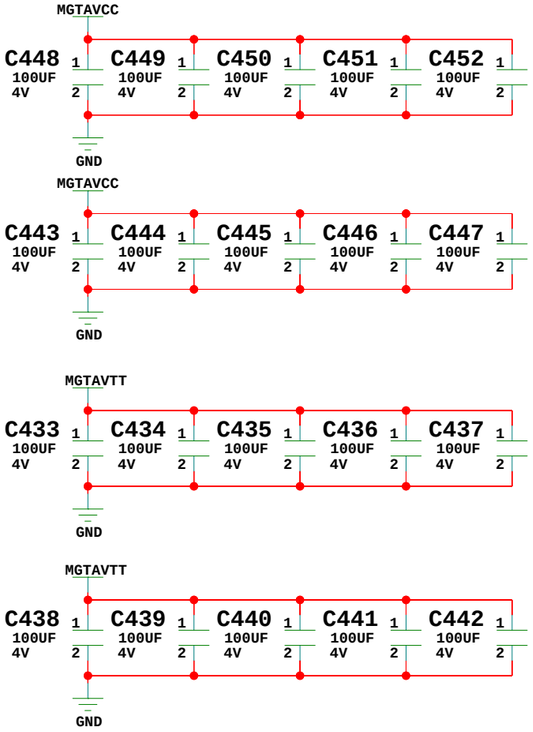


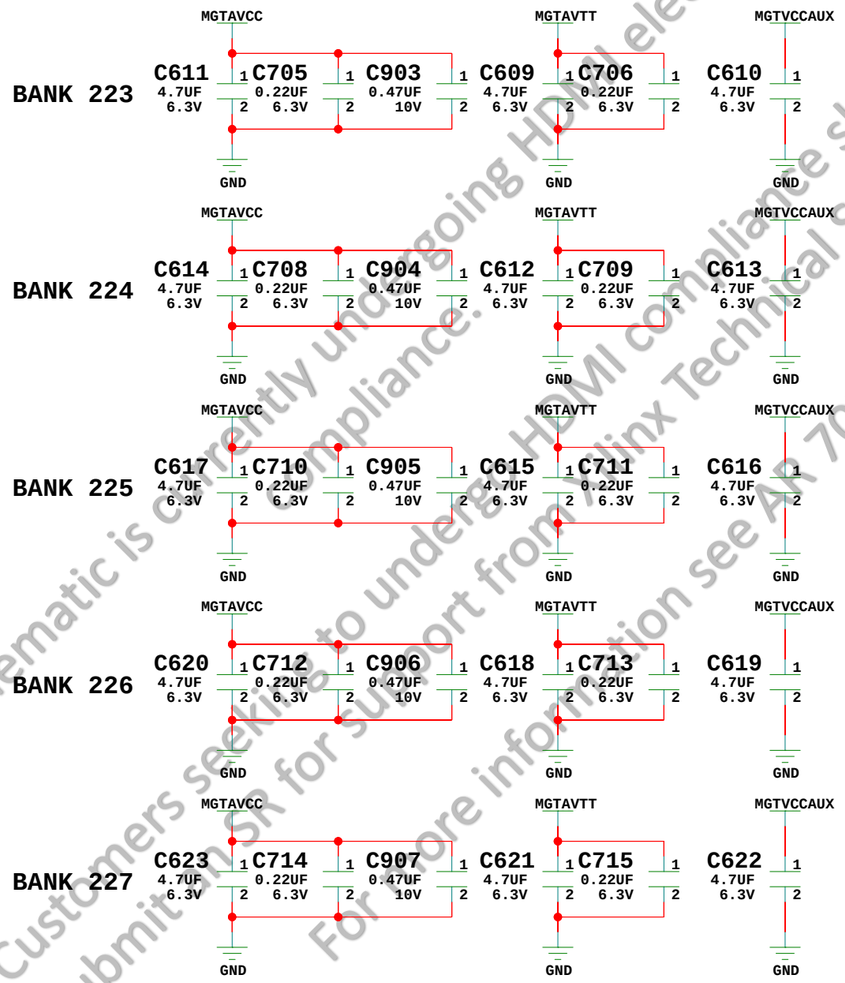


Zynq Decoupling 1

XILINX

TITLE: Zynq Decoupling 1 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 19 OF 95	DRAWN BY: BF



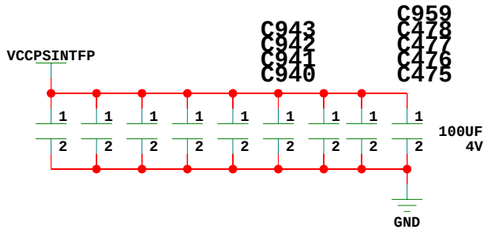


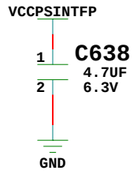
This schematic is currently undergoing HDMI electrical compliance. Customers seeking to ... HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

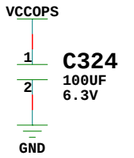
Zynq Decoupling 2

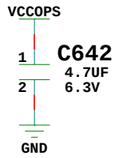
XILINX

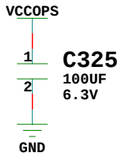
TITLE: Zynq Decoupling 2 SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 20 OF 95	DRAWN BY: BF

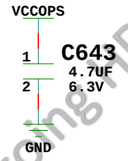


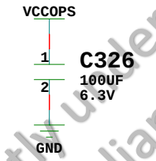


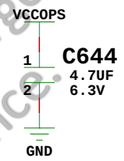


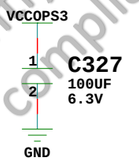


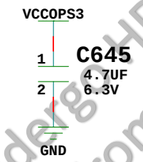


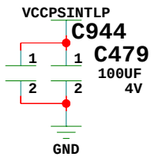


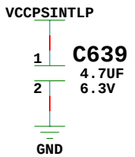


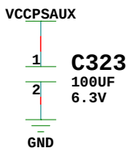


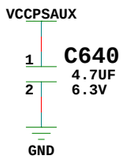


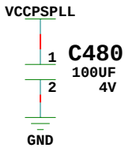




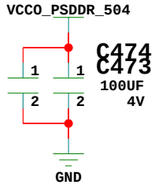


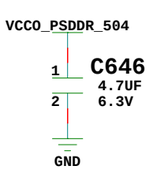


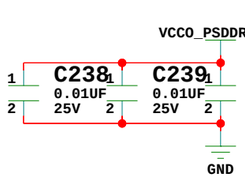


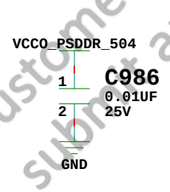


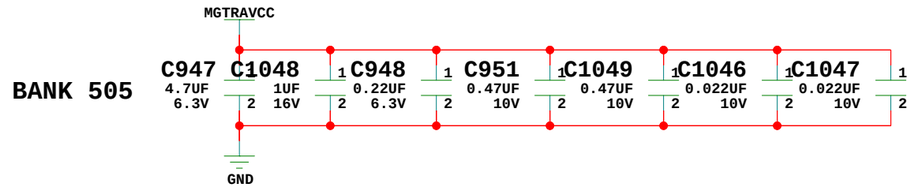




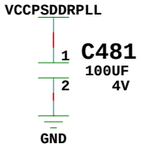


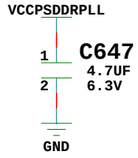






Place this round pad capacitor directly under U1





This schematic is currently undergoing HDMI electrical compliance. Customers seeking to understand HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

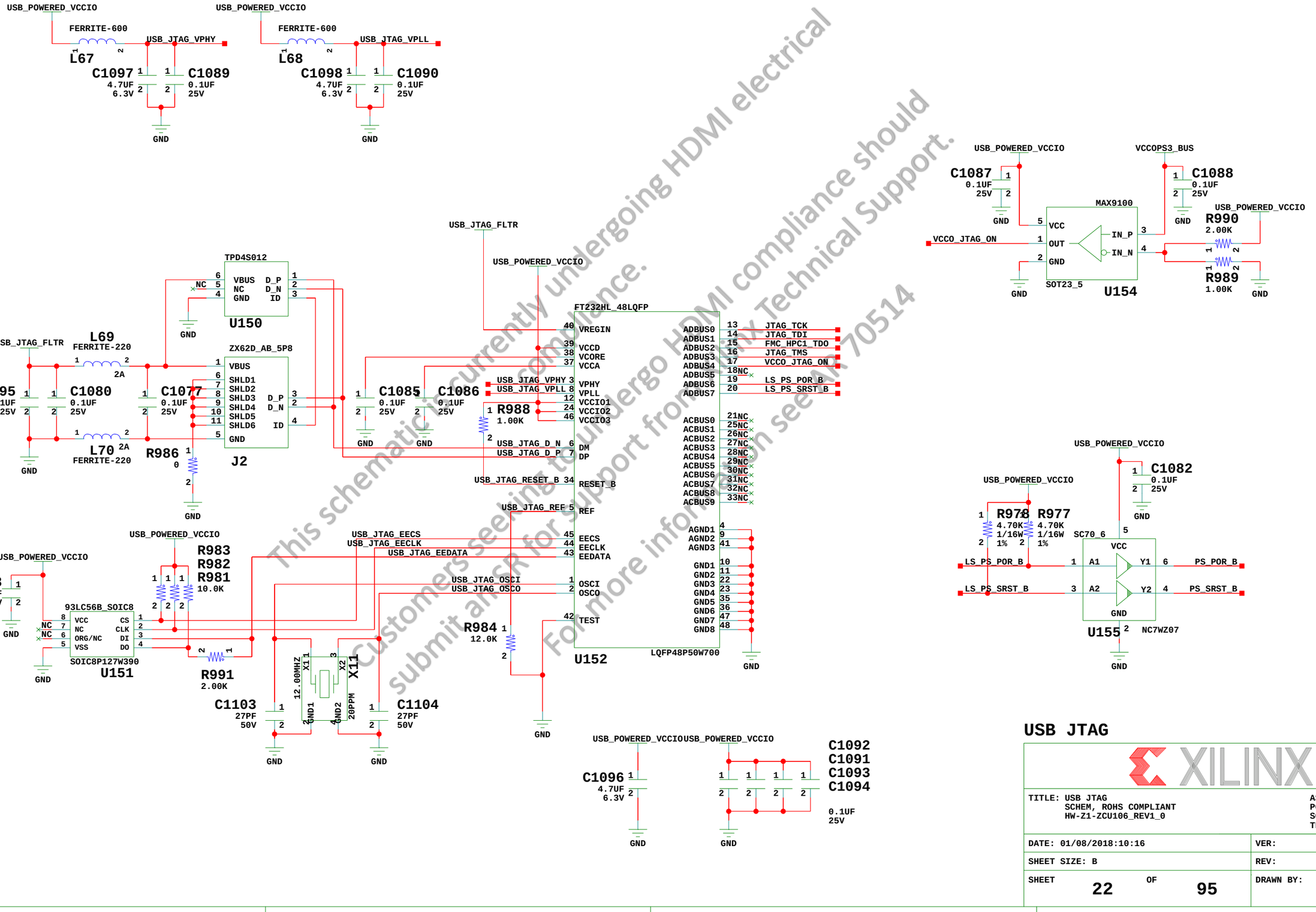
Zynq Decoupling 3

XILINX

TITLE: Zynq Decoupling 3
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0

ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186

DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 21 OF 95	DRAWN BY: BF

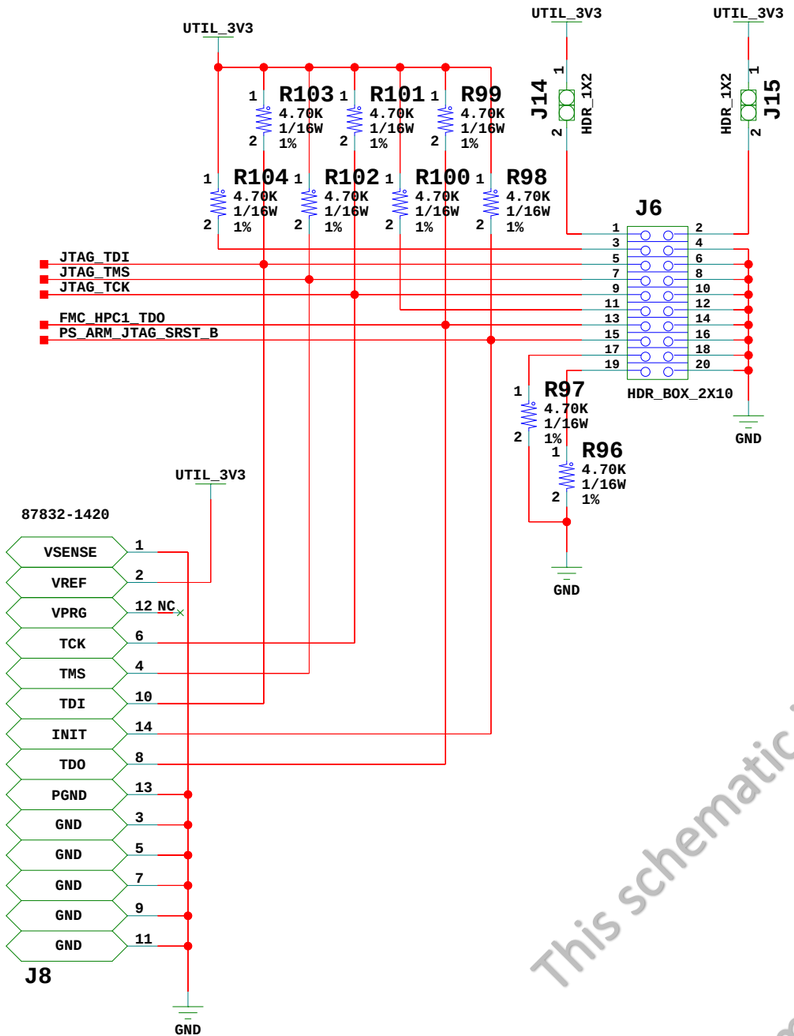


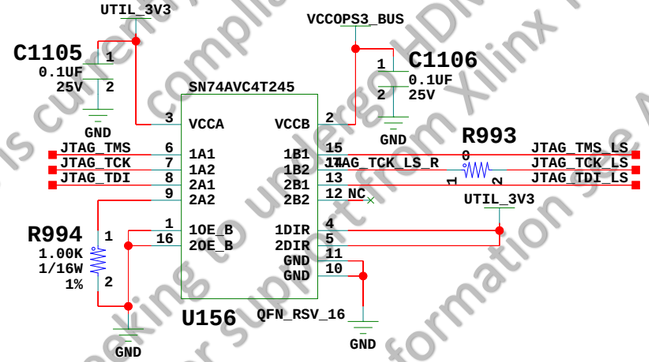
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

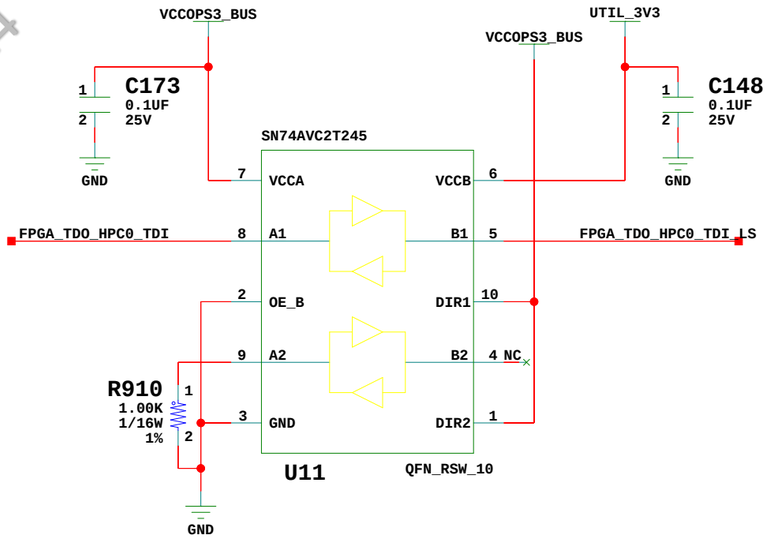
USB JTAG

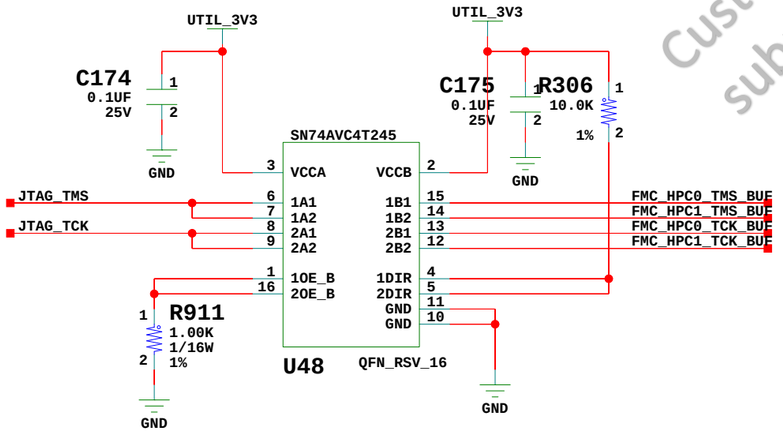
XILINX

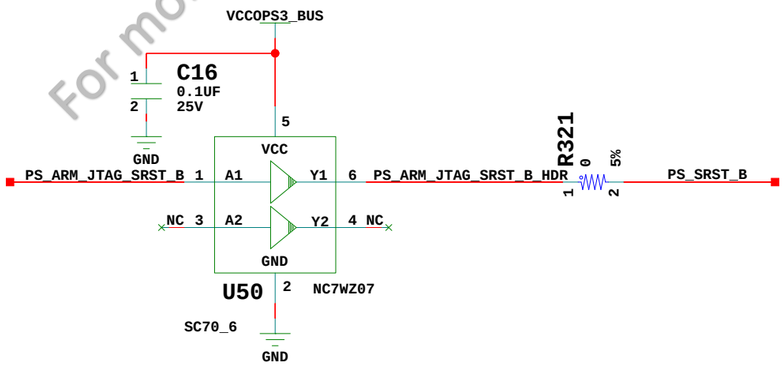
TITLE: USB JTAG SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 22 OF 95	DRAWN BY: BF











This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

JTAG Headers

XILINX

TITLE: JTAG Headers SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 23 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical
Customers needing to use HDMI compliance should
submit an SR for support from Xilinx Technical Support
For more information see UG 70514

PS_DDR4_VTT

DDR4_DIMM_VDDQ

DDR4_SO-DIMM_KEY_SHIELDED

J1

PS_DDR4_VPP_2V5

UTIL_3V3

GND

IIC Address = 0b1010001 (0x51)
SODIMM SA[2:0]=001

DDR4_DIMM_VDDQ

C2 0.1UF 25V

R108 100 1/10W 1%

VREF_SODIMM

C45 0.1UF 25V

R90 100 1/10W 1%

Layout Directive:
Place 0.1UF cap under memory adjacent to pin 25

UTIL_3V3

R506 4.70K 1/16W 1%
R505 4.70K 1/16W 1%
R38 4.70K 1/16W 1%
R37 4.70K 1/16W 1%
R32 4.70K 1/16W 1%

DDR4_SODIMM_CKE0
DDR4_SODIMM_CKE1
DDR4_SODIMM_SA2
DDR4_SODIMM_SA1
DDR4_SODIMM_SA0

DDR4_DIMM_VDDQ

R243 4.70K 1/16W 1%
R246 4.70K 1/16W 1%
R247 4.70K 1/16W 1%

DDR4_SODIMM_EVENT_B
DDR4_SODIMM_CS3_B
DDR4_SODIMM_CS2_B

DIMMCAGE1

SODIMM_CAGE

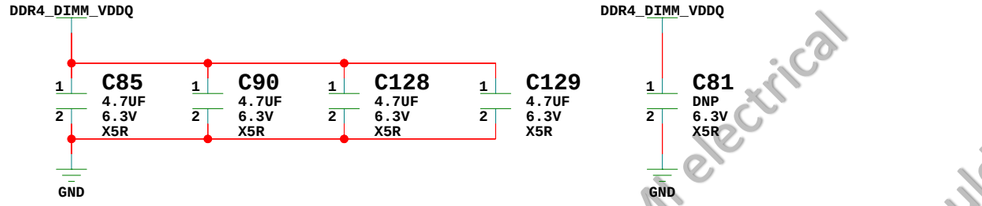
GND1 GND2 GND3 GND4 GND5 GND6 GND7 GND8 GND9 GND10 GND11 GND12 GND13 GND14 GND15 GND16 GND17 GND18 GND19

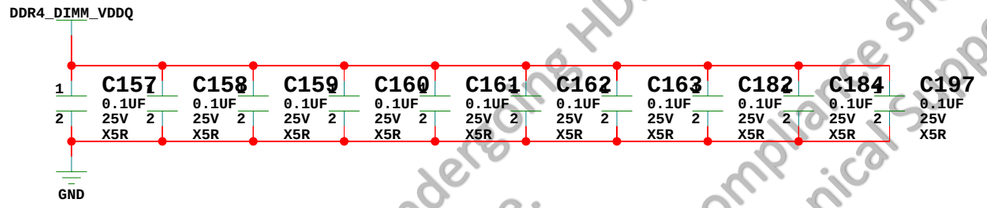
GND

PS DDR4 SODIMM 72 bit

XILINX

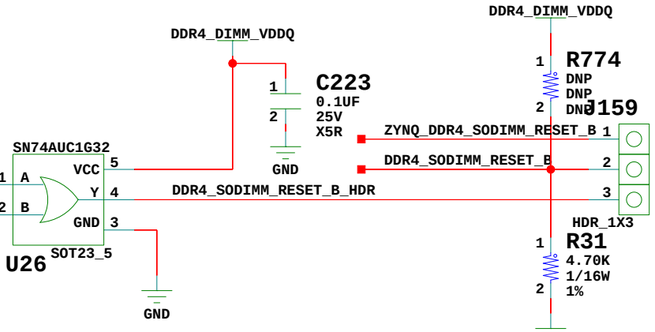
TITLE: PS DDR4 SODIMM 72 bit SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 24 OF 95	DRAWN BY: BF



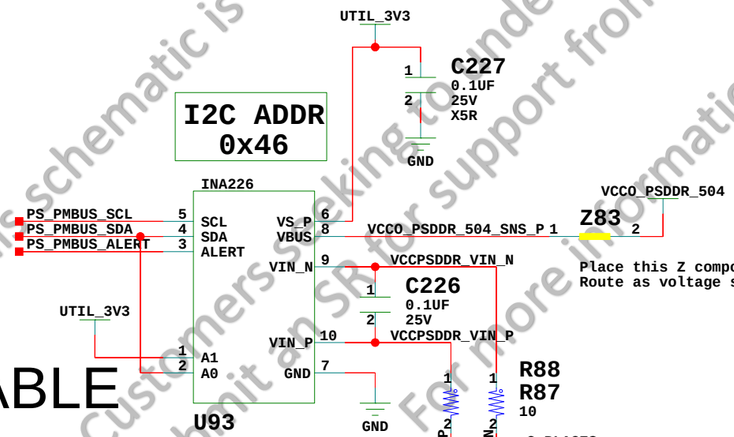


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70531

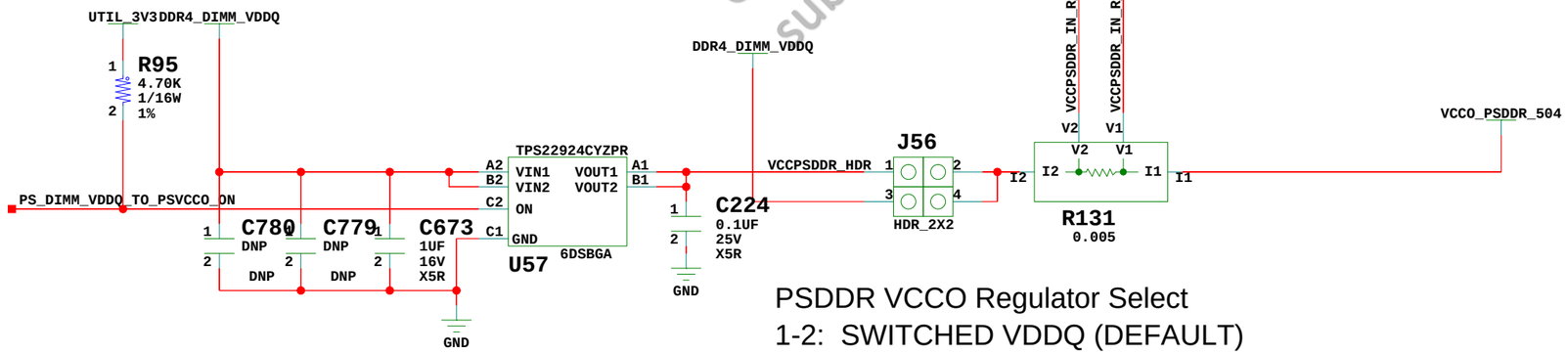
PLACE ABOVE CAPS BENEATH/CLOSE TO DIMM SOCKET



DDR DIMM RESET_B CONTROL



DDR DIMM VCCO DISABLE

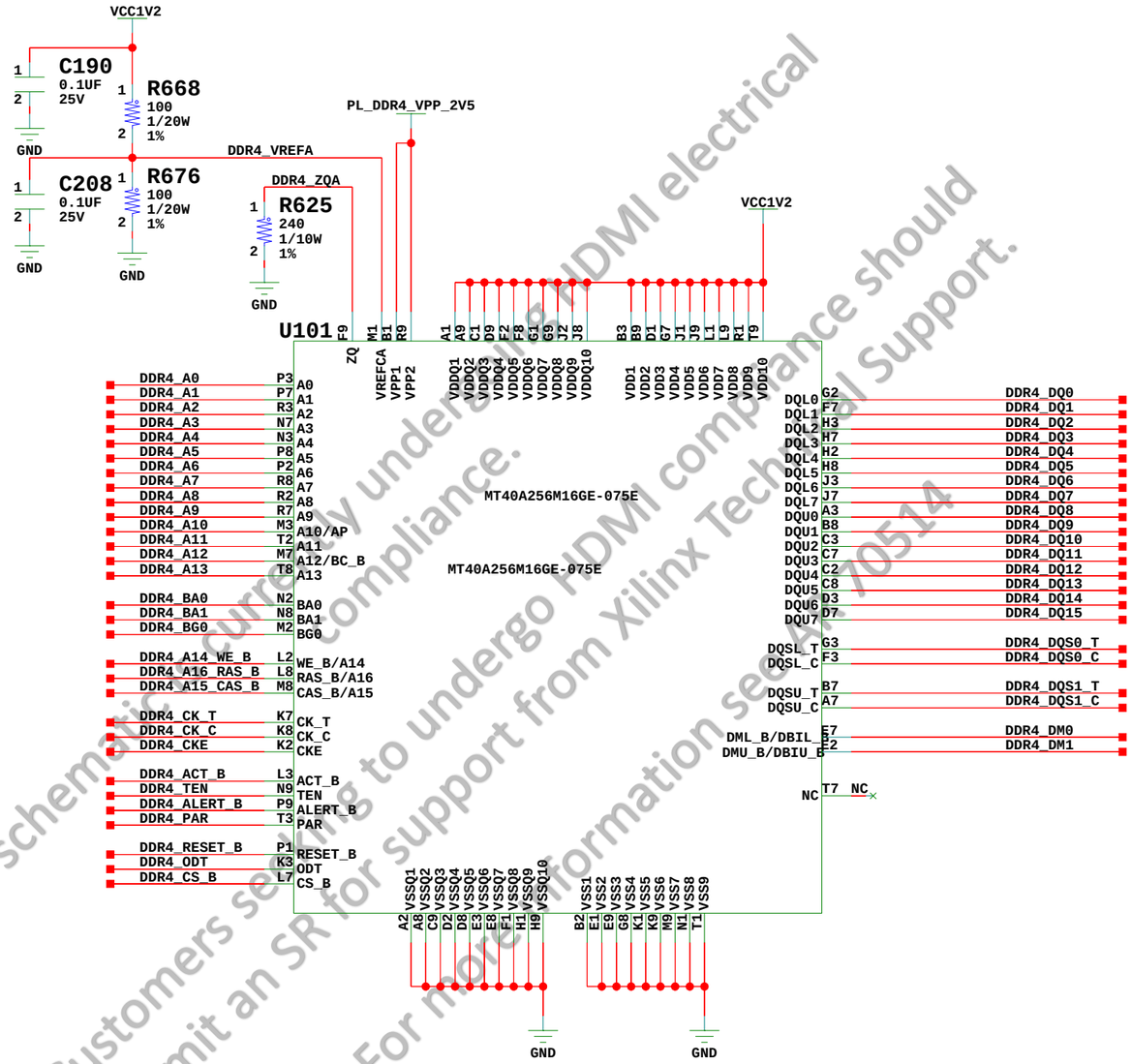


PSDDR VCCO Regulator Select
1-2: SWITCHED VDDQ (DEFAULT)
3-4: DIRECT VDDQ

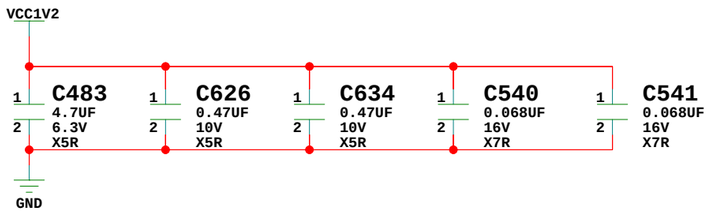
PS DDR4 SODIMM Decoupling

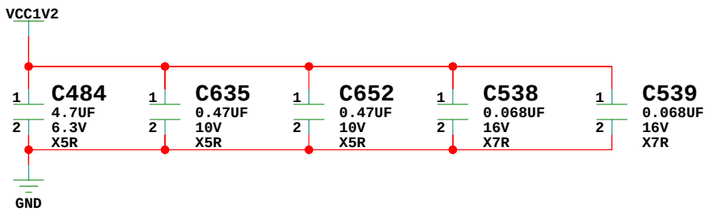
XILINX

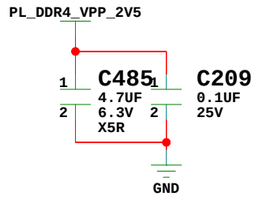
TITLE: PS DDR4 SODIMM Decoupling SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 25 OF 95	DRAWN BY: BF



This schematic currently undergoing HDMI electrical
Customers seeking to undergo HDMI compliance.
submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

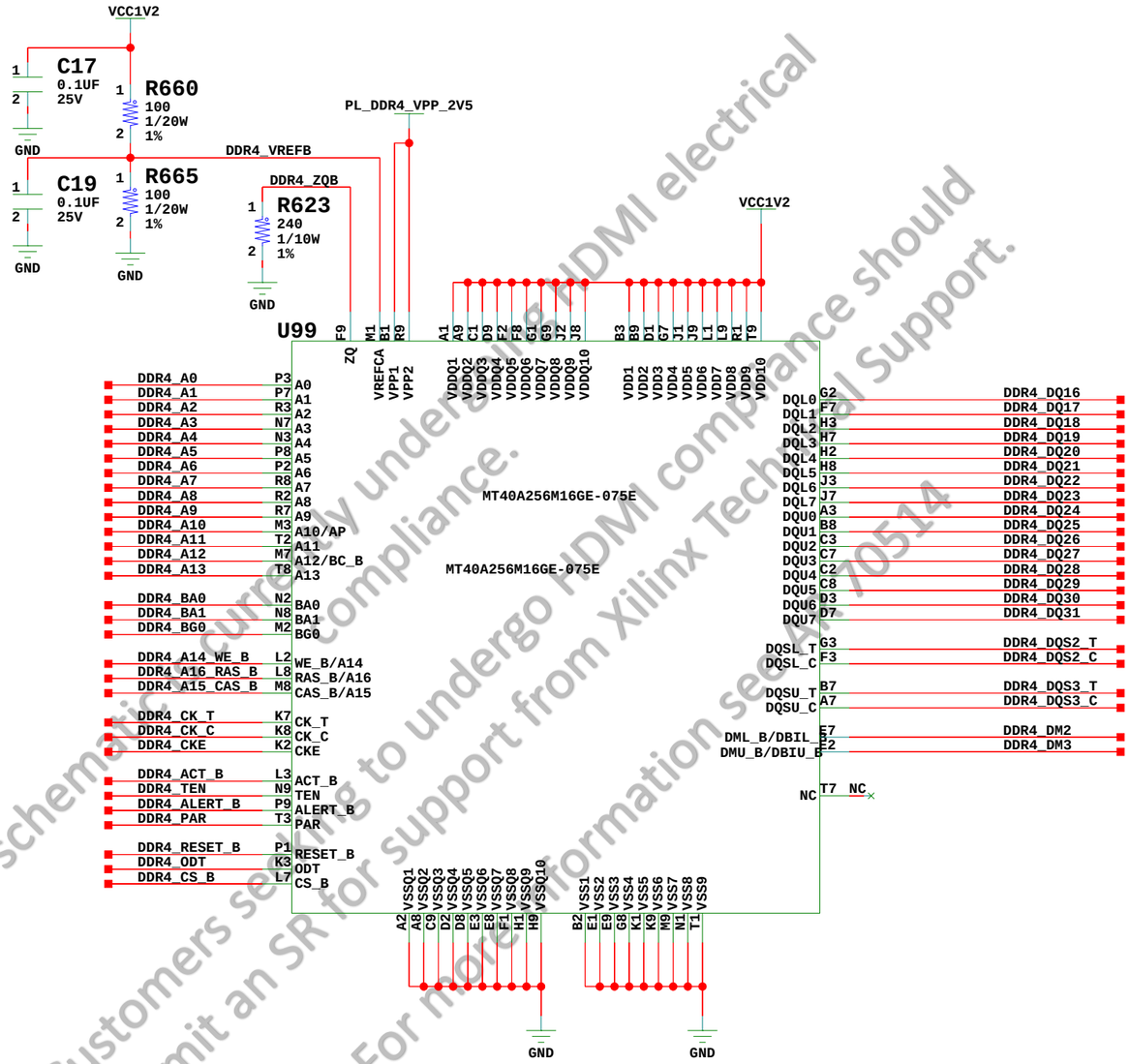




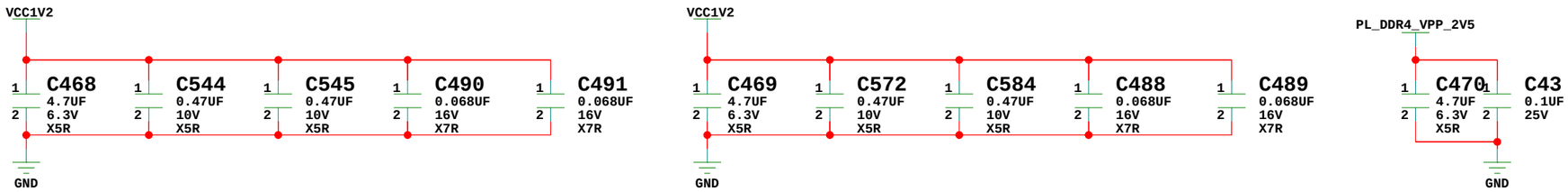


PL Component DDR4 Data [15-0]

TITLE: PL Component DDR4 Data [15-0] SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 26 OF 95	DRAWN BY: BF



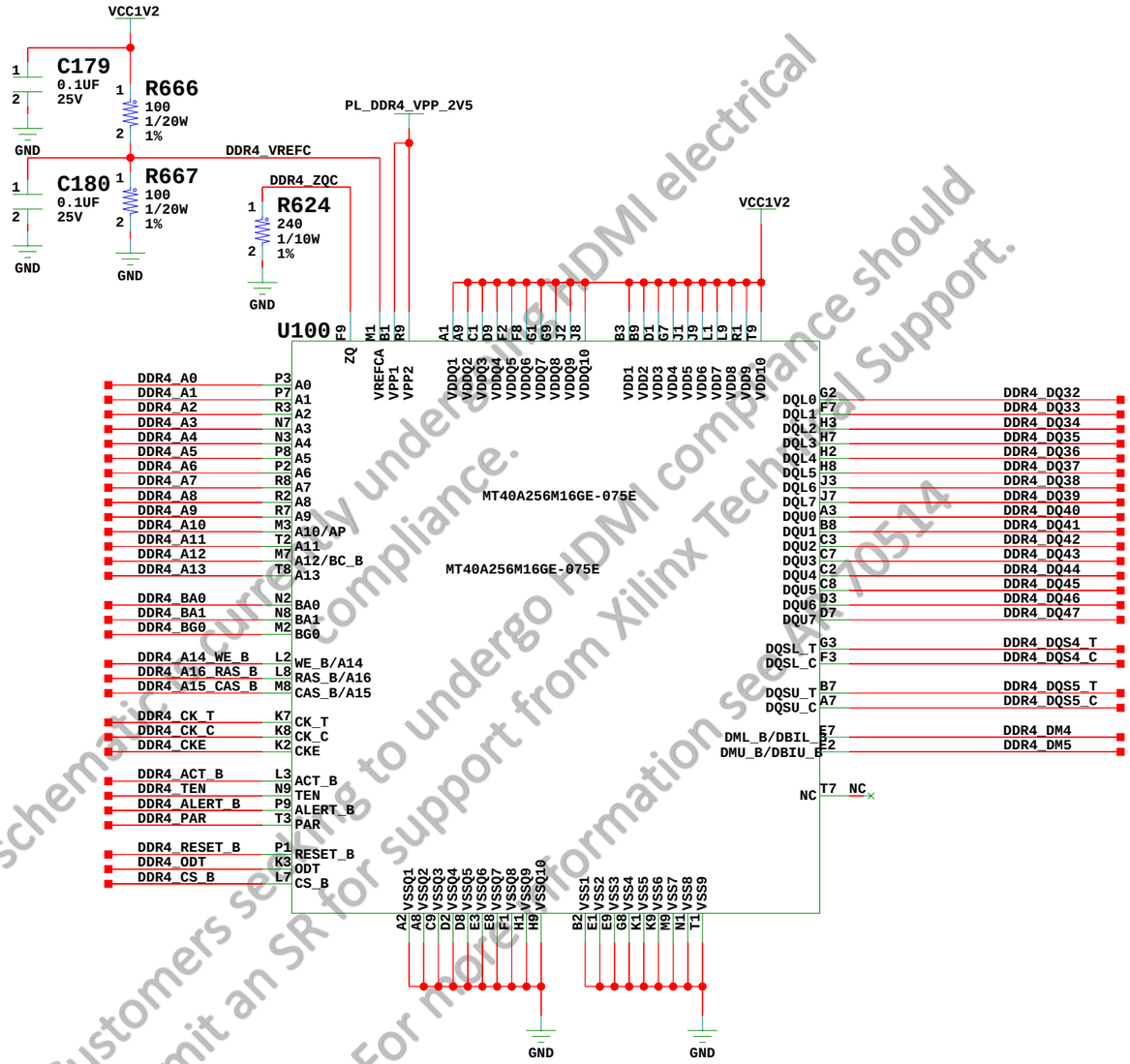
This schematic currently undergoing HDMI electrical compliance.
Customers seeking to use HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



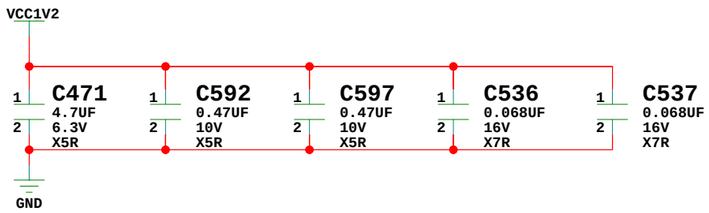
PL Component DDR4 Data [31-16]

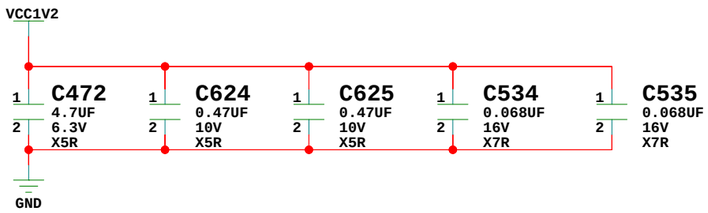
XILINX

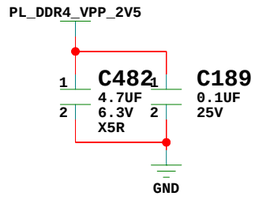
TITLE: PL Component DDR4 Data [31-16] SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 27 OF 95	DRAWN BY: BF



This schematic currently undergoing HDMI electrical compliance.
Customers seeking to use HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



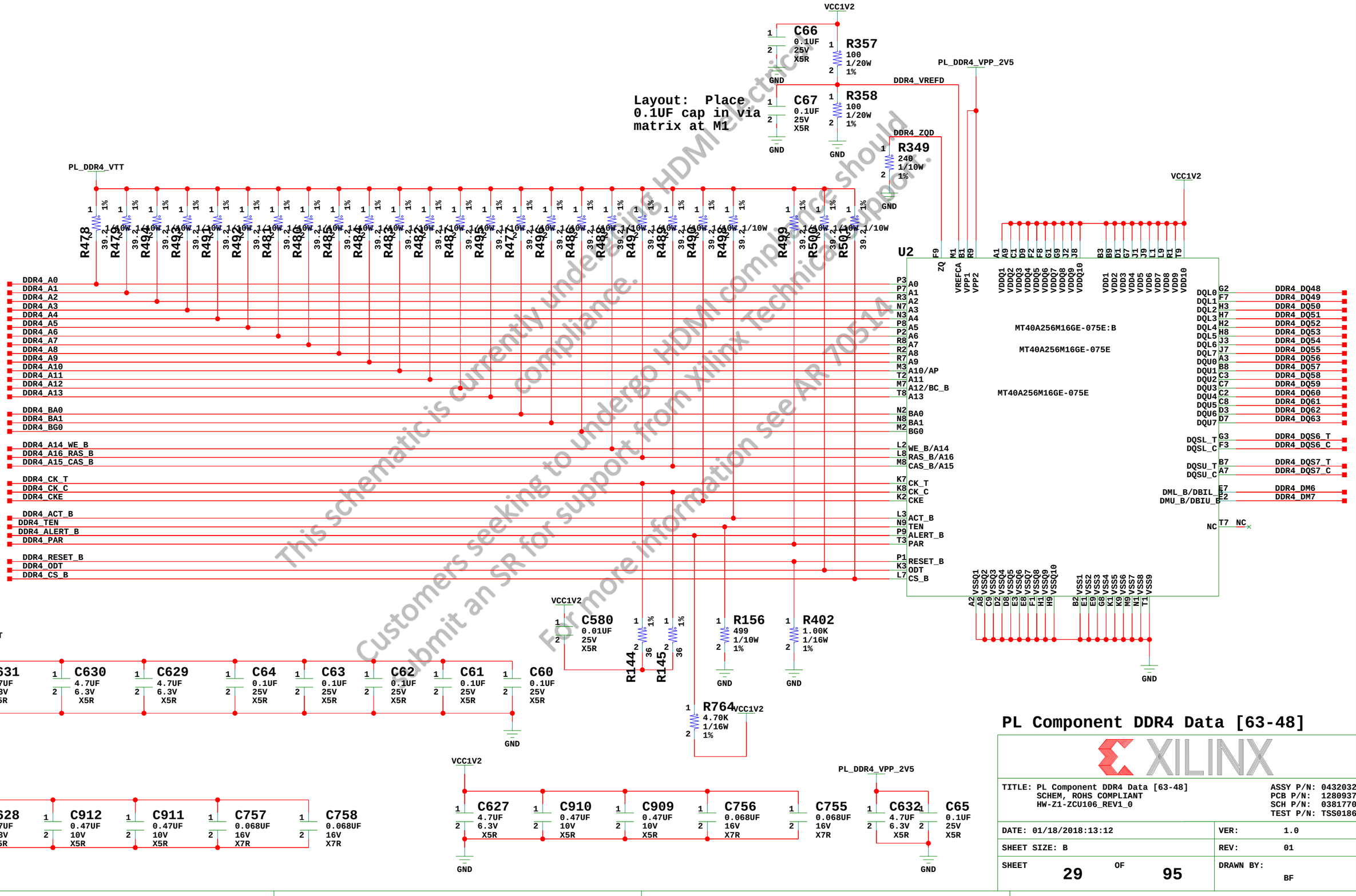




PL Component DDR4 Data [47-32]

XILINX

TITLE: PL Component DDR4 Data [47-32] SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 28 OF 95	DRAWN BY: BF

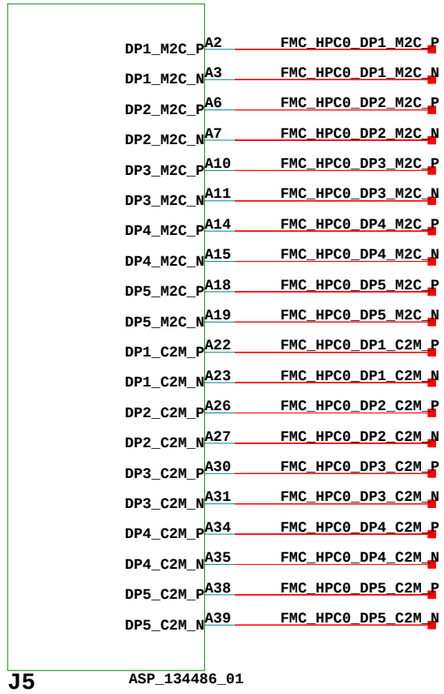


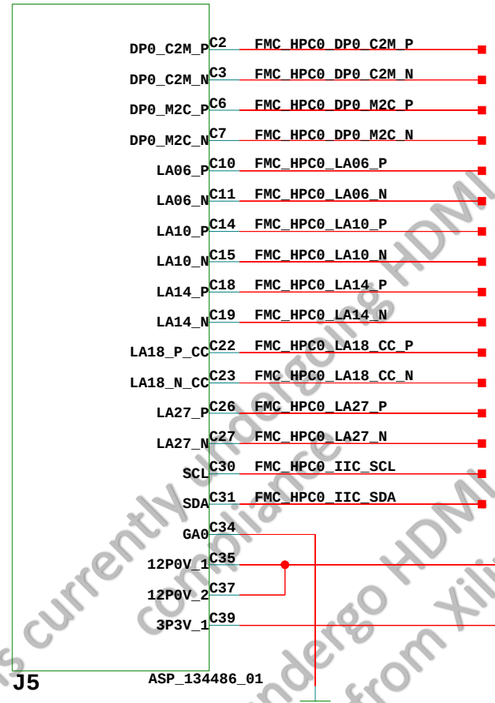
This schematic is currently undergoing HDMI electrical compliance. Customers seeking to use HDMI should submit an SR for support from Xilinx Technical Support. For more information see AR 70514.

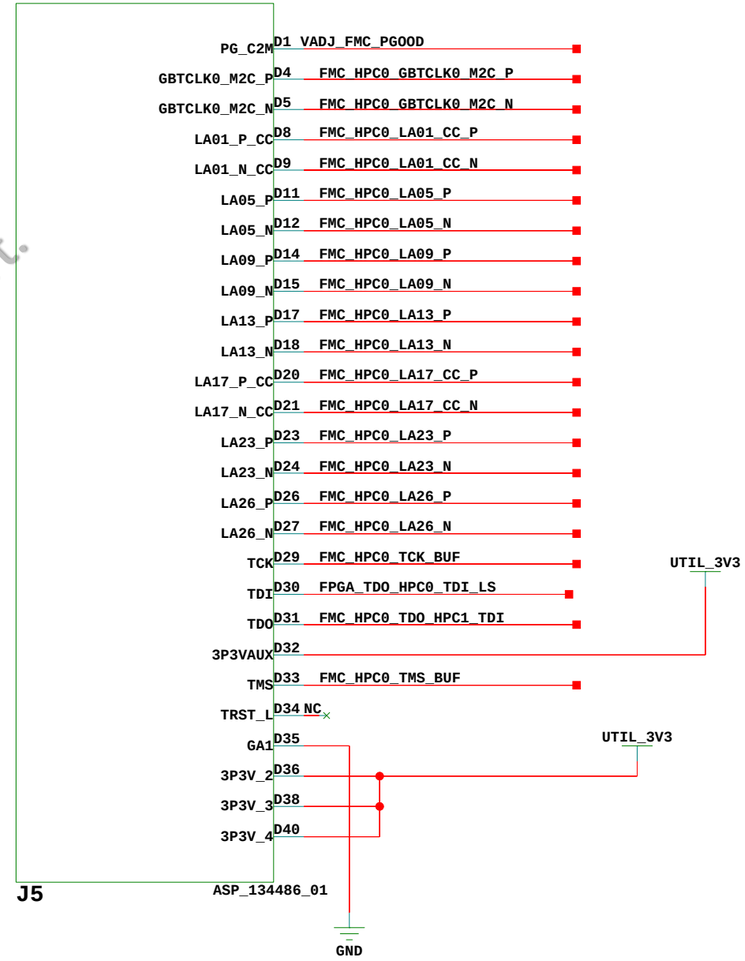
PL Component DDR4 Data [63-48]

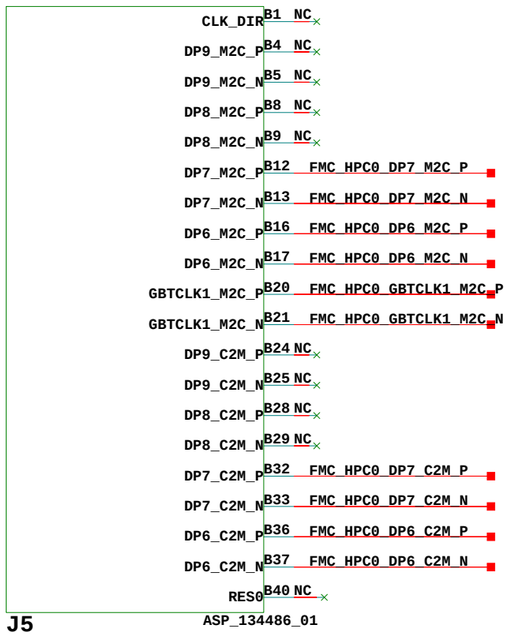
XILINX

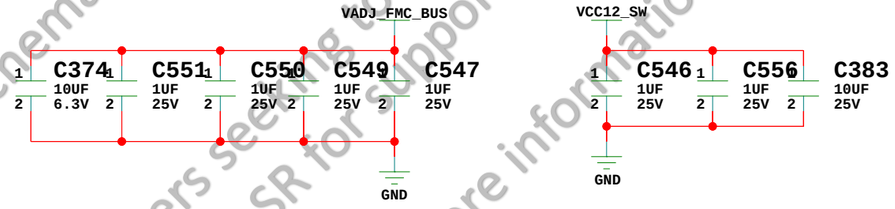
TITLE: PL Component DDR4 Data [63-48] SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/18/2018:13:12	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 29 OF 95	DRAWN BY: BF

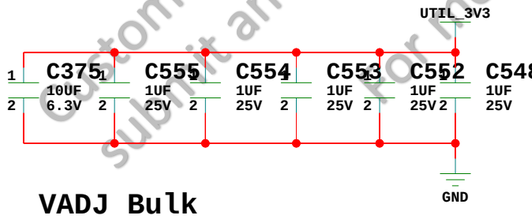


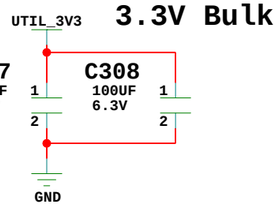






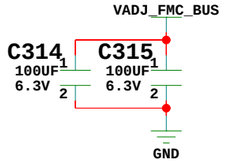


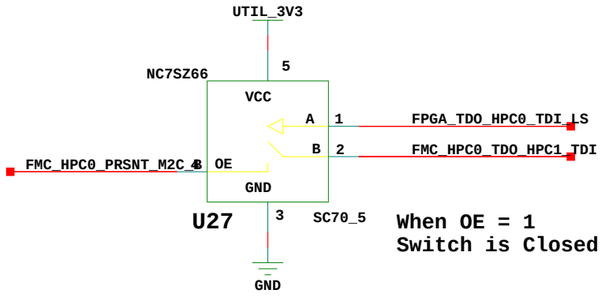




3.3V Bulk

VADJ Bulk





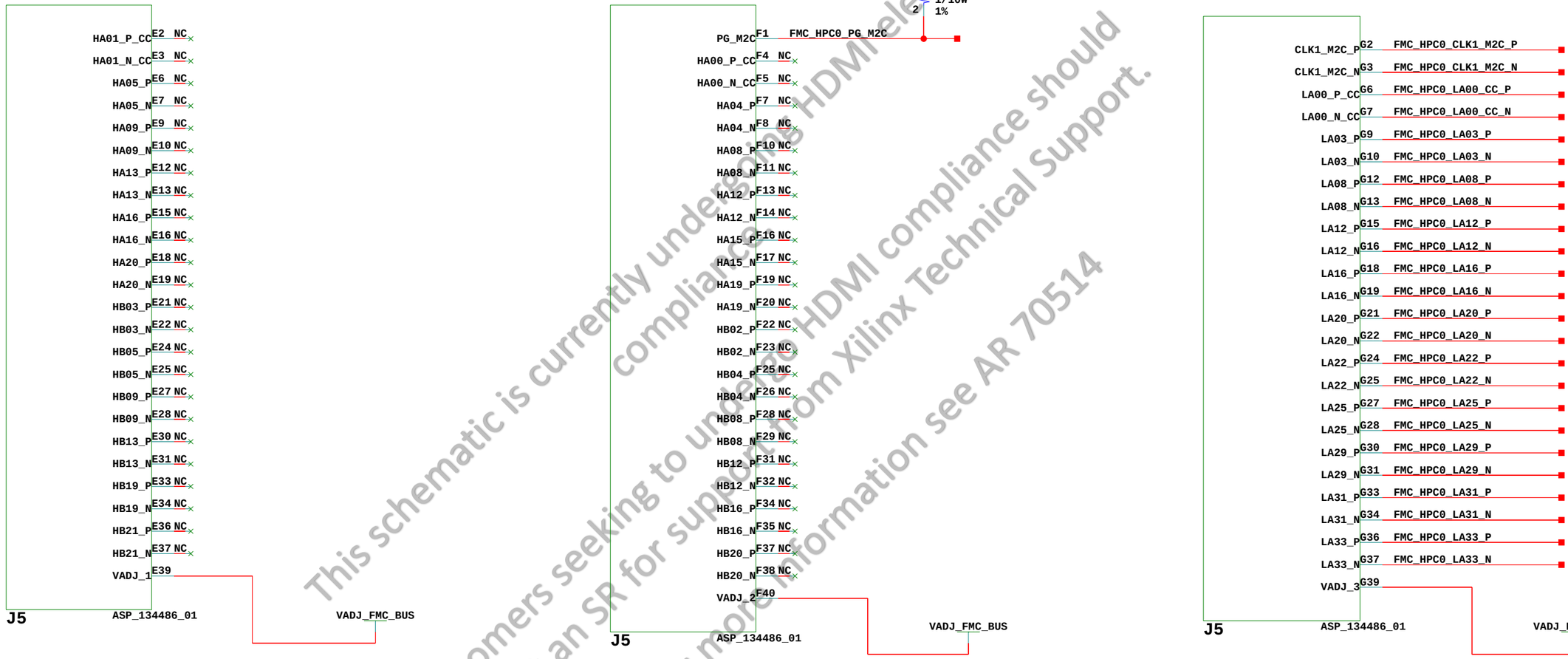
When OE = 1
Switch is Closed

This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC0 Header Rows A B C D

XILINX

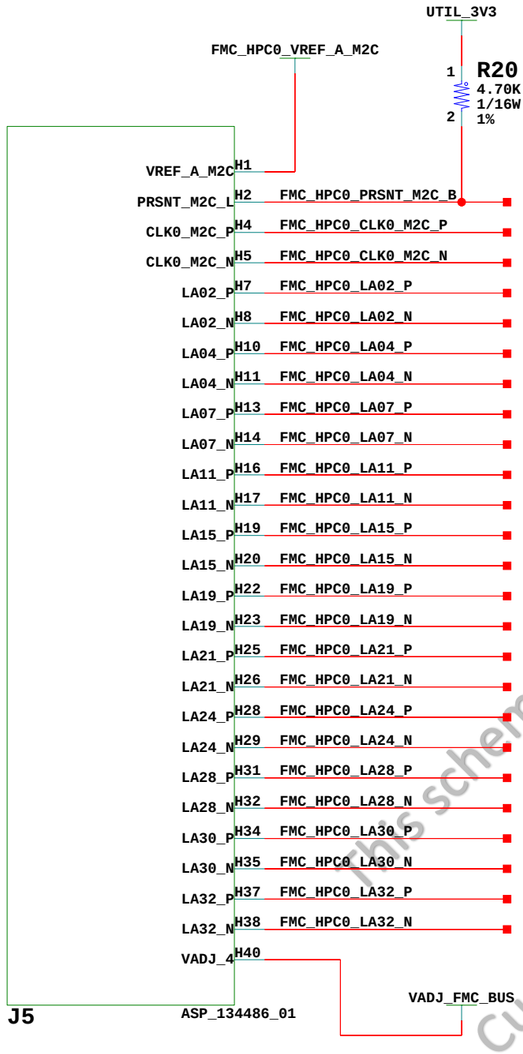
TITLE: PL FMC HPC0 Header Rows A B C D SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 30 OF 95	DRAWN BY: BF

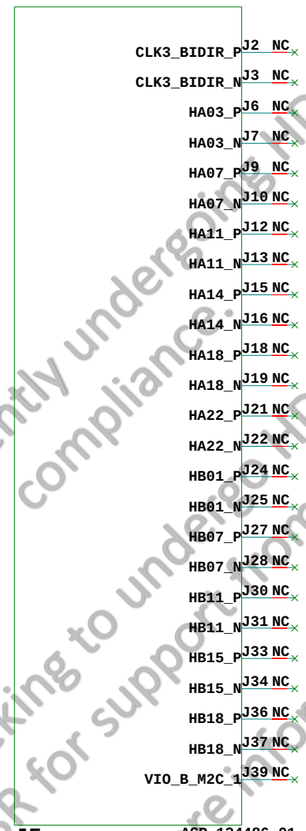


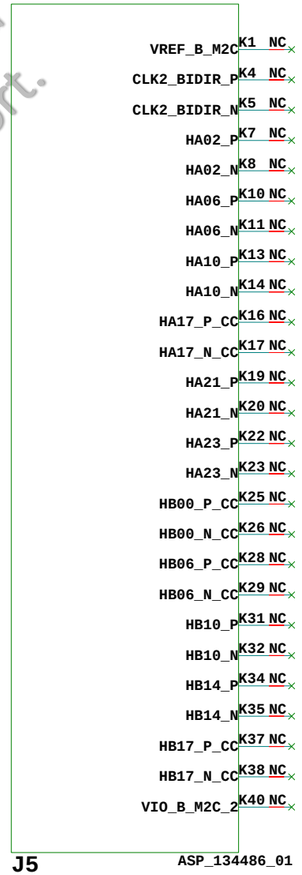
This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to use the HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC0 Header Rows E F G

TITLE: PL FMC HPC0 Header Rows E F G SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 31 OF 95	DRAWN BY: BF







This schematic is currently undergoing HDMI electrical compliance. Customers seeking to underpin HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC0 Header Rows H J K

XILINX

TITLE: PL FMC HPC0 Header Rows H J K SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 32 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 71514

J5

ASP_134486_01

GND_80 E17 GND_79 E14 GND_78 E11 GND_77 E8 GND_76 E5 GND_75 E4 GND_74 E1 GND_73 D39 GND_72 D37 GND_71 D28 GND_70 D25 GND_69 D22 GND_68 D19 GND_67 D16 GND_66 D13 GND_65 D10 GND_64 D7 GND_63 D6 GND_62 D3 GND_61 D2 GND_60 C40 GND_59 C38 GND_58 C36 GND_57 C33 GND_56 C32 GND_55 C29 GND_54 C28 GND_53 C25 GND_52 C24 GND_51 C21 GND_50 C20 GND_49 C17 GND_48 C16 GND_47 C13 GND_46 C12 GND_45 C9 GND_44 C8 GND_43 C5 GND_42 C4 GND_41 C1 GND_40 B39 GND_39 B38 GND_38 B35 GND_37 B34 GND_36 B31 GND_35 B30 GND_34 B27 GND_33 B26 GND_32 B23 GND_31 B22 GND_30 B19 GND_29 B18 GND_28 B15 GND_27 B14 GND_26 B11 GND_25 B10 GND_24 B7 GND_23 B6 GND_22 B3 GND_21 B2 GND_20 A40 GND_19 A37 GND_18 A36 GND_17 A33 GND_16 A32 GND_15 A29 GND_14 A28 GND_13 A25 GND_12 A24 GND_11 A21 GND_10 A20 GND_9 A17 GND_8 A16 GND_7 A13 GND_6 A12 GND_5 A9 GND_4 A8 GND_3 A5 GND_2 A4 GND_1 A1 GND

ST2 GND_ST2 ST1 GND_ST1 K39 GND_159 K36 GND_158 K33 GND_157 K30 GND_156 K27 GND_155 K24 GND_154 K21 GND_153 K18 GND_152 K15 GND_151 K12 GND_150 K9 GND_149 K6 GND_148 K3 GND_147 K2 GND_146 J40 GND_145 J38 GND_144 J35 GND_143 J32 GND_142 J29 GND_141 J26 GND_140 J23 GND_139 J20 GND_138 J17 GND_137 J14 GND_136 J11 GND_135 J8 GND_134 J5 GND_133 J4 GND_132 J1 GND_131 H39 GND_130 H36 GND_129 H33 GND_128 H30 GND_127 H27 GND_126 H24 GND_125 H21 GND_124 H18 GND_123 H15 GND_122 H12 GND_121 H9 GND_120 H6 GND_119 H3 GND_118 G40 GND_117 G38 GND_116 G35 GND_115 G32 GND_114 G29 GND_113 G26 GND_112 G23 GND_111 G20 GND_110 G17 GND_109 G14 GND_108 G11 GND_107 G8 GND_106 G5 GND_105 G4 GND_104 G1 GND_103 F39 GND_102 F36 GND_101 F33 GND_100 F30 GND_99 F27 GND_98 F24 GND_97 F21 GND_96 F18 GND_95 F15 GND_94 F12 GND_93 F9 GND_92 F6 GND_91 F3 GND_90 F2 GND_89 E40 GND_88 E38 GND_87 E35 GND_86 E32 GND_85 E29 GND_84 E26 GND_83 E23 GND_82 E20 GND_81 GND

C869 100PF 500V NP0 R320 DNP DNP DNP

C870 100PF 500V NP0 R319 DNP DNP DNP

GND GND

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC0 Header GND

XILINX

TITLE: PL FMC HPC0 Header GND SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 33 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

J4 — ASP_134486_01

DP1_M2C_P	A2	NC	
DP1_M2C_N	A3	NC	
DP2_M2C_P	A6	NC	
DP2_M2C_N	A7	NC	
DP3_M2C_P	A10	NC	
DP3_M2C_N	A11	NC	
DP4_M2C_P	A14	NC	
DP4_M2C_N	A15	NC	
DP5_M2C_P	A18	NC	
DP5_M2C_N	A19	NC	
DP1_C2M_P	A22	NC	
DP1_C2M_N	A23	NC	
DP2_C2M_P	A26	NC	
DP2_C2M_N	A27	NC	
DP3_C2M_P	A30	NC	
DP3_C2M_N	A31	NC	
DP4_C2M_P	A34	NC	
DP4_C2M_N	A35	NC	
DP5_C2M_P	A38	NC	
DP5_C2M_N	A39	NC	

J4 — ASP_134486_01

DP0_C2M_P	C2	FMC_HPC1_DP0_C2M_P	
DP0_C2M_N	C3	FMC_HPC1_DP0_C2M_N	
DP0_M2C_P	C6	FMC_HPC1_DP0_M2C_P	
DP0_M2C_N	C7	FMC_HPC1_DP0_M2C_N	
LA06_P	C10	FMC_HPC1_LA06_P	
LA06_N	C11	FMC_HPC1_LA06_N	
LA10_P	C14	FMC_HPC1_LA10_P	
LA10_N	C15	FMC_HPC1_LA10_N	
LA14_P	C18	FMC_HPC1_LA14_P	
LA14_N	C19	FMC_HPC1_LA14_N	
LA18_P_CC	C22	NC	
LA18_N_CC	C23	NC	
LA27_P	C26	NC	
LA27_N	C27	NC	
SCL	C30	FMC_HPC1_IIC_SCL	
SDA	C31	FMC_HPC1_IIC_SDA	
GA0	C34		VCC12_SW
12P0V_1	C35		
12P0V_2	C37		UTIL_3V3
3P3V_1	C39		

GND

VADJ_FMC_BUS

C364 10UF 6.3V · C500 1UF 25V · C499 1UF 25V · C498 1UF 25V · C507 1UF 25V

GND

VCC12_SW

C506 1UF 25V · C505 1UF 25V · C381 10UF 25V

GND

J4 — ASP_134486_01

PG_C2M	D1	VADJ_FMC_PGOOD	
GBTCLK0_M2C_P	D4	FMC_HPC1_GBTCLK0_M2C_P	
GBTCLK0_M2C_N	D5	FMC_HPC1_GBTCLK0_M2C_N	
LA01_P_CC	D8	FMC_HPC1_LA01_CC_P	
LA01_N_CC	D9	FMC_HPC1_LA01_CC_N	
LA05_P	D11	FMC_HPC1_LA05_P	
LA05_N	D12	FMC_HPC1_LA05_N	
LA09_P	D14	FMC_HPC1_LA09_P	
LA09_N	D15	FMC_HPC1_LA09_N	
LA13_P	D17	FMC_HPC1_LA13_P	
LA13_N	D18	FMC_HPC1_LA13_N	
LA17_P_CC	D20	NC	
LA17_N_CC	D21	NC	
LA23_P	D23	NC	
LA23_N	D24	NC	
LA26_P	D26	NC	
LA26_N	D27	NC	
TCK	D29	FMC_HPC1_TCK_BUF	UTIL_3V3
TDI	D30	FMC_HPC0_TDO_HPC1_TDI	
TDO	D31	FMC_HPC1_TDO	
3P3VAUX	D32		
TMS	D33	FMC_HPC1_TMS_BUF	
TRST_L	D34	NC	
GA1	D35		UTIL_3V3
3P3V_2	D36		
3P3V_3	D38		
3P3V_4	D40		

GND

J4 — ASP_134486_01

CLK_DIR	B1	NC	
DP9_M2C_P	B4	NC	
DP9_M2C_N	B5	NC	
DP8_M2C_P	B8	NC	
DP8_M2C_N	B9	NC	
DP7_M2C_P	B12	NC	
DP7_M2C_N	B13	NC	
DP6_M2C_P	B16	NC	
DP6_M2C_N	B17	NC	
GBTCLK1_M2C_P	B20	NC	
GBTCLK1_M2C_N	B21	NC	
DP9_C2M_P	B24	NC	
DP9_C2M_N	B25	NC	
DP8_C2M_P	B28	NC	
DP8_C2M_N	B29	NC	
DP7_C2M_P	B32	NC	
DP7_C2M_N	B33	NC	
DP6_C2M_P	B36	NC	
DP6_C2M_N	B37	NC	
RES0	B40	NC	

UTIL_3V3

C365 10UF 6.3V · C504 1UF 25V · C503 1UF 25V · C502 1UF 25V · C501 1UF 25V · C508 1UF 25V · C274 100UF 6.3V

GND

3.3V Bulk

UTIL_3V3

C275 100UF 6.3V

GND

UTIL_3V3

C321 100UF 6.3V

GND

VADJ Bulk

VADJ_FMC_BUS

C312 100UF 6.3V · C313 100UF 6.3V

GND

UTIL_3V3

NC7SZ66 — U24 — SC70_5

	5	VCC	
	1	A	FMC_HPC0_TDO_HPC1_TDI
	2	B	FMC_HPC1_TDO
FMC_HPC1_PRSNT_M2C_B	4	OE	
	3	GND	

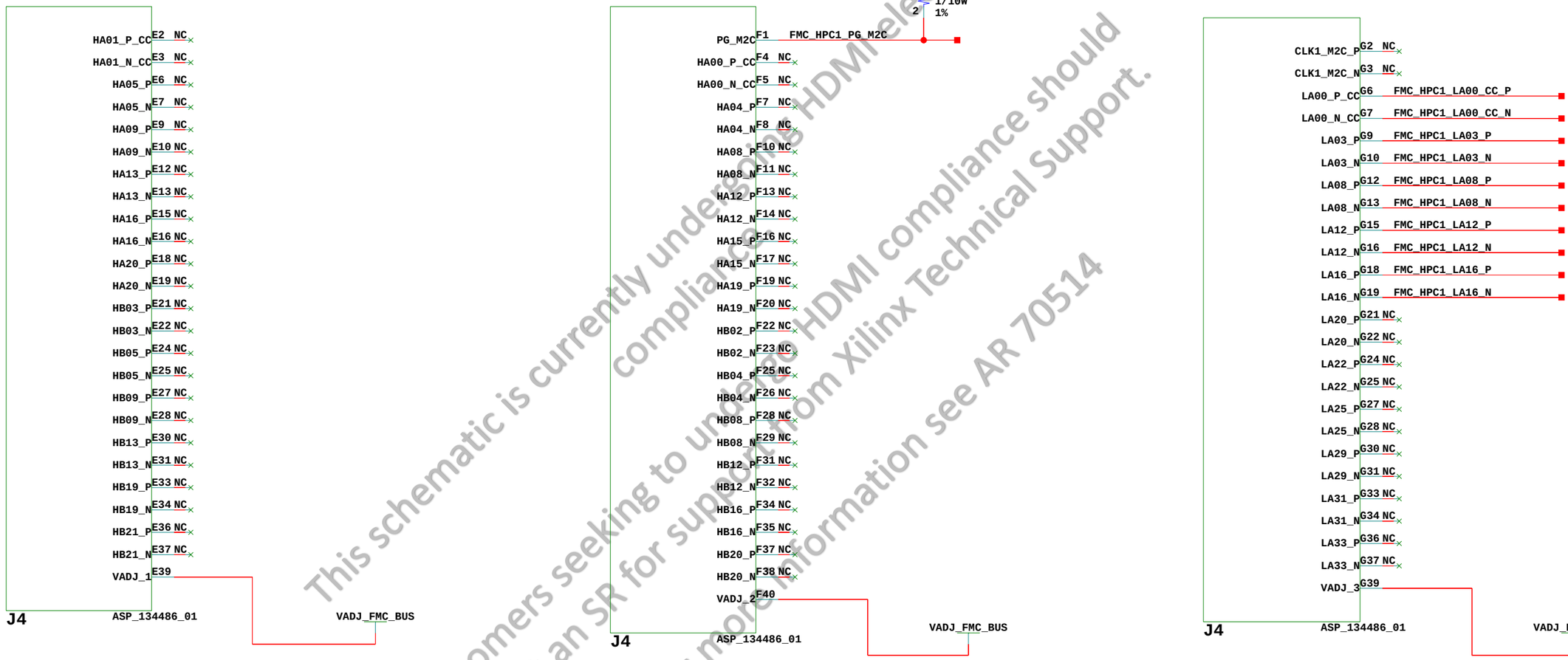
When OE = 1
Switch is Closed

GND

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC1 Header Rows A B C D

XILINX

TITLE: PL FMC HPC1 Header Rows A B C D SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 34 OF 95	DRAWN BY: BF

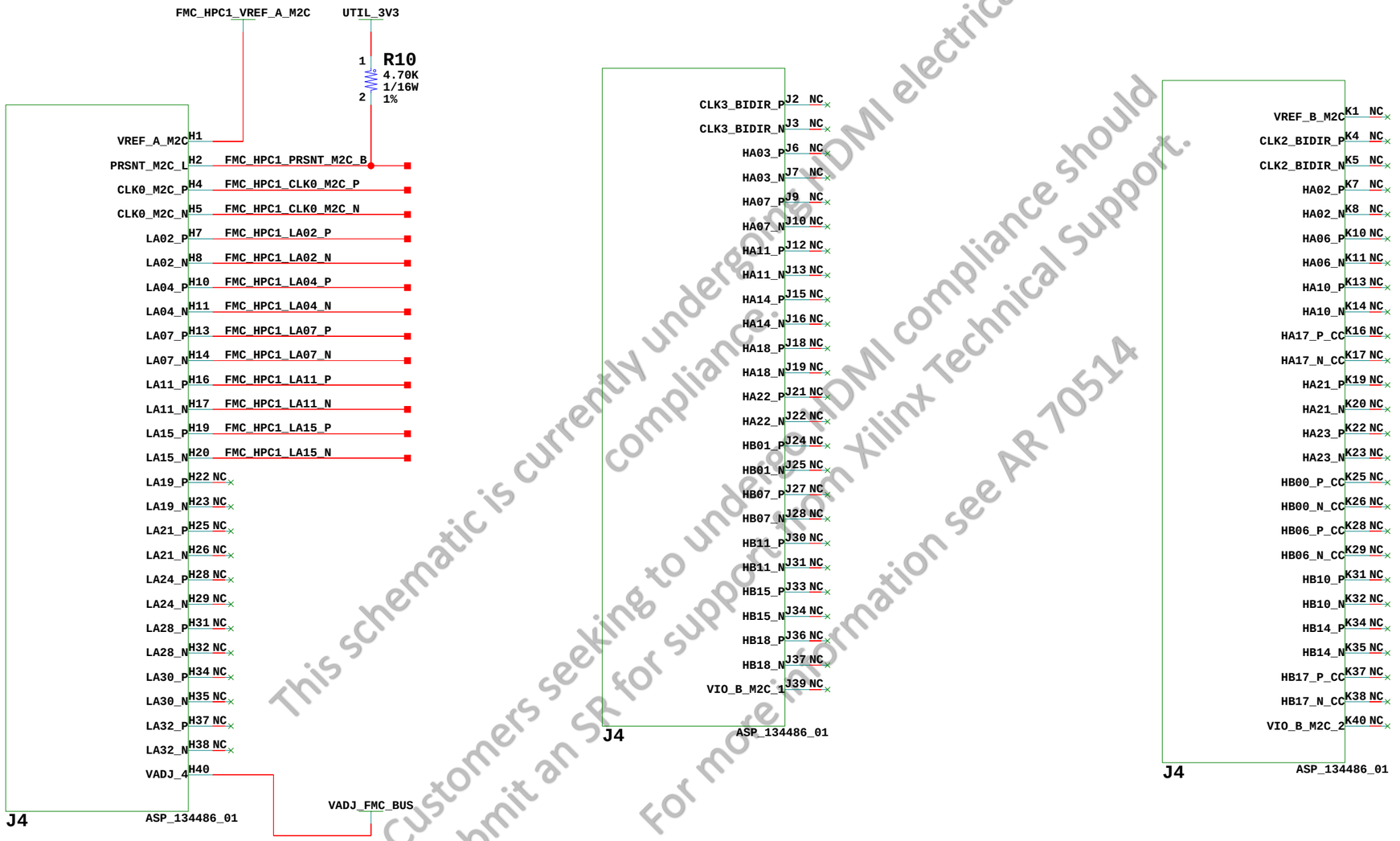


This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to use HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC1 Header Rows E F G

XILINX

TITLE: PL FMC HPC1 Header Rows E F G SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 35 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance. Customers seeking to underpin HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC1 Header Rows H J K

XILINX

TITLE: PL FMC HPC1 Header Rows H J K SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 36 OF 95	DRAWN BY: BF

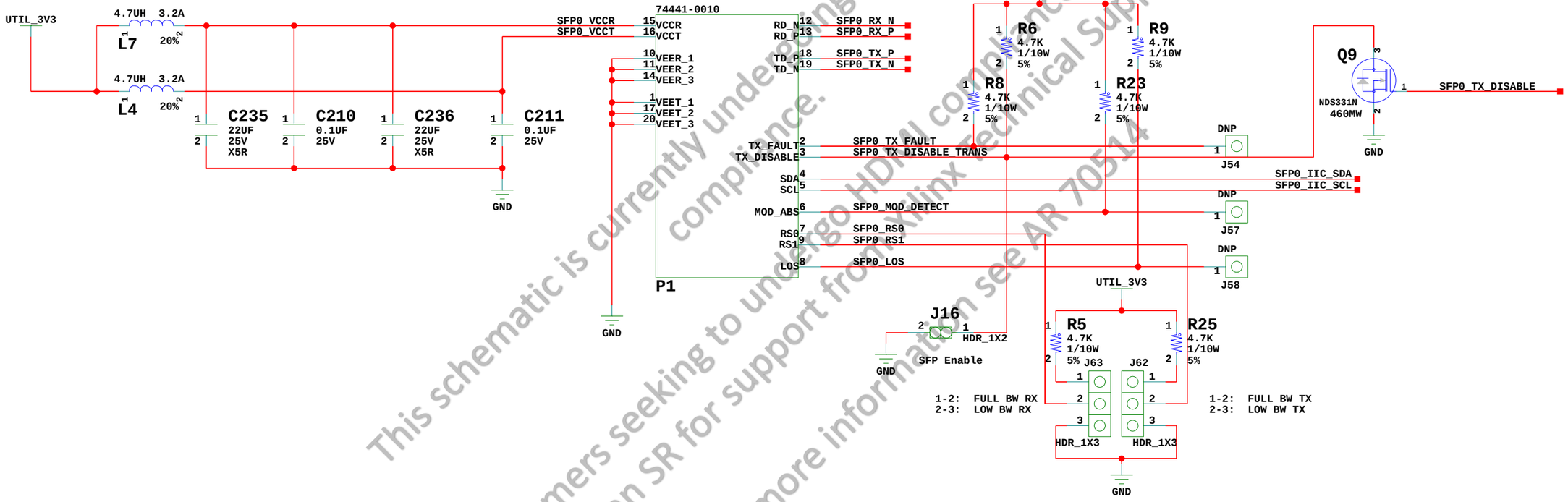


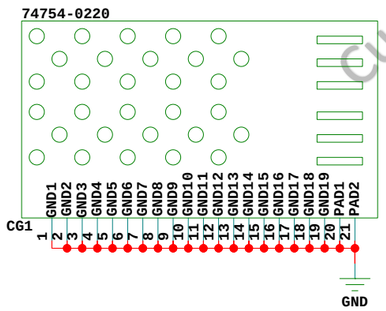
This schematic is currently undergoing HDMI electrical compliance testing.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 71514

ANSI/VITA 57.1 - Revised 2010
PL FMC HPC1 Header GND

XILINX

TITLE: PL FMC HPC1 Header GND SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 37 OF 95	DRAWN BY: BF



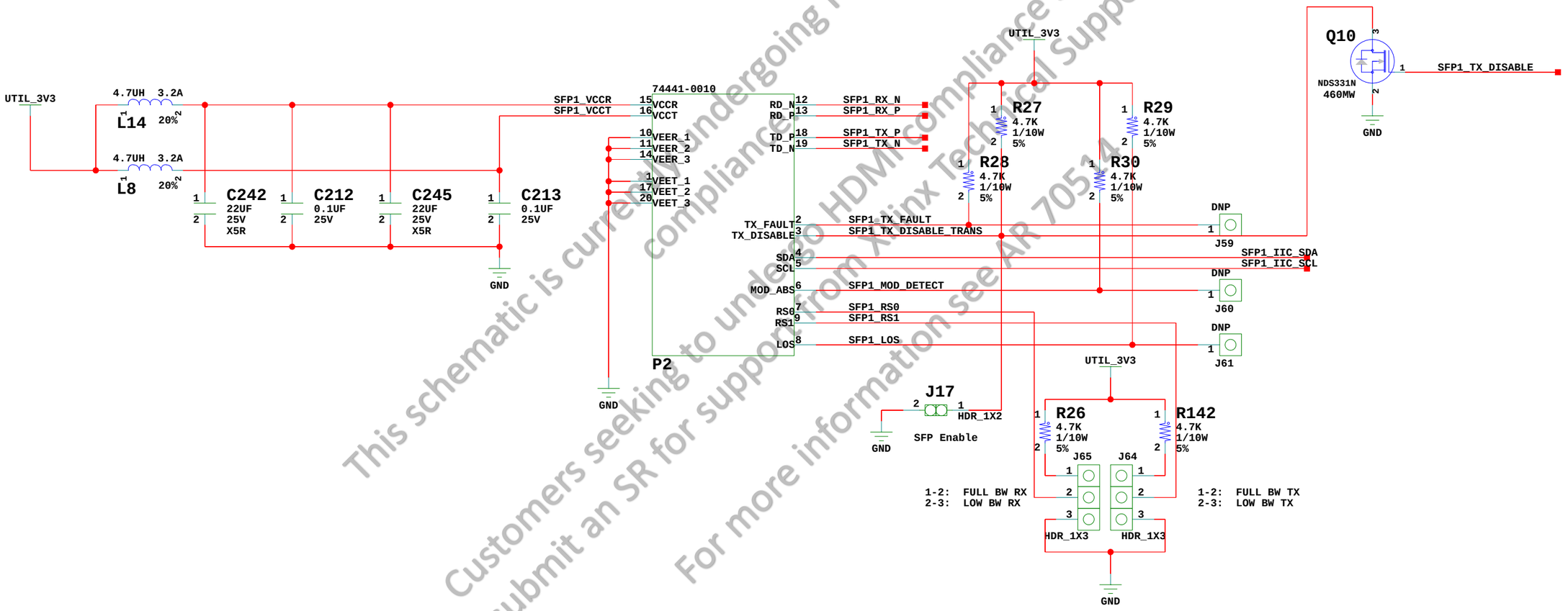


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL SFP0 - SFP+ Connector and Dual Cage

XILINX

TITLE: PL SFP0 - SFP+ Connector and Dual Cage SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 38 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to use 3rd party HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

PL SFP1 - SFP+ Connector

XILINX

TITLE: PL SFP1 - SFP+ Connector SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 39 OF 95	DRAWN BY: BF

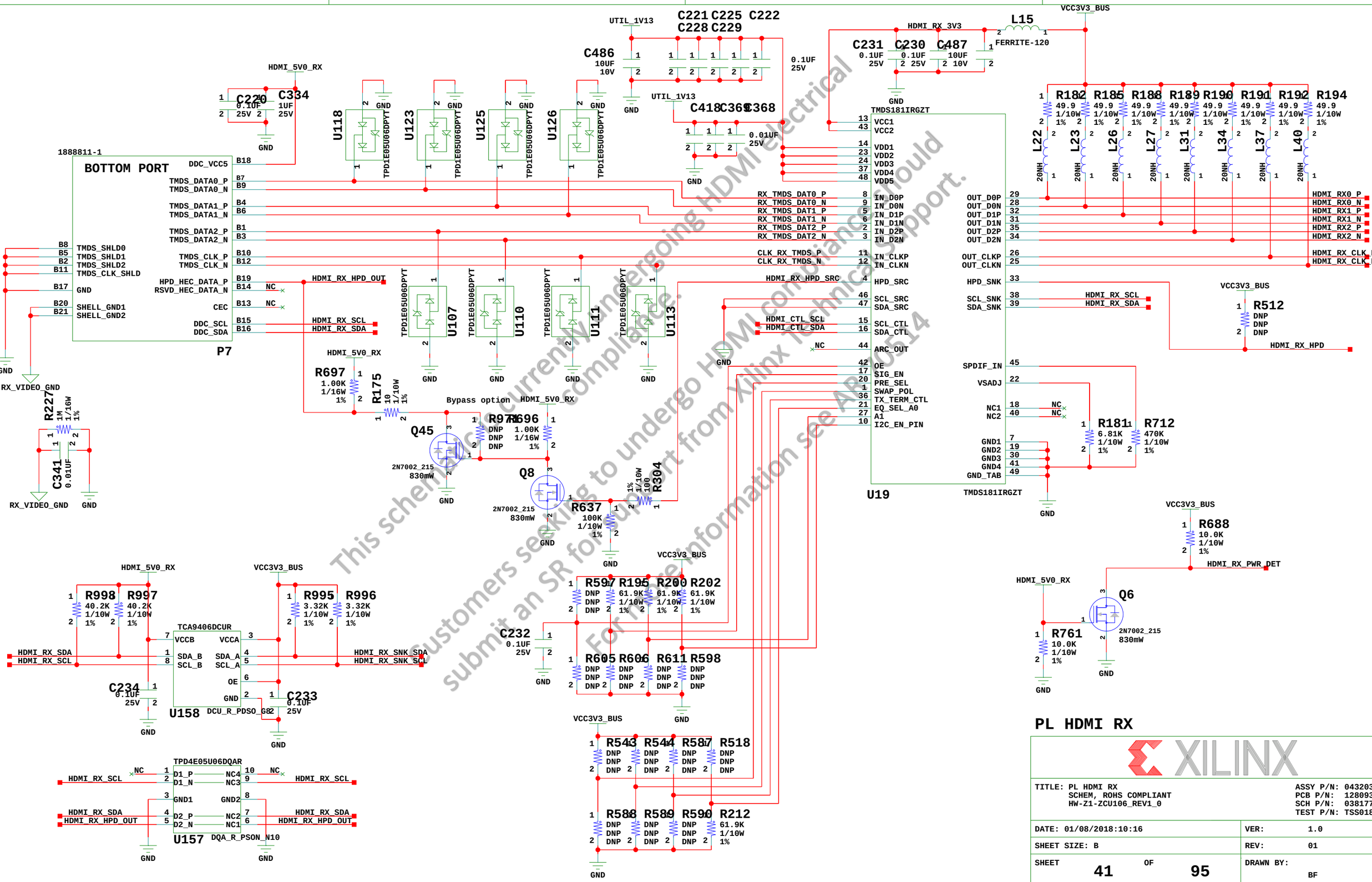
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to understand HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

C560 C562 C561
C563 C564
0.1UF
25V
UTIL_1V13
C571
10UF
10V
GND
HDMI_TX_3V3
L61
FERRITE-120
VCC3V3_BUS
C559 C558 C570
0.1UF 0.1UF 10UF
25V 2 25V 2 10V
GND
UTIL_1V13
C569 C568 C567
0.01UF
25V
GND
C565 C516 C515 C518 C517 C519 C578 C577
0.1UF 25V
SN65DP159RGZ
U94
VCC1
VCC2
VDD1
VDD2
VDD3
VDD4
VDD5
IN_D0P
IN_D0N
IN_D1P
IN_D1N
IN_D2P
IN_D2N
IN_CLKP
IN_CLKN
HPD_SRC
SCL_SRC
SDA_SRC
SCL_CTL
SDA_CTL
VSADJ
I2C_EN_PIN
PRE_SEL
EQ_SEL_A0
HDMI_SEL_B_A1
SLEW_CTL
OE
SWAP_POL
CEC_EN
TX_TERM_CTL
AUX_SRC_N
AUX_SRC_P
GND1
GND2
GND3
GND4
GND_TAB
HDMI_TX0_P
HDMI_TX0_N
HDMI_TX1_P
HDMI_TX1_N
HDMI_TX2_P
HDMI_TX2_N
HDMI_TX_LVDS_OUT_P
HDMI_TX_LVDS_OUT_N
HDMI_TX0_C_P
HDMI_TX0_C_N
HDMI_TX1_C_P
HDMI_TX1_C_N
HDMI_TX2_C_P
HDMI_TX2_C_N
HDMI_TX_LVDS_OUT_C_P
HDMI_TX_LVDS_OUT_C_N
NC
HDMI_TX_SRC_SCL
HDMI_TX_SRC_SDA
HDMI_CTL_SCL
HDMI_CTL_SDA
HDMI_TX_EN
R557
64.9K
1/10W
1%
R554
6.49K
1/10W
1%
R558 R559
64.9K
1/10W
1%
GND
OUT_D0P
OUT_D0N
OUT_D1P
OUT_D1N
OUT_D2P
OUT_D2N
OUT_CLKP
OUT_CLKN
HPD_SNK
SCL_SNK
SDA_SNK
TX_TMDS_DAT0_P
TX_TMDS_DAT0_N
TX_TMDS_DAT1_P
TX_TMDS_DAT1_N
TX_TMDS_DAT2_P
TX_TMDS_DAT2_N
CLK_TX_TMDS_P
CLK_TX_TMDS_N
TX_HPD_SINK
R563
1/10W
5%
HDMI_TX_SNK_SCL
HDMI_TX_SNK_SDA
U91
U71
U72
U73
TPD1E05U06DPYT
GND
U82
U83
U89
U90
TPD1E05U06DPYT
VCC3V3
C588
0.1UF
25V
R919 R634
10.0K 10.0K
1/10W 1/10W
1%
UTIL_5V0
C576
0.1UF
25V
TPD12S016RKTR
VCCA
VCC5V
CEC_A
SCL_A
SDA_A
HPD_A
LS_OE
CT_HPD
D2_P
D2_N
D1_P
D1_N
D0_P
D0_N
CLK_P
CLK_N
5V_OUT
CEC_B
SCL_B
SDA_B
HPD_B
GND3
GND2
GND1
HDMI_TX_CEC
HDMI_TX_HPD
CEC_SOURCE
HDMI_TX_5V0_CONN
1888811-1
TOP PORT
P7
DDC_VCC5
TMDS_DATA0_P
TMDS_DATA0_N
TMDS_DATA1_P
TMDS_DATA1_N
TMDS_DATA2_P
TMDS_DATA2_N
TMDS_CLK_P
TMDS_CLK_N
HPD_HEC_DATA_P
RSVD_HEC_DATA_N
CEC
DDC_SCL
DDC_SDA
TMDS_SHLD0
TMDS_SHLD1
TMDS_SHLD2
TMDS_CLK_SHLD
GND
SHELL_GND1
SHELL_GND2
C520
0.1UF
25V
C566
1UF
25V
R881
0
1/10W
5%
C573
0.01UF
TX_VIDEO_GND
C557
0.1UF
25V
M24C64-W
TSSOP_8
U109
R644
10.0K
1/10W
1%
R645 R651 R652
10.0K
1/10W
1%
U70
TPD12S016RKTR

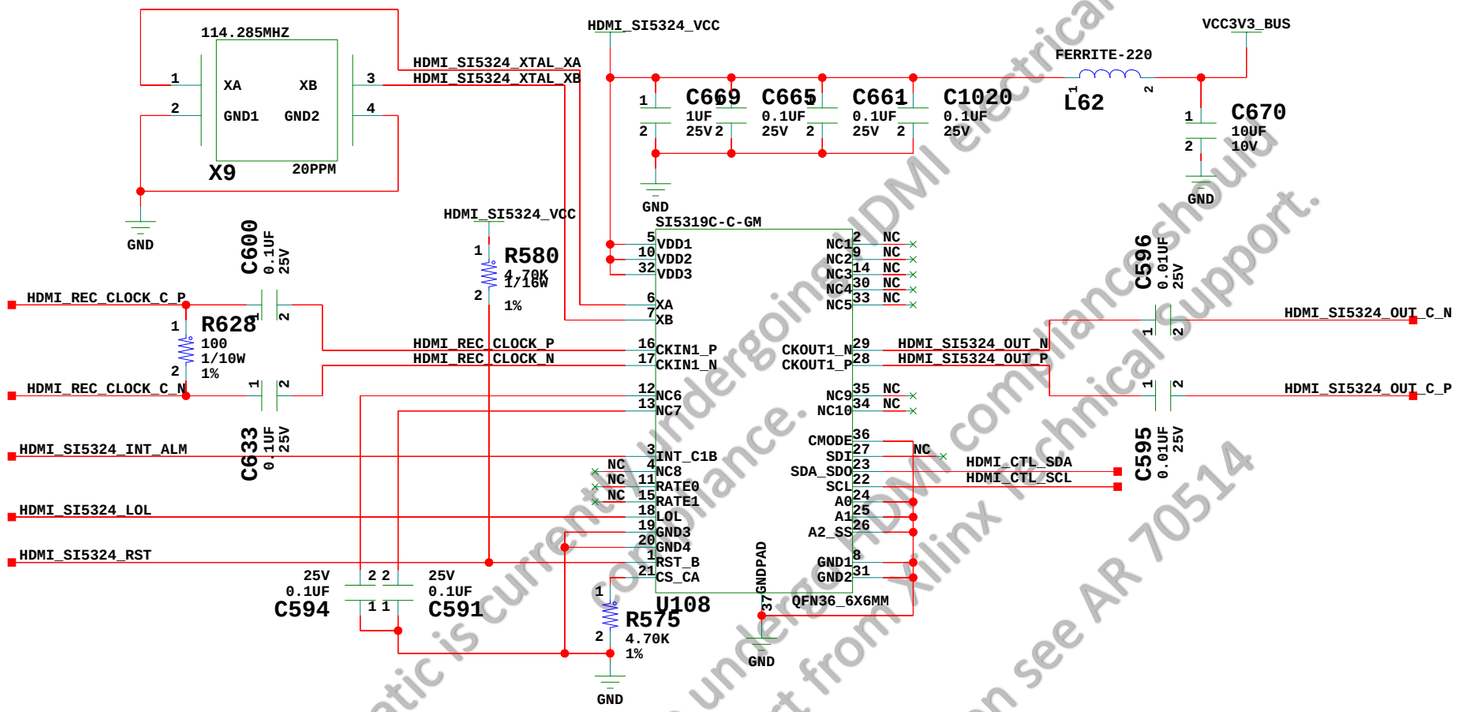
PL HDMI TX

XILINX

TITLE: PL HDMI TX SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 40 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR#65214

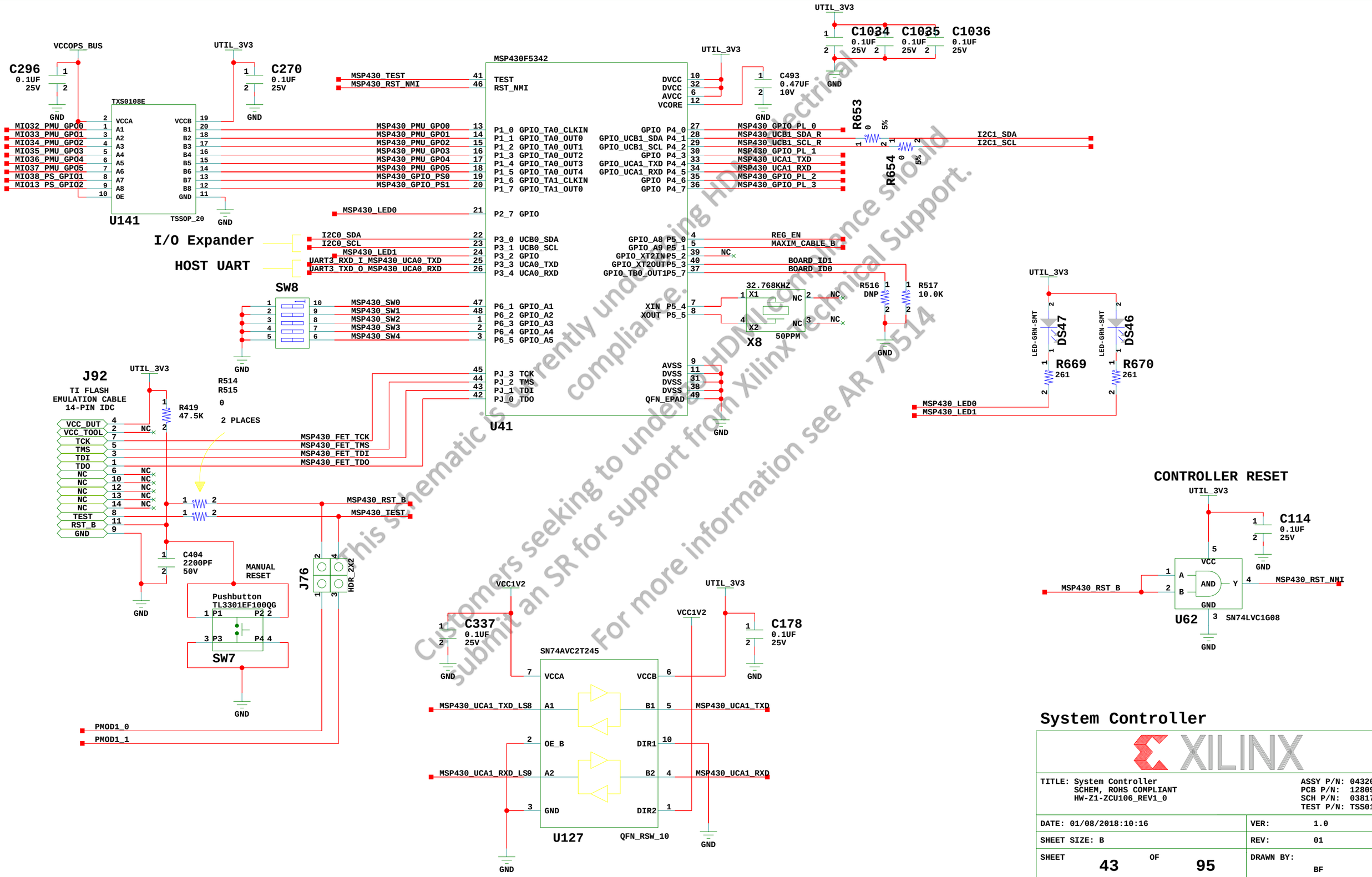


This schematic is currently undergoing HDMI electrical compliance. Customers seeking to understand HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

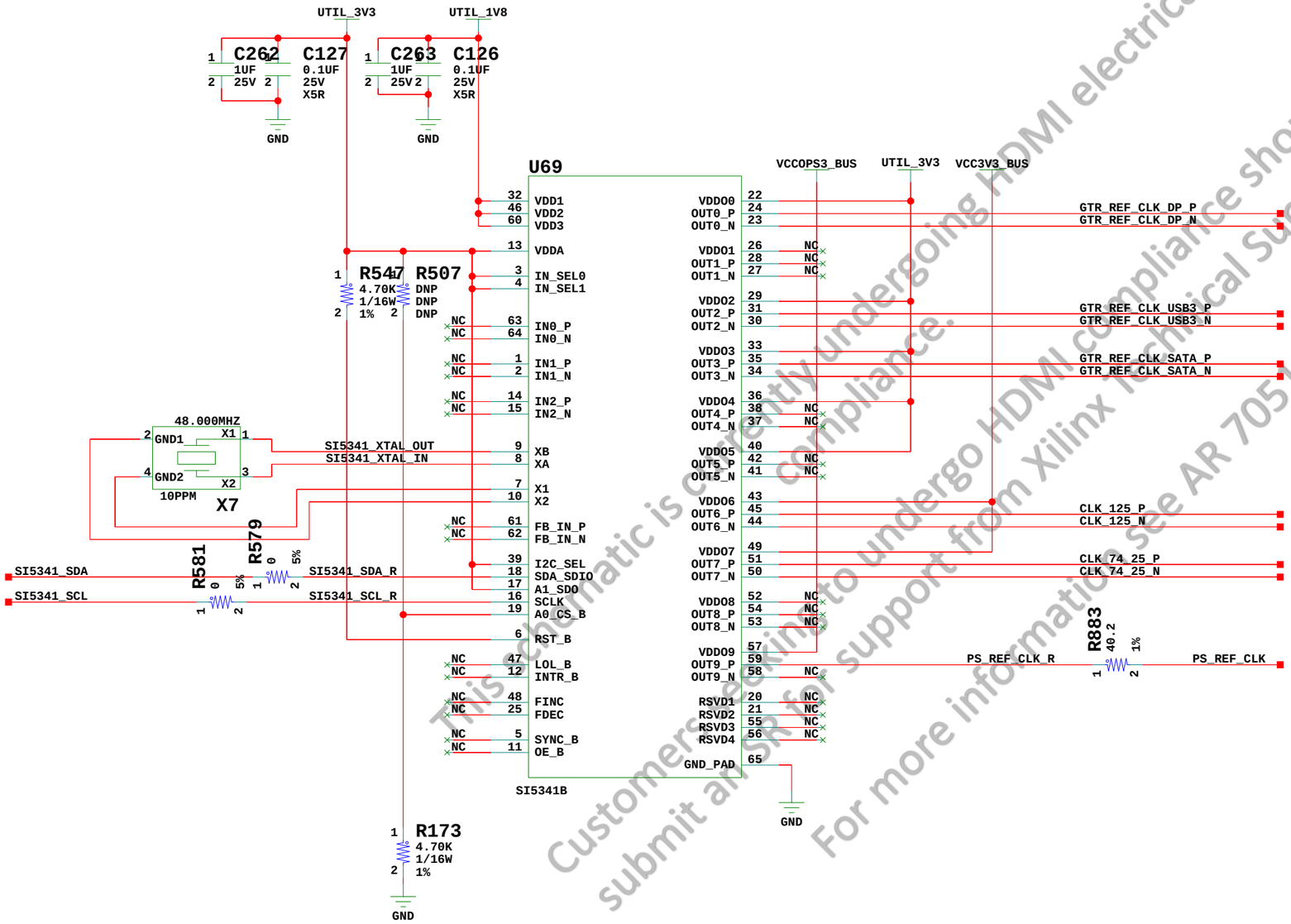
PL HDMI Clock Recovery

XILINX

TITLE: PL HDMI Clock Recovery SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 42 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance. Customers seeking to use HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

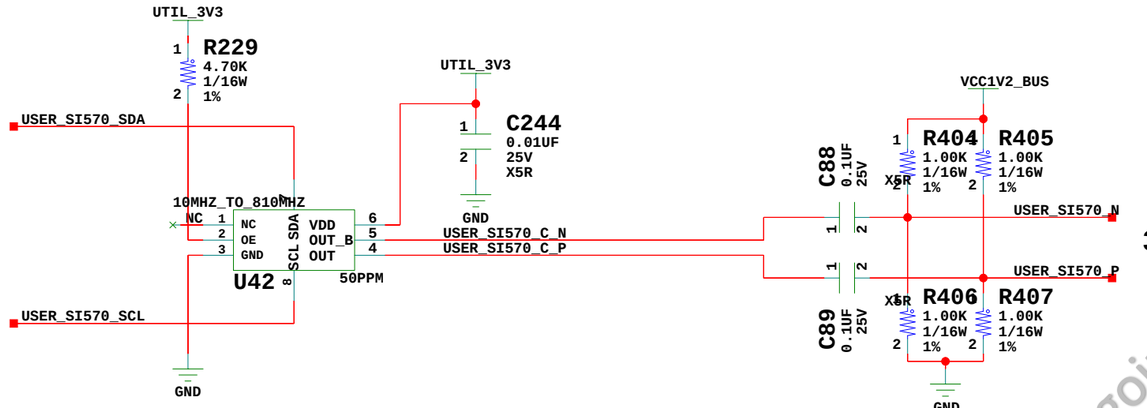


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SP for support from Xilinx Technical Support.
For more information see AR 70514

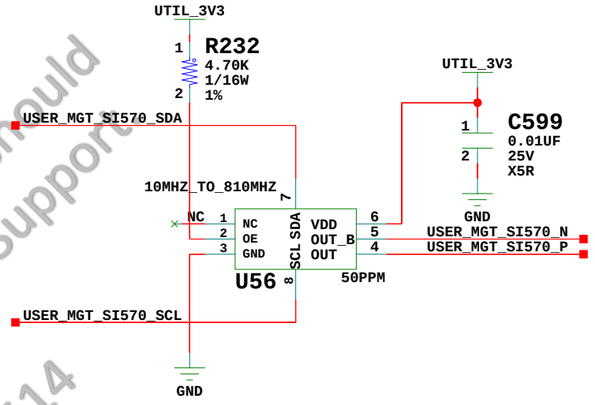
Fixed Clocks

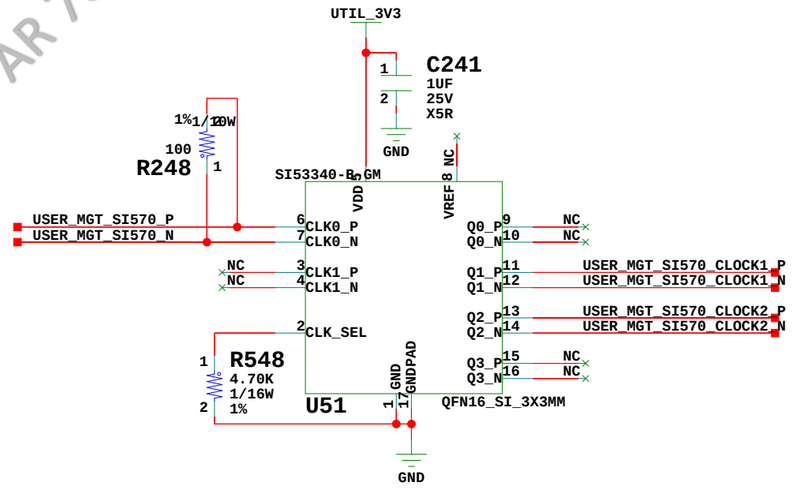
XILINX

TITLE: Fixed Clocks SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 44 OF 95	DRAWN BY: BF



300MHz LVDS



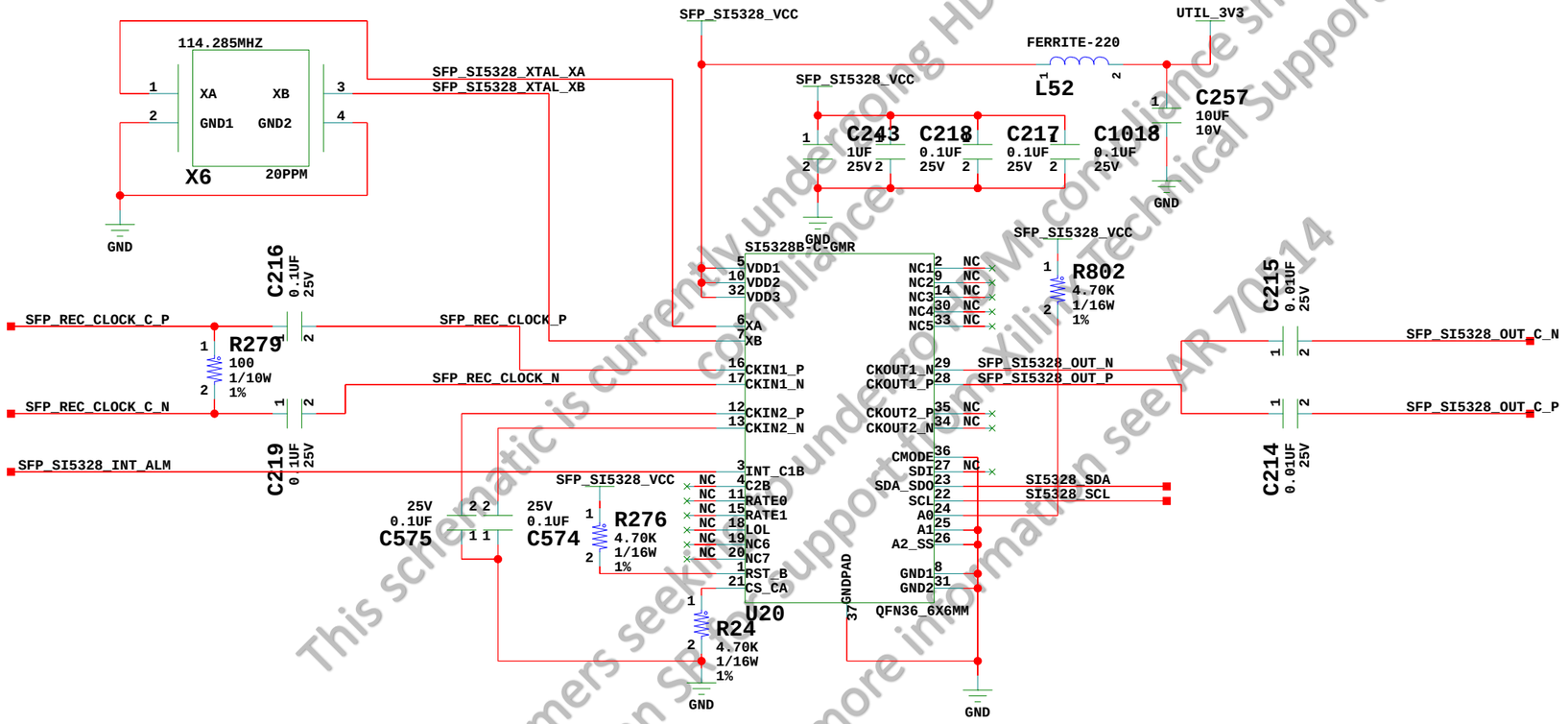


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support
For more information see AR 70514

PL Programmable Clocks - SMAs

XILINX

TITLE: PL Programmable Clocks - SMAs SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 45 OF 95	DRAWN BY: BF



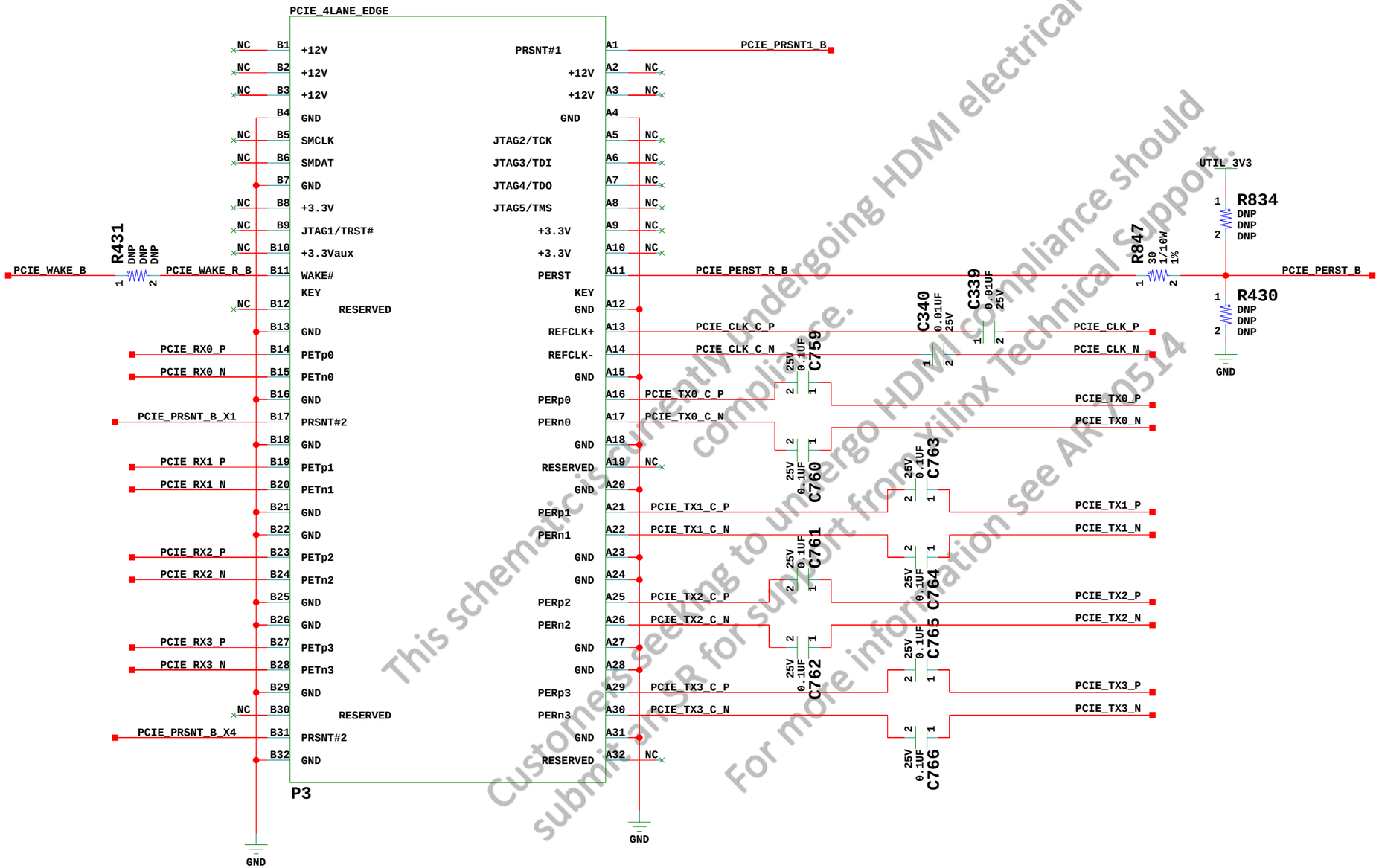
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking support should submit an SDSS request through MyXilinx Technical Support.
For more information see AR 70514

PL SFP Clock Recovery

XILINX

TITLE: PL SFP Clock Recovery SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 46 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance. Customers seeking to use Avnet Avnet Avnet



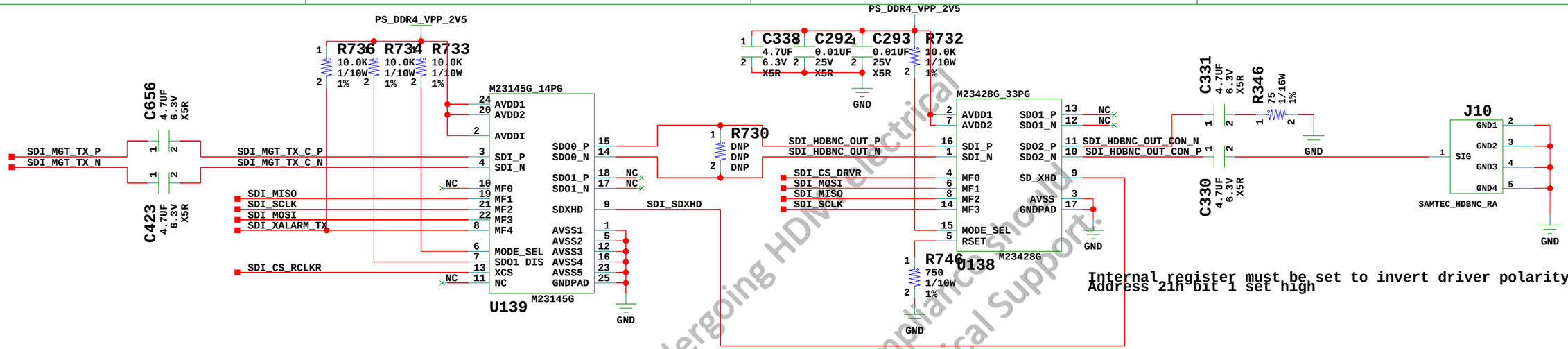
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL PCIe x4 Edge Connector

XILINX

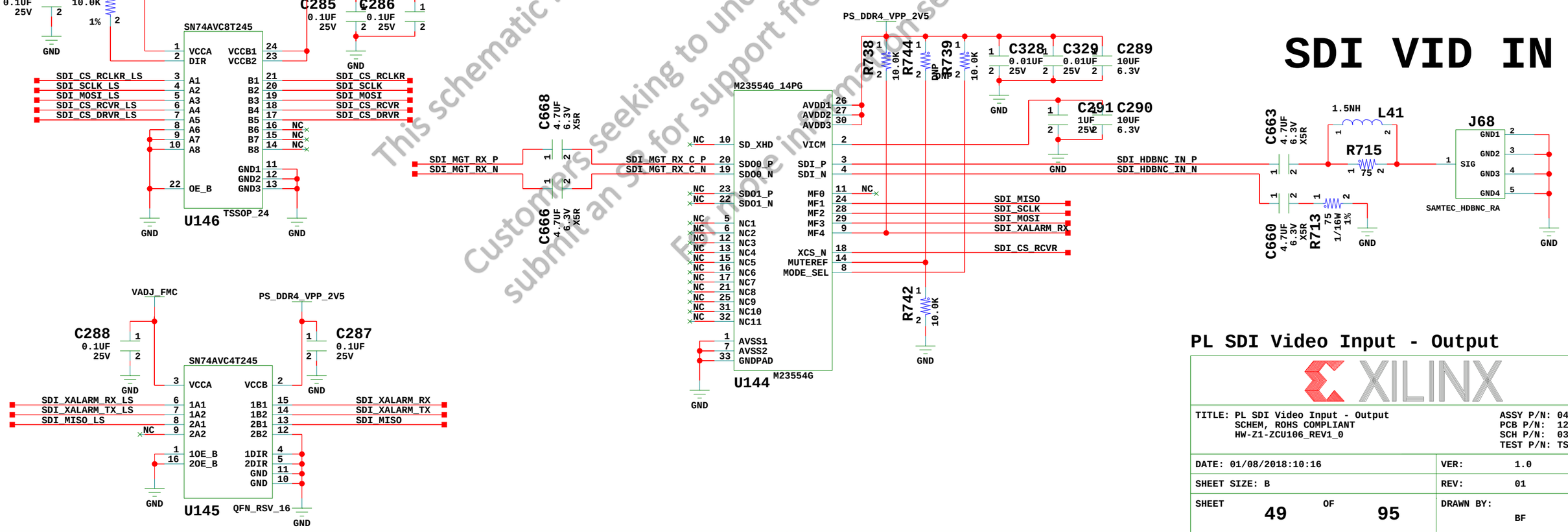
TITLE: PL PCIe x4 Edge Connector SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 48 OF 95	DRAWN BY: BF

SDI VID OUT



Internal register must be set to invert driver polarity high
Address 21h bit 1 set high

SDI VID IN

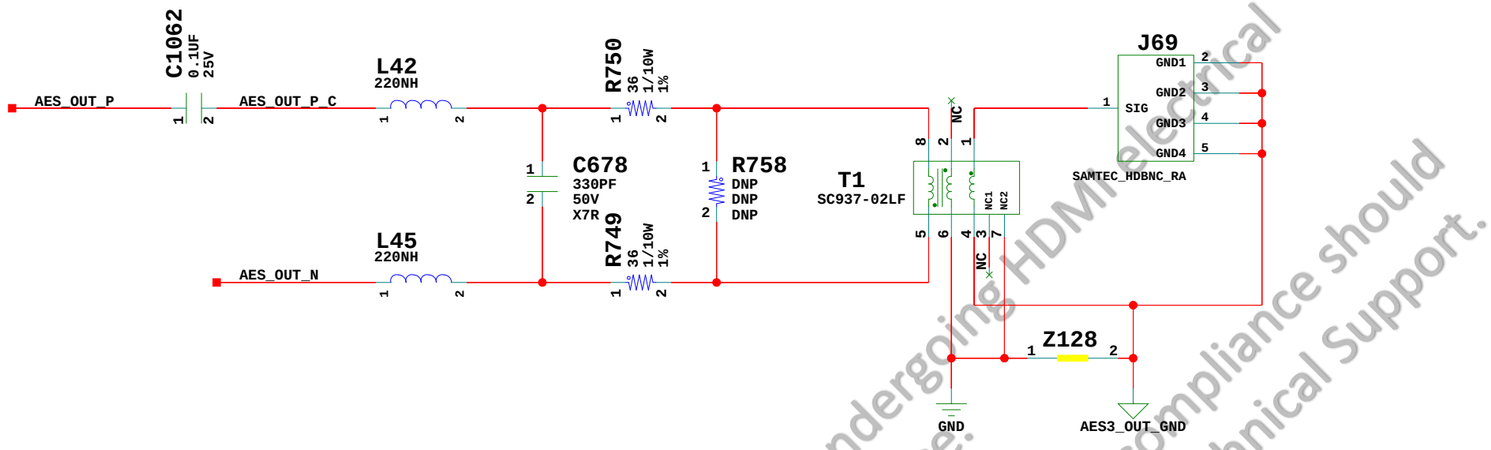


PL SDI Video Input - Output

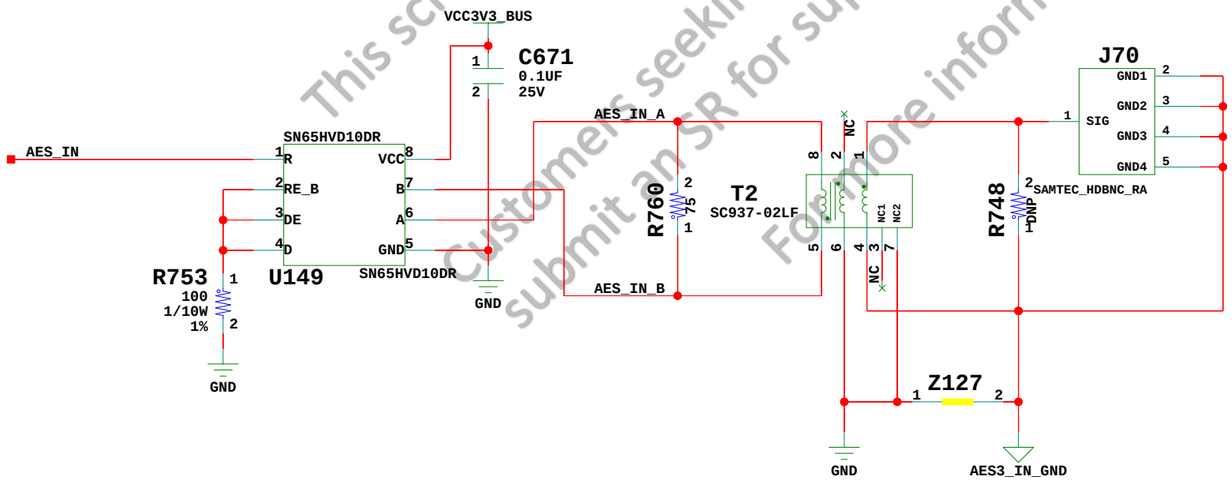
XILINX

TITLE: PL SDI Video Input - Output SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 49 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



AES3 OUT



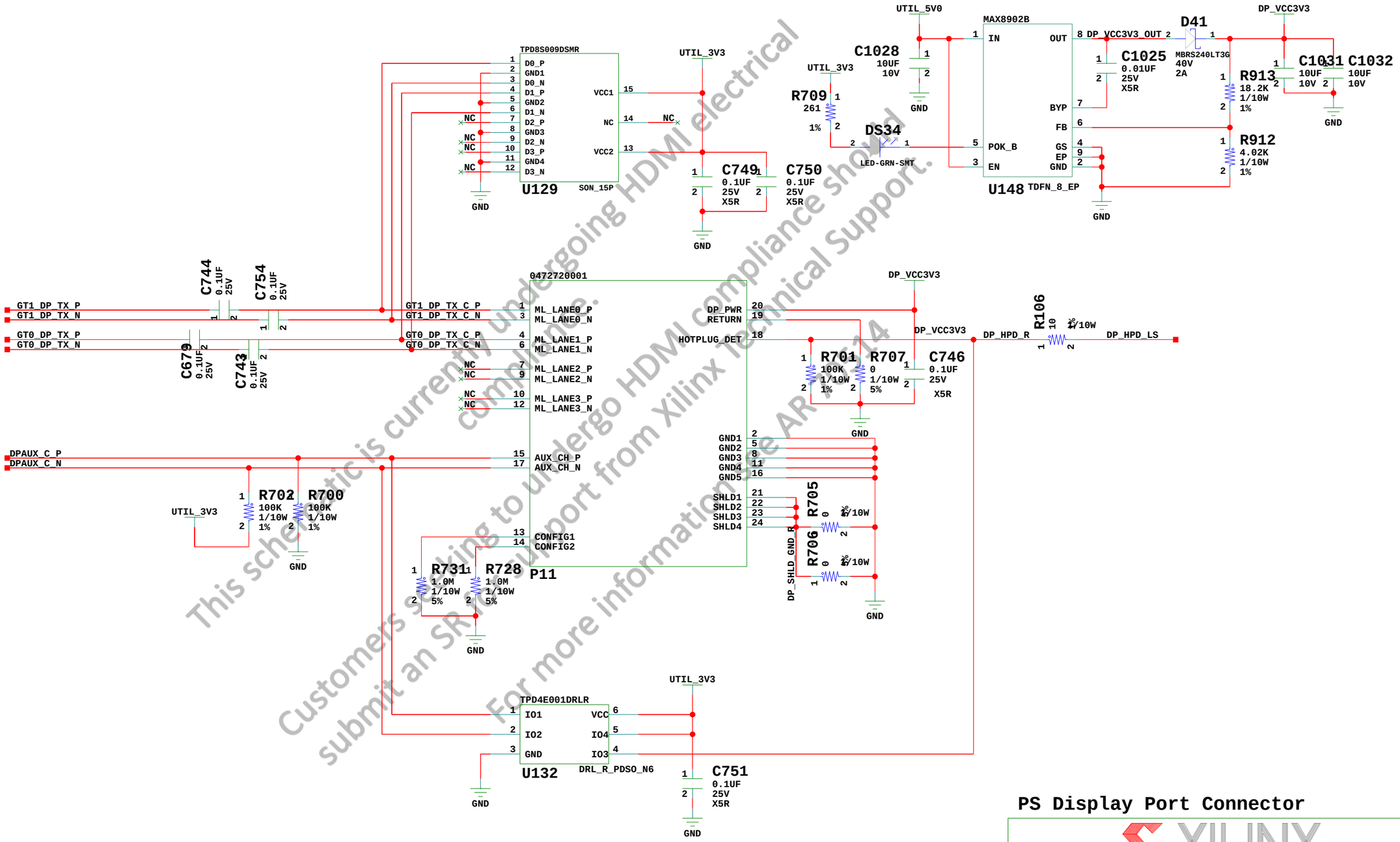
AES3 IN

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL AES3 Audio Input - Output

XILINX

TITLE: PL AES3 Audio Input - Output SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/18/2018:13:12	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 50 OF 95	DRAWN BY: BF

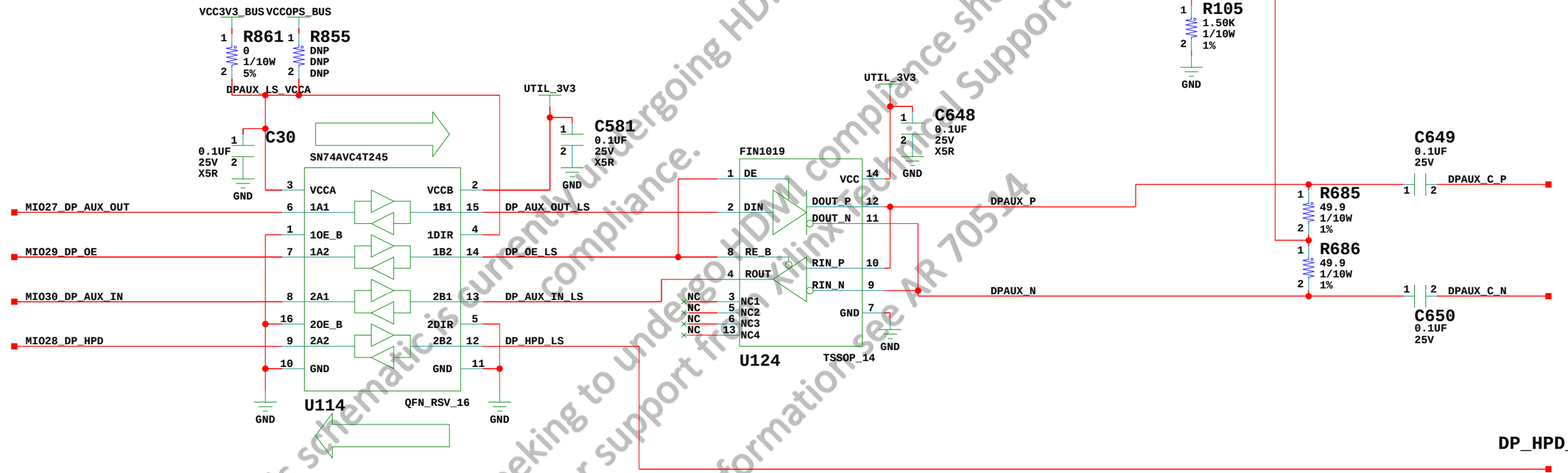


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to use this design must undergo HDMI compliance should submit an SR to support from Xilinx Technical Support.
For more information see AR 57214

PS Display Port Connector

XILINX

TITLE: PS Display Port Connector SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 51 OF 95	DRAWN BY: BF

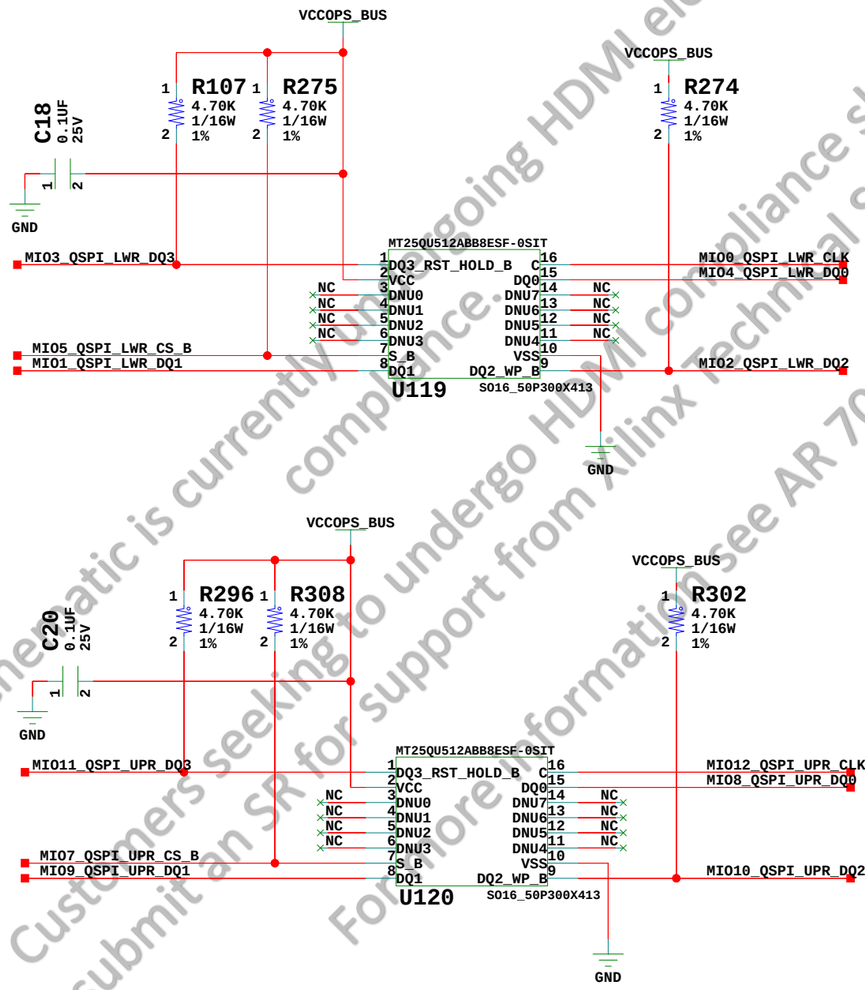


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PS Display Port IO

XILINX

TITLE: PS Display Port IO SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 52 OF 95	DRAWN BY: BF

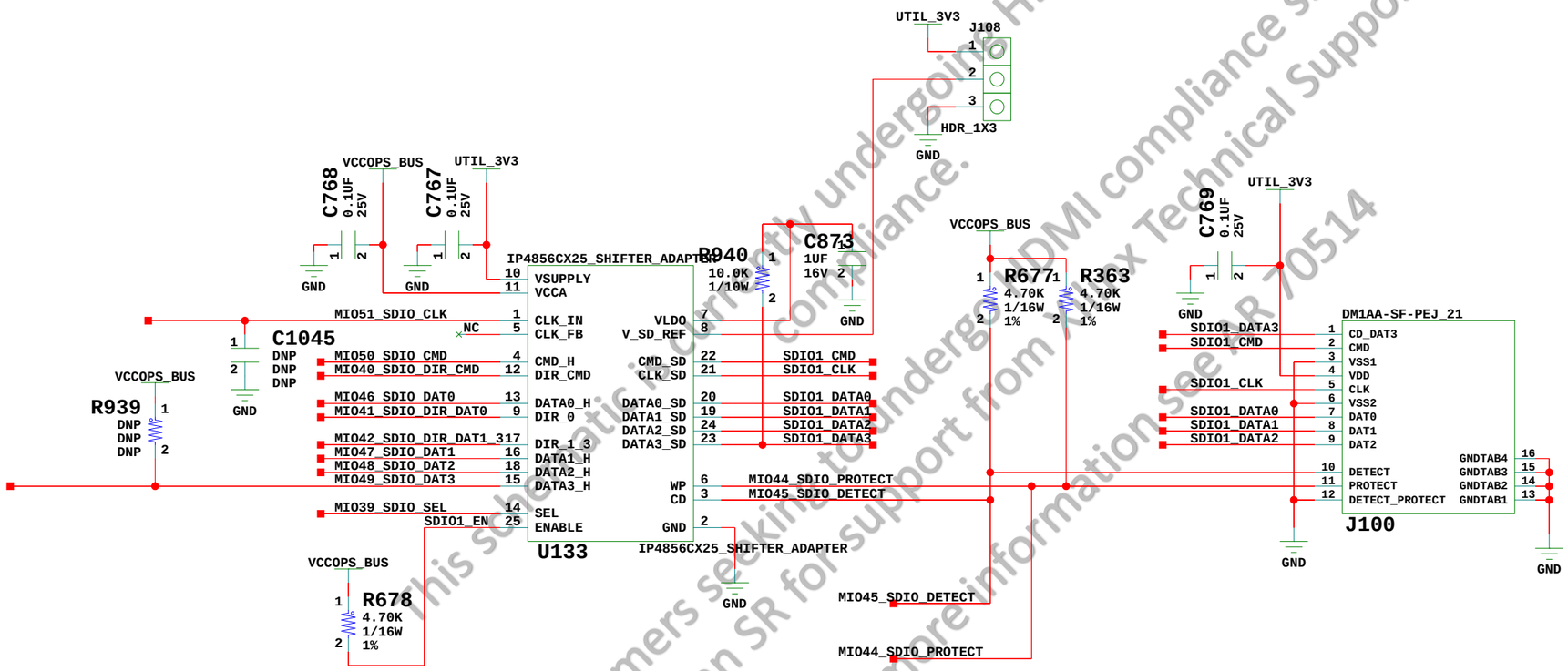


This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

PS QSPI

XILINX

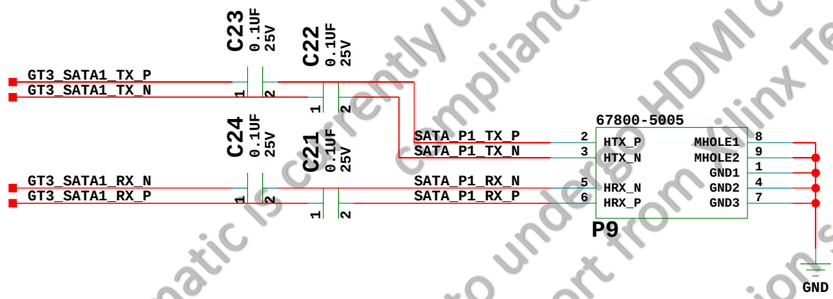
TITLE: PS QSPI SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 53 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see XTP 70514

PS SD Card Connector

TITLE: PS SD Card Connector SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/18/2018:13:11	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 54 OF 95	DRAWN BY: BF



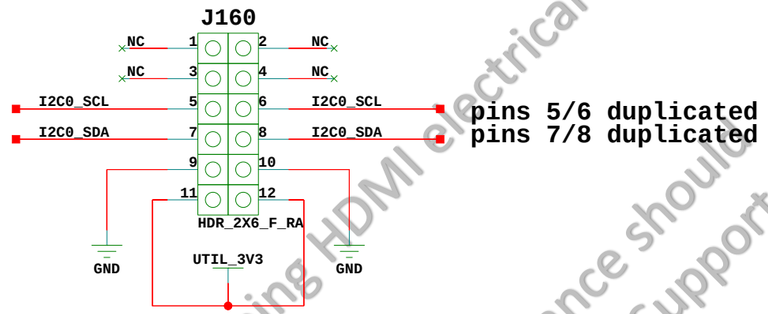
This schematic is currently undergoing HDMI electrical Compliance.
Customers seeking to understand HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

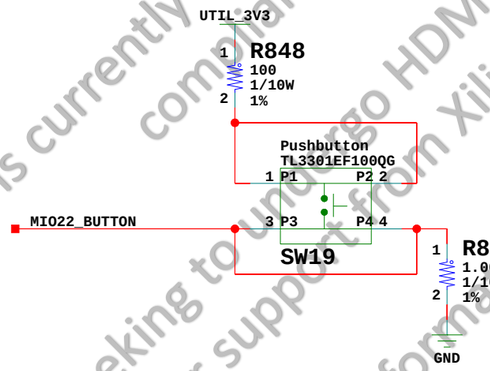
PS SATA

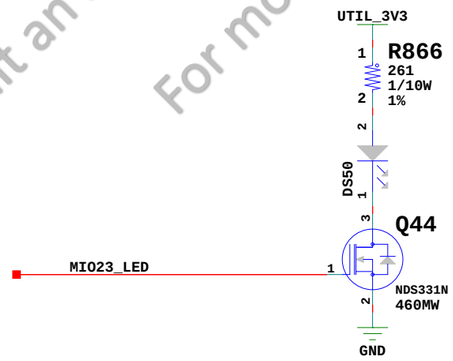
XILINX

TITLE: PS SATA SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 55 OF 95	DRAWN BY: BF

Connects to R.A. Female 2x6 PMOD receptacle





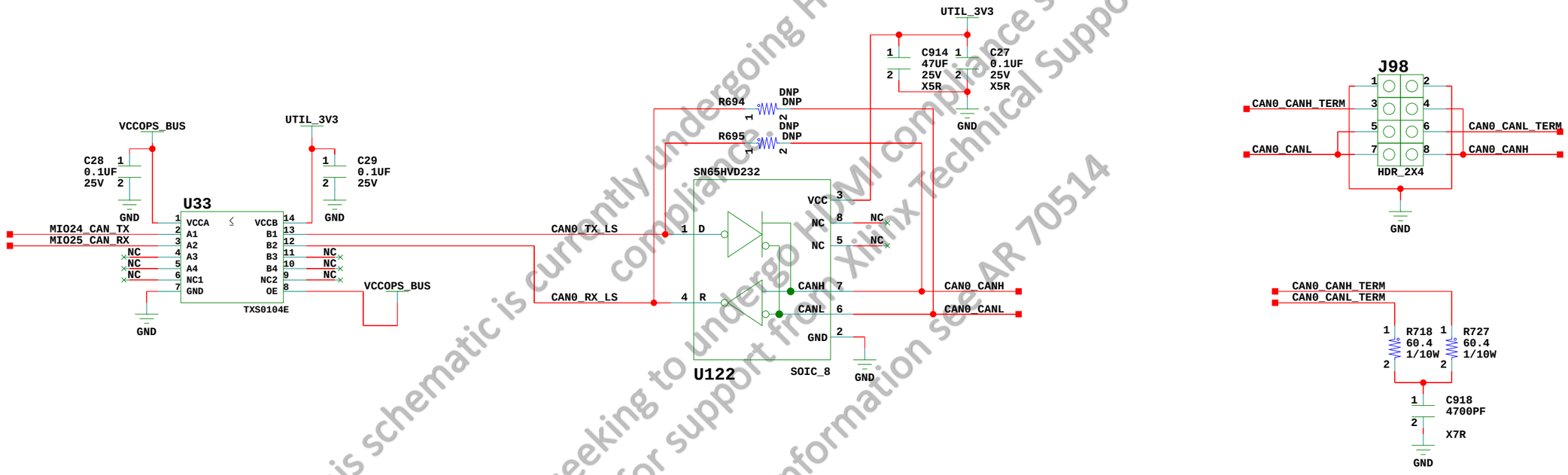


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to ensure HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PS MIO PMOD - Button - LED

XILINX

TITLE: PS MIO PMOD - Button - LED SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 56 OF 95	DRAWN BY: BF

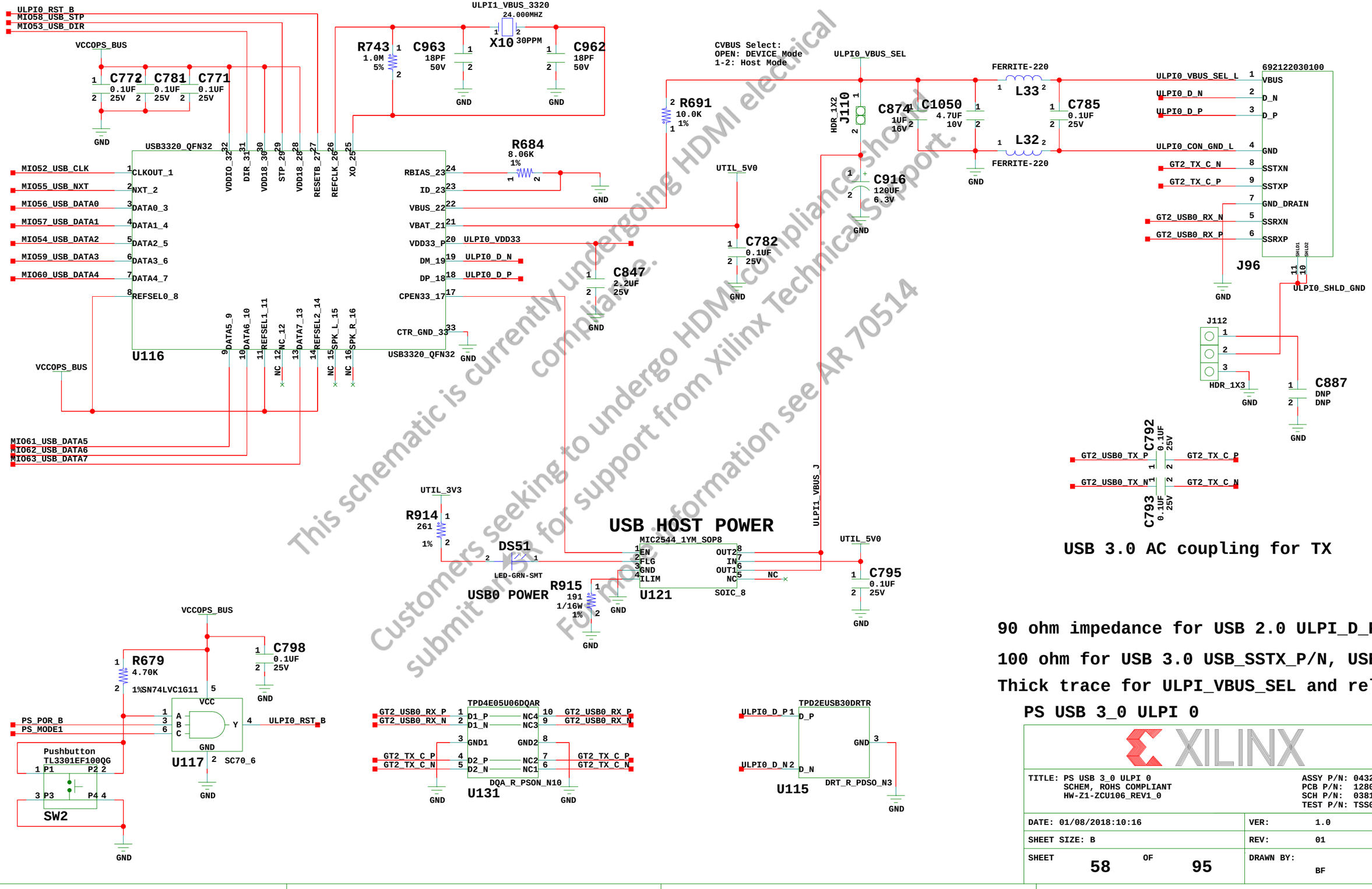


This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

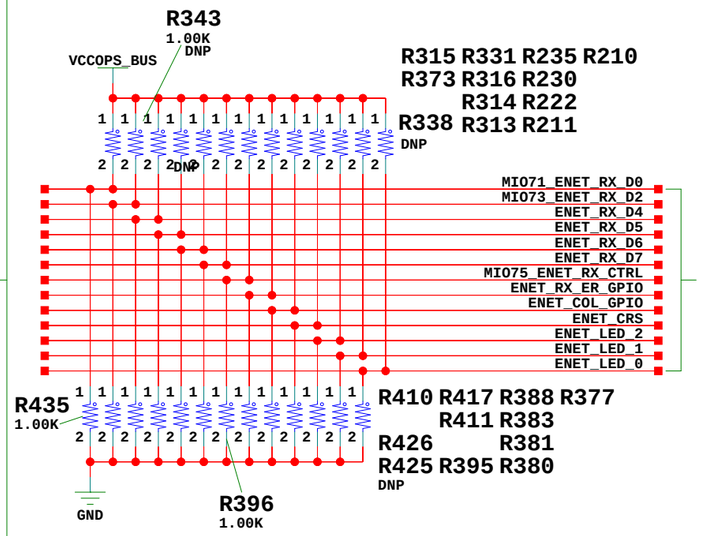
PS CAN Bus

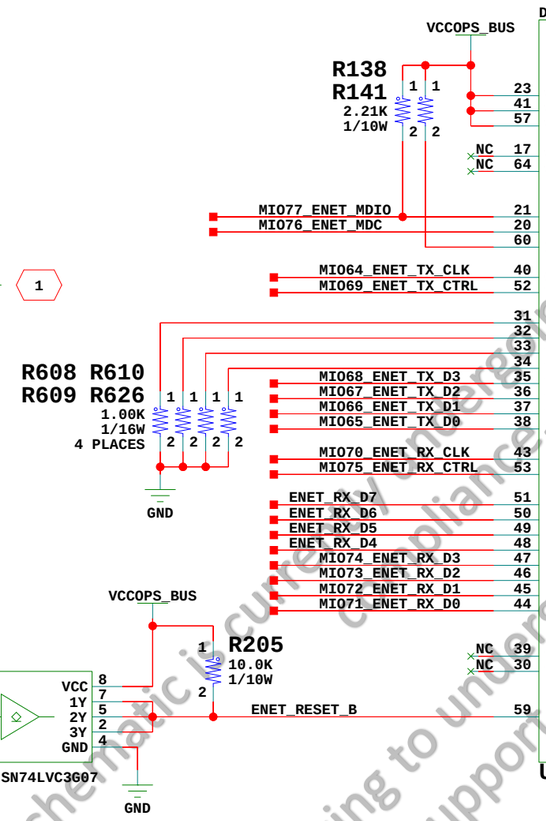
XILINX

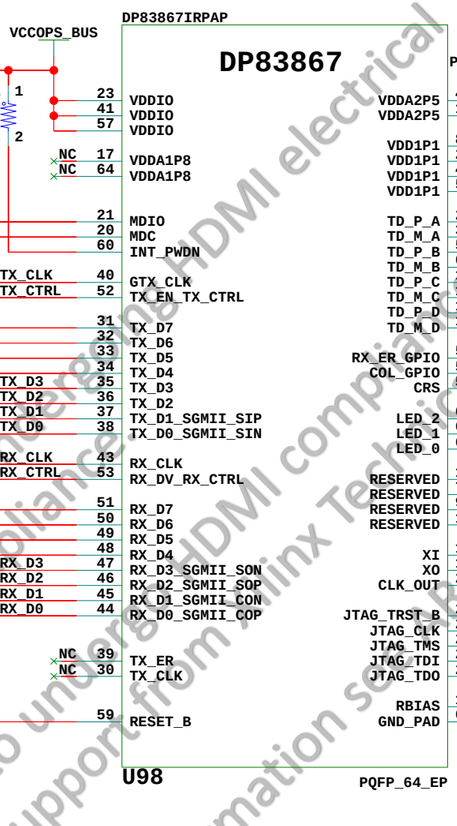
TITLE: PS CAN Bus SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 57 OF 95	DRAWN BY: BF

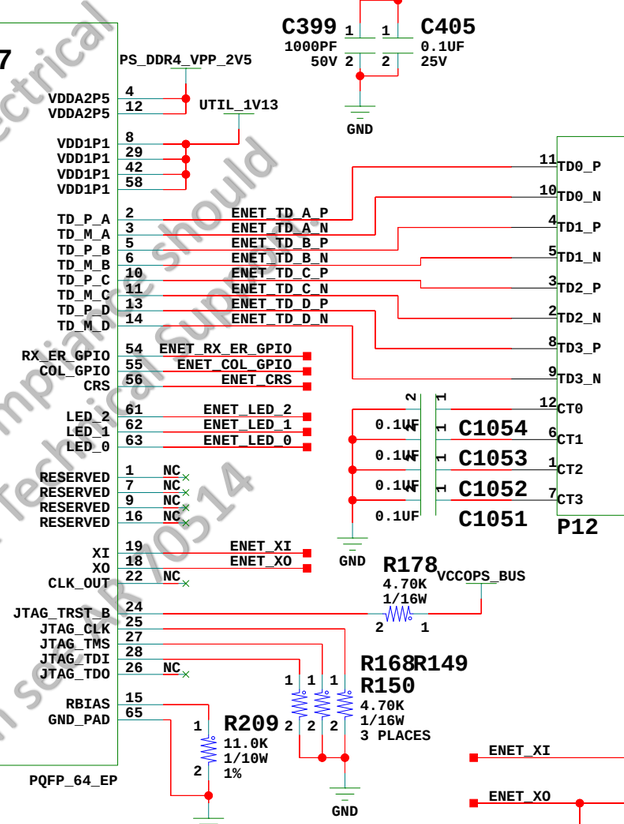


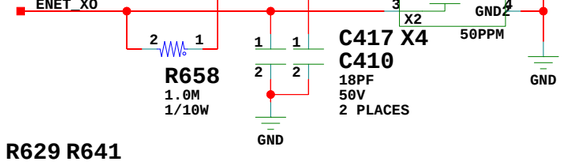
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



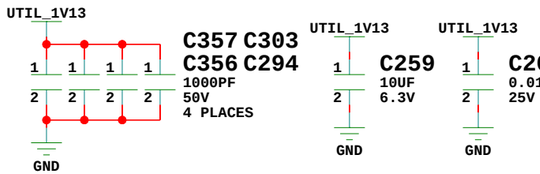


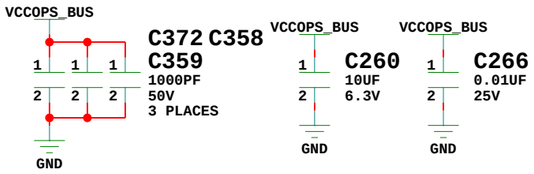


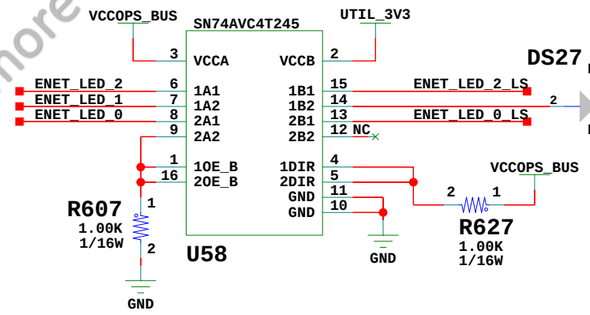


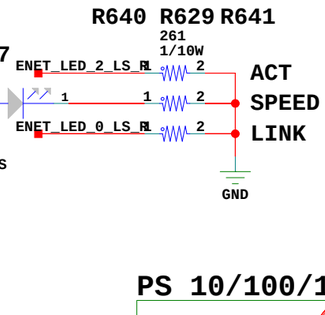


2 ENET PHY DECOUPLING

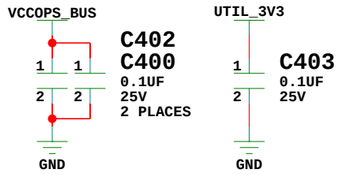








PERIPHERAL DECOUPLING



2 THESE CAPACITORS ARE PLACED AS CLOSE AS POSSIBLE TO THEIR RESPECTIVE PINS ON THE DP83867.

1 STRAP CONFIGURATION RESISTORS. SEE DP83867 DATASHEET FOR DETAILS.

This schematic is current. Customers seeking to use compliance HDMI electrical submit an SR for support from Xilinx Technical Support. For more information see UG 0314

PS 10/100/1000 Ethernet PHY

XILINX

TITLE: PS 10/100/1000 Ethernet PHY SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 59 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

VCC1V2
UTIL_3V3
C11 0.1UF 25V
R305 10.0K 1%
C12 0.1UF 25V
C13 0.1UF 25V
GND
SN74AVC8T245
VCCA
DIR
VCCB1
VCCB2
OE_B
GND1
GND2
GND3
A1 A2 A3 A4 A5 A6 A7 A8
B1 B2 B3 B4 B5 B6 B7 B8
GPIO_LED_0_LS
GPIO_LED_1_LS
GPIO_LED_2_LS
GPIO_LED_3_LS
GPIO_LED_4_LS
GPIO_LED_5_LS
GPIO_LED_6_LS
GPIO_LED_7_LS
GPIO_LED_0
GPIO_LED_1
GPIO_LED_2
GPIO_LED_3
GPIO_LED_4
GPIO_LED_5
GPIO_LED_6
GPIO_LED_7
U106
TSSOP_24

LEDs near top right edge

GPIO_LED_0
GPIO_LED_1
GPIO_LED_2
GPIO_LED_3
GPIO_LED_4
GPIO_LED_5
GPIO_LED_6
GPIO_LED_7
LED-GRN-SMT
DS44 R366 261 1/10W
DS43 R378 261 1/10W
DS42 R382 261 1/10W
DS41 R385 261 1/10W
DS40 R386 261 1/10W
DS39 R387 261 1/10W
DS37 R390 261 1/10W
DS38 R392 261 1/10W
GND

VCC1V2
R287 100 1/10W 1%
Pushbutton TL3301EF100QG
GPIO_SW_N
SW18
R350 1.00K 1/16W 1%
GND

VCC1V2
R292 100 1/10W 1%
Pushbutton TL3301EF100QG
GPIO_SW_W
SW14
R341 1.00K 1/16W 1%
GND

VCC1V2
R285 100 1/10W 1%
Pushbutton TL3301EF100QG
GPIO_SW_C
SW15
R342 1.00K 1/16W 1%
GND

VCC1V2
R295 100 1/10W 1%
Pushbutton TL3301EF100QG
GPIO_SW_E
SW17
R348 1.00K 1/16W 1%
GND

VCC1V2
R283 100 1/10W 1%
Pushbutton TL3301EF100QG
GPIO_SW_S
SW16
R347 1.00K 1/16W 1%
GND

VADJ_FMC
Pushbutton TL3301EF100QG
CPU_RESET
SW20
R918 1.00K 1/16W 1%
GND

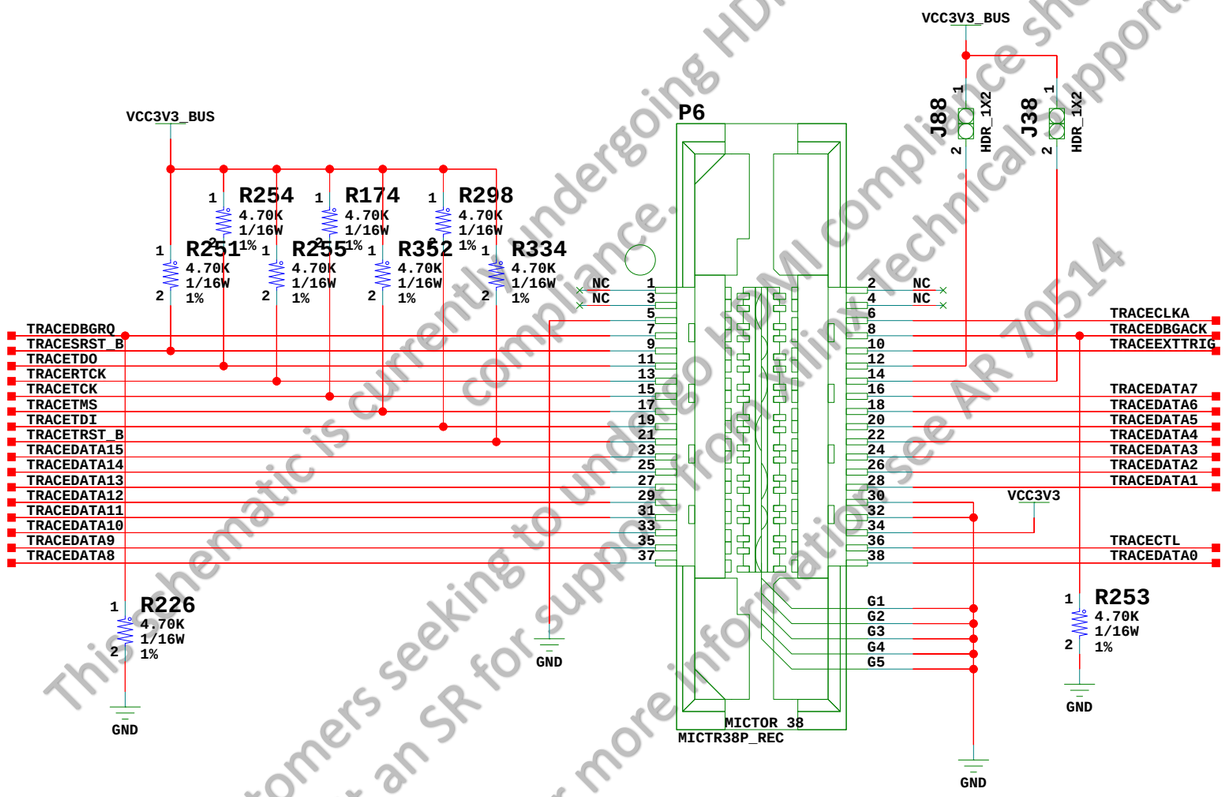
VADJ_FMC
SDA08H1SBD
GPIO_DIP_SW7
GPIO_DIP_SW6
GPIO_DIP_SW5
GPIO_DIP_SW4
GPIO_DIP_SW3
GPIO_DIP_SW2
GPIO_DIP_SW1
GPIO_DIP_SW0
SW13
R340 R360 R324 R335 1.00K 1/16W 1%
R337 R359 R322 R325 1.00K 1/16W 1%
GND
SDA08H1SBD

PL Buttons - Switches - LEDs

XILINX

TITLE: PL Buttons - Switches - LEDs SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 60 OF 95	DRAWN BY: BF

EMIO ARM Trace/Debug

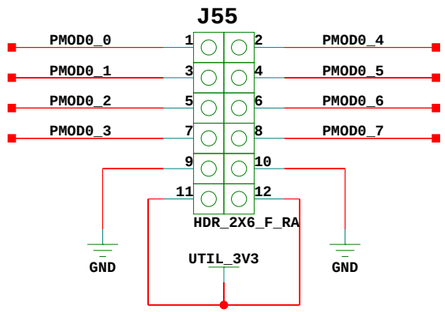


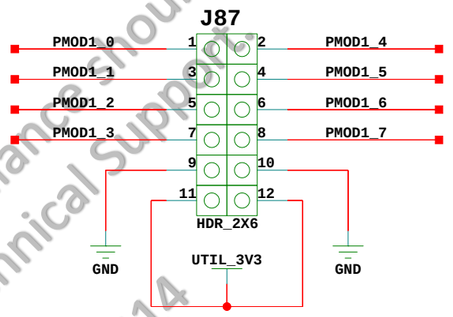
This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical support. For more information see AR 70514

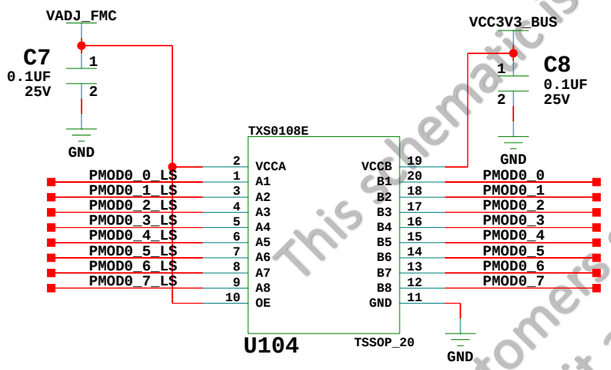
PL EMIO ARM Trace Debug

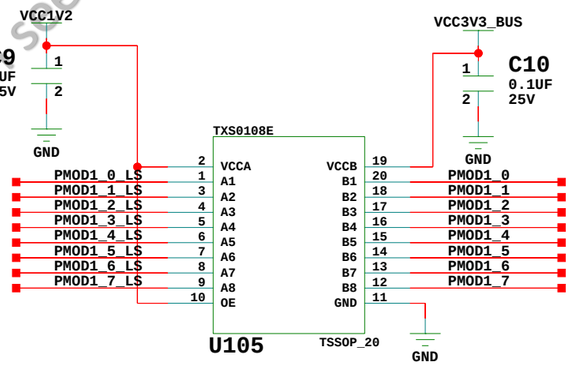
XILINX

TITLE: PL EMIO ARM Trace Debug SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 61 OF 95	DRAWN BY: BF









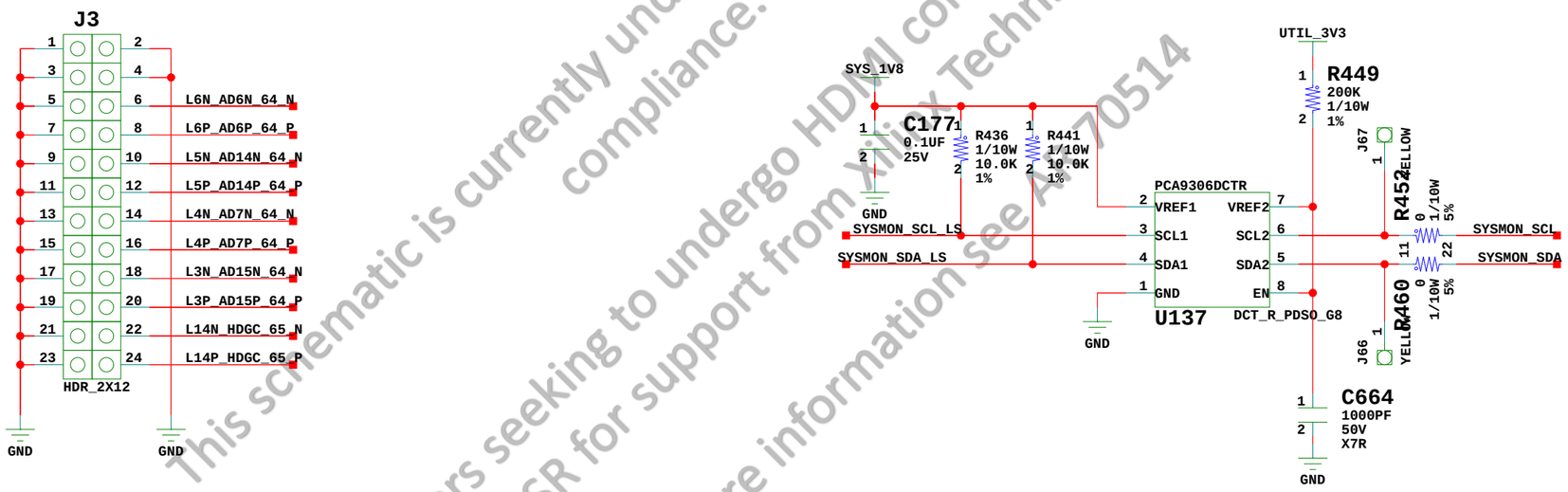
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL PMODs

XILINX

TITLE: PL PMODs SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 62 OF 95	DRAWN BY: BF

Prototype Header



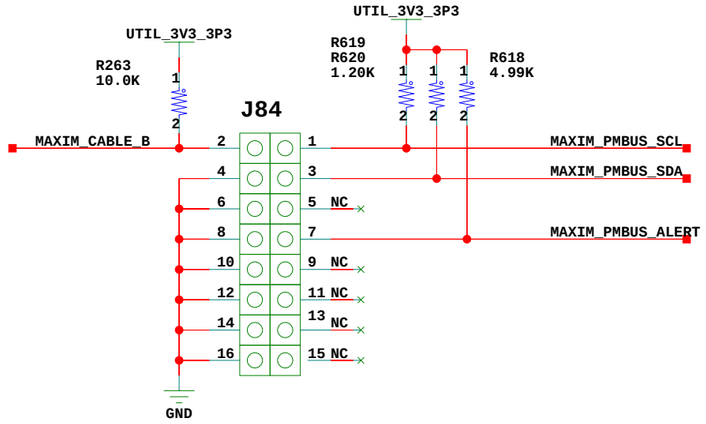
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL Prototype Header - SYSMON Level Shi

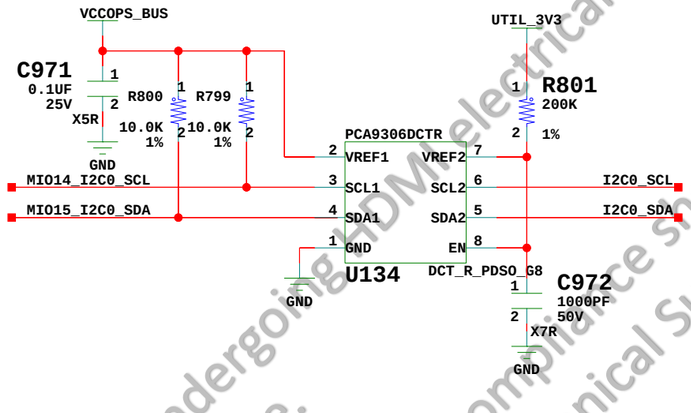
XILINX

TITLE: PL Prototype Header - SYSMON Level Shifter SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 63 OF 95	DRAWN BY: BF

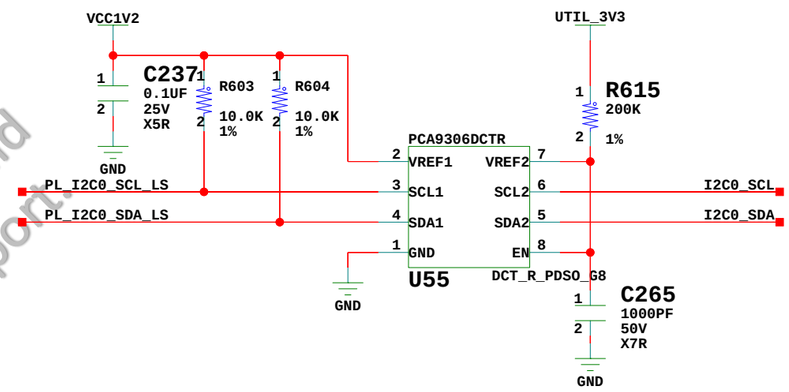
MAXIM PMBUS PROGRAMMING CABLE

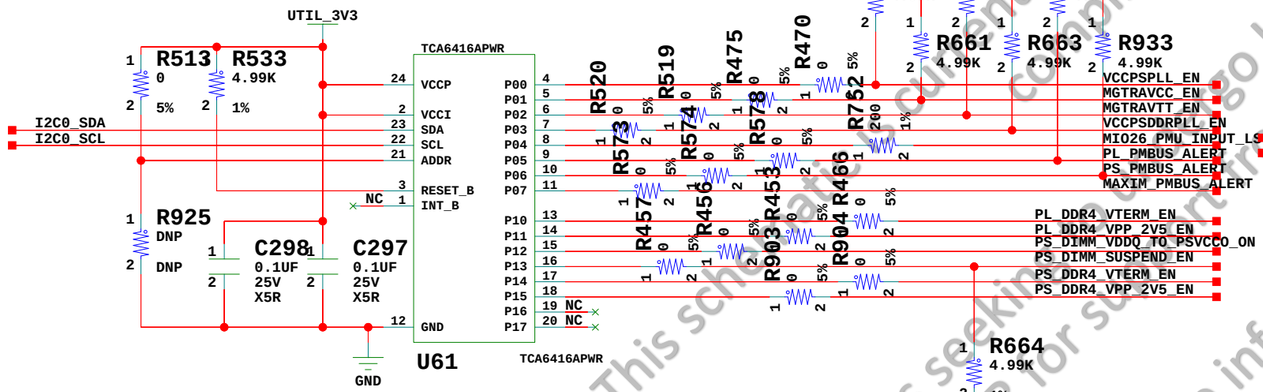


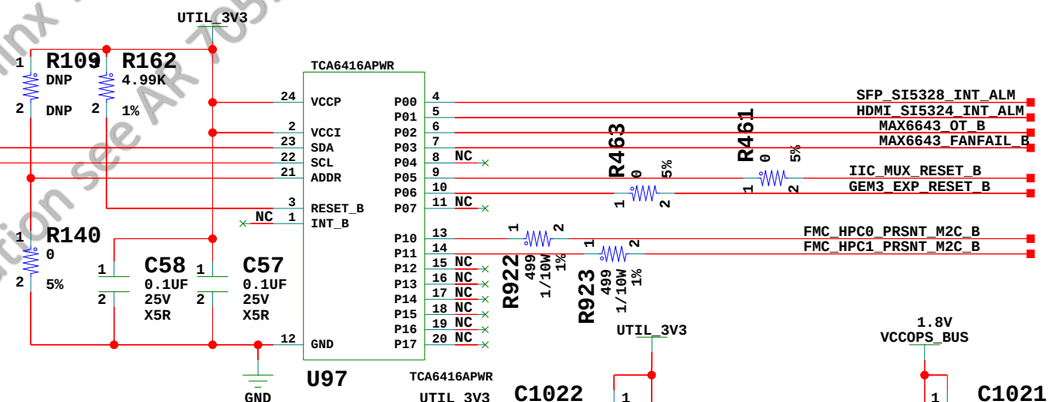
PS I2C Level Shifter (I2C0)

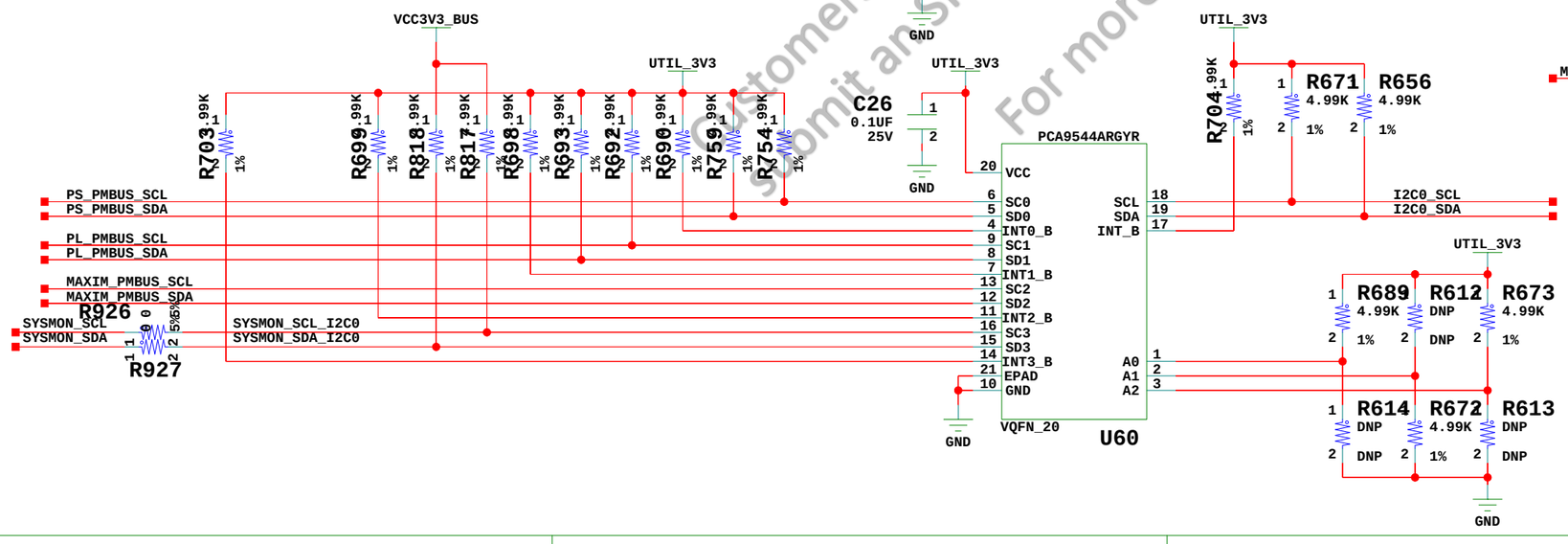


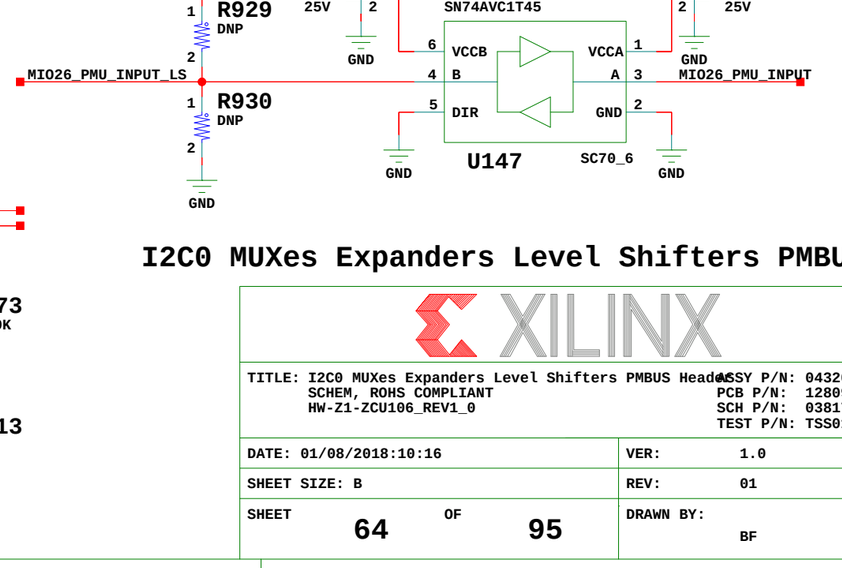
PL I2C Level Shifter (disabled if VREF1 is off)









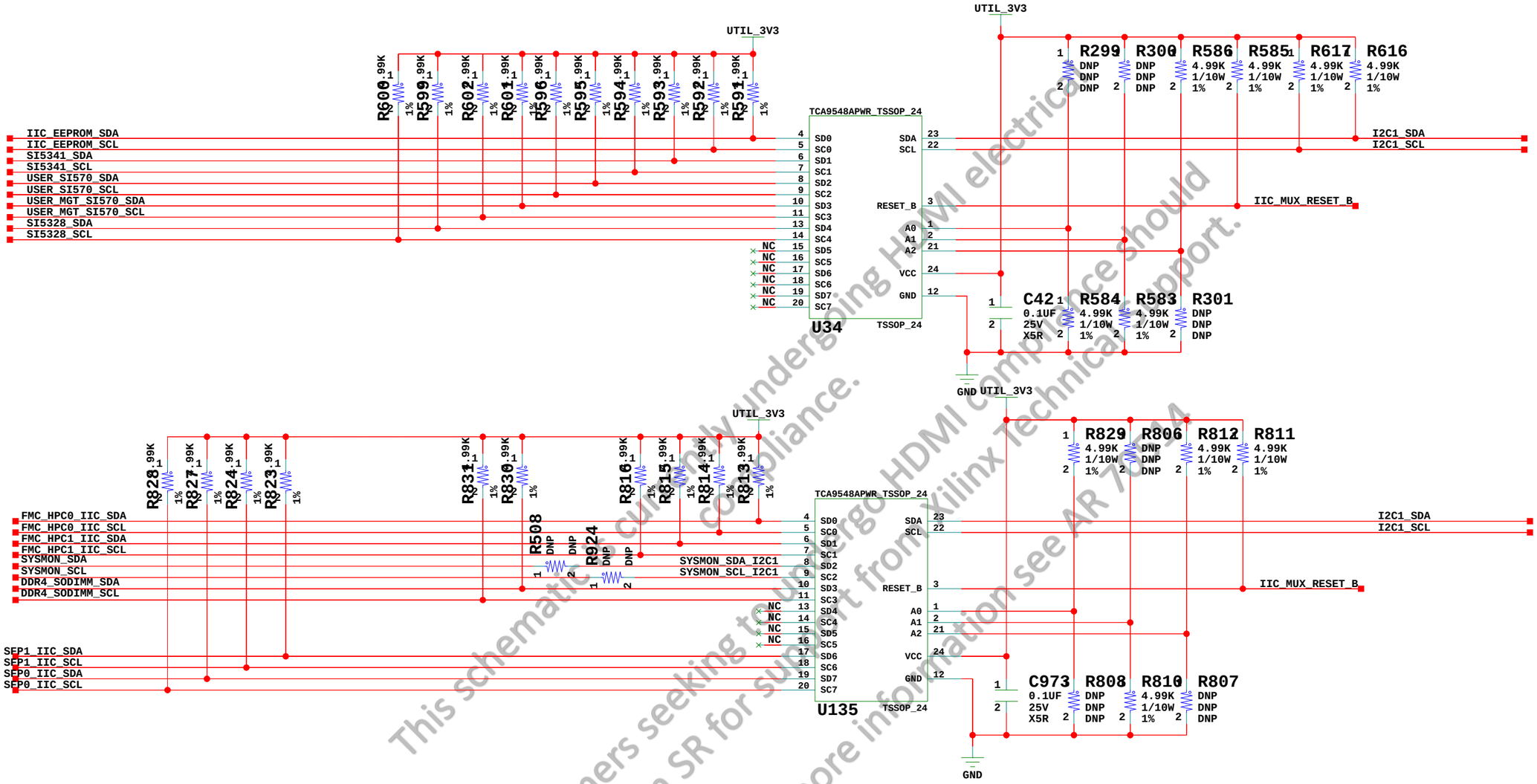


I2C0 MUXes Expanders Level Shifters PMBUS H

XILINX

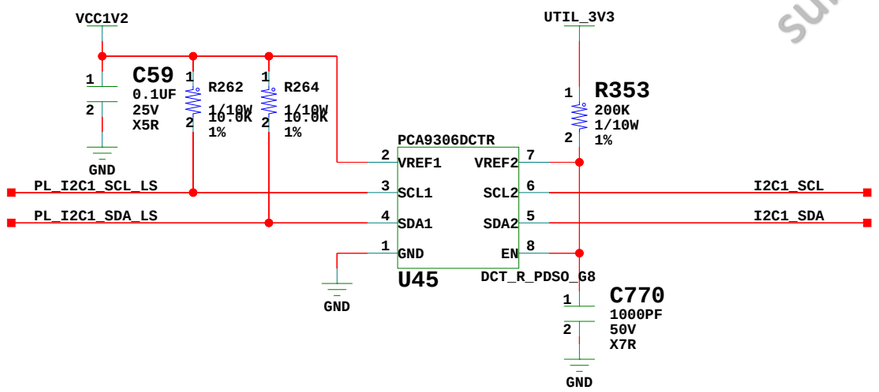
TITLE: I2C0 MUXes Expanders Level Shifters PMBUS Hea SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 64 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to design to HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

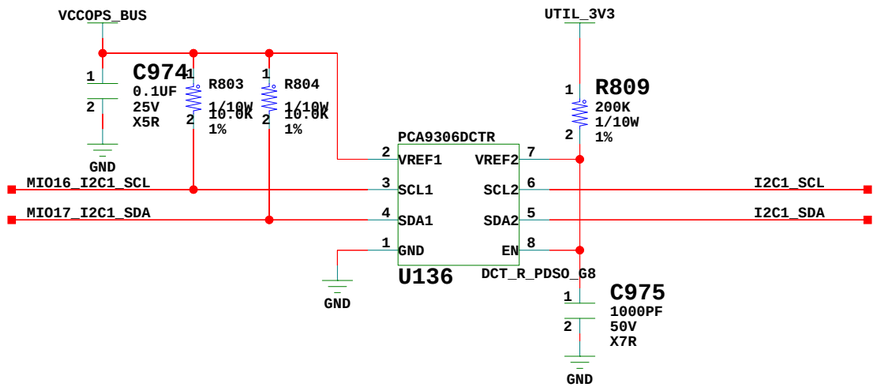


This schematic currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical support.
For more information see AR 70114

PL I2C Level Shifter (disabled if VREF1 is off)



PS I2C Level Shifter (I2C1)



I2C1 MUXes and Level Shifters

XILINX

TITLE: I2C1 MUXes and Level Shifters SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 65 OF 95	DRAWN BY: BF

6-PIN MINI-FIT AC ADAPTER (BRICK)

J52: 12V (1), 12V (4), COM (2), 12V (5), COM (3), COM (6) → INPUT_GND

VCC12_P → R448 2.00K 1/16W → MASTER_SW_12V

C382 10UF 25V X5R

Q7 SI7157DP (S, D)

R269 10.0K 1/10W

C657 1UF 16V

R412 1.00K 1/16W → Q4 BSS138 360MW

SW1 ON / OFF (SH1, SH2) → INPUT_GND

C533 1UF 25V X5R → INPUT_GND

U25 50MOHM (B, CB, X5R, PSG, CG1, CG2, CG3) → GND

J28 DNP → VCC12_SW

C922 330UF 25V ELEC → GND

DS2 LED-GRN-SMT → MASTER POWER

R268 2.00K 1/16W 1% → GND

J29 DNP, J30 DNP, J31 DNP, J32 DNP, J33 DNP, J34 DNP → GND

GND Test points

UTIL_3V3 → R657 10.0K 1/10W → EN

R650 3.32K 1/10W 1%

MAX16052 U103 (EN, VCC, OUT, IN, CDELAY, GND)

VCC12_P → R711 4.99K 1/10W 1% → VCC

C923 0.1UF 25V X5R → INPUT_GND

REG_EN

C931 0.01UF 25V X5R

R649 1.50K 1/10W 1%

R710 1.00K 1/16W 1%

J153 HDR_1X2

C932 DNP DNP DNP → INPUT_GND

1 Maxim Regulator Inhibit Jumper

C278 0.1UF 25V → GND

R958 R957 R956 R955 4.70K 1/16W 1%

R962 R961 R960 R959 DNP

R236 4.70K 1/16W 1%

R278 4.70K 1/16W 1%

R717 4.70K 1/16W 1%

R964 R963 DNP

R966 R965 DNP

U128 MAX6643 (TACHSET, TH1, TH2, TL1, TL2, DXP1, GND, VDD, OT1, OT2, OT_B, FANFAIL_B, FULLSPEED, PWM_OUT, FAN_IN1, FAN_IN2) MAX6643_QSOP16

SYSMON_DXP, SYSMON_DXN → C279 2200PF 50V X7R → Z5

MAX6643_OT_B, MAX6643_FANFAIL_B, FAN_PWM, FAN_TACH_12V

Q23 SI2300DS 1.7W → FAN_NEG

Fan Controller

Keyed Fan Header

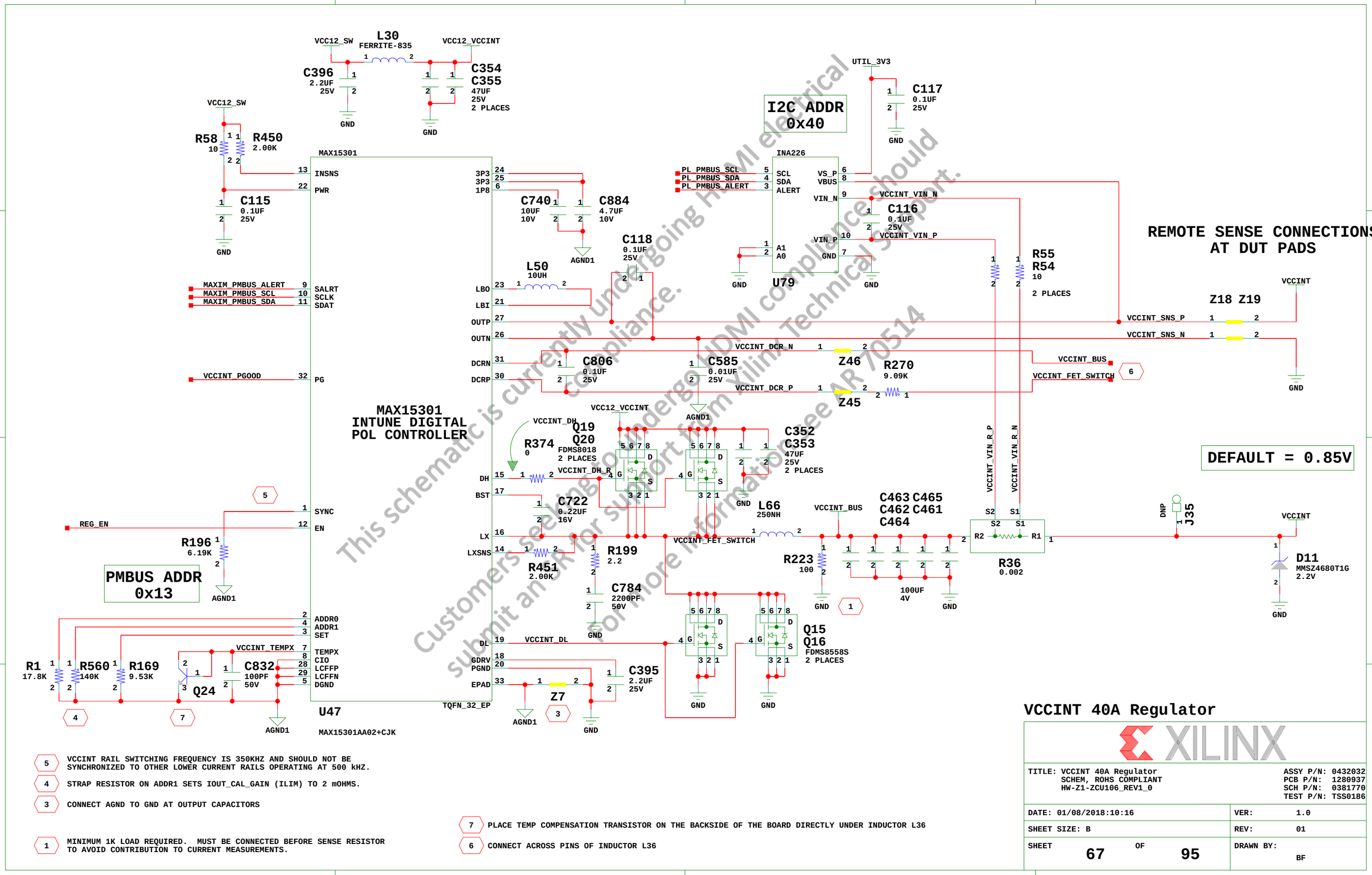
J71 22_11_2032: VCC12_SW, FAN_NEG, FAN_TACH_12V → R970 4.70K 1/16W 1%

12V Power Connectors Switch

XILINX

TITLE: 12V Power Connectors Switch SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 66 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514



5 VCCINT RAIL SWITCHING FREQUENCY IS 350KHZ AND SHOULD NOT BE SYNCHRONIZED TO OTHER LOWER CURRENT RAILS OPERATING AT 500 kHZ.

4 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS.

3 CONNECT AGND TO GND AT OUTPUT CAPACITORS

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

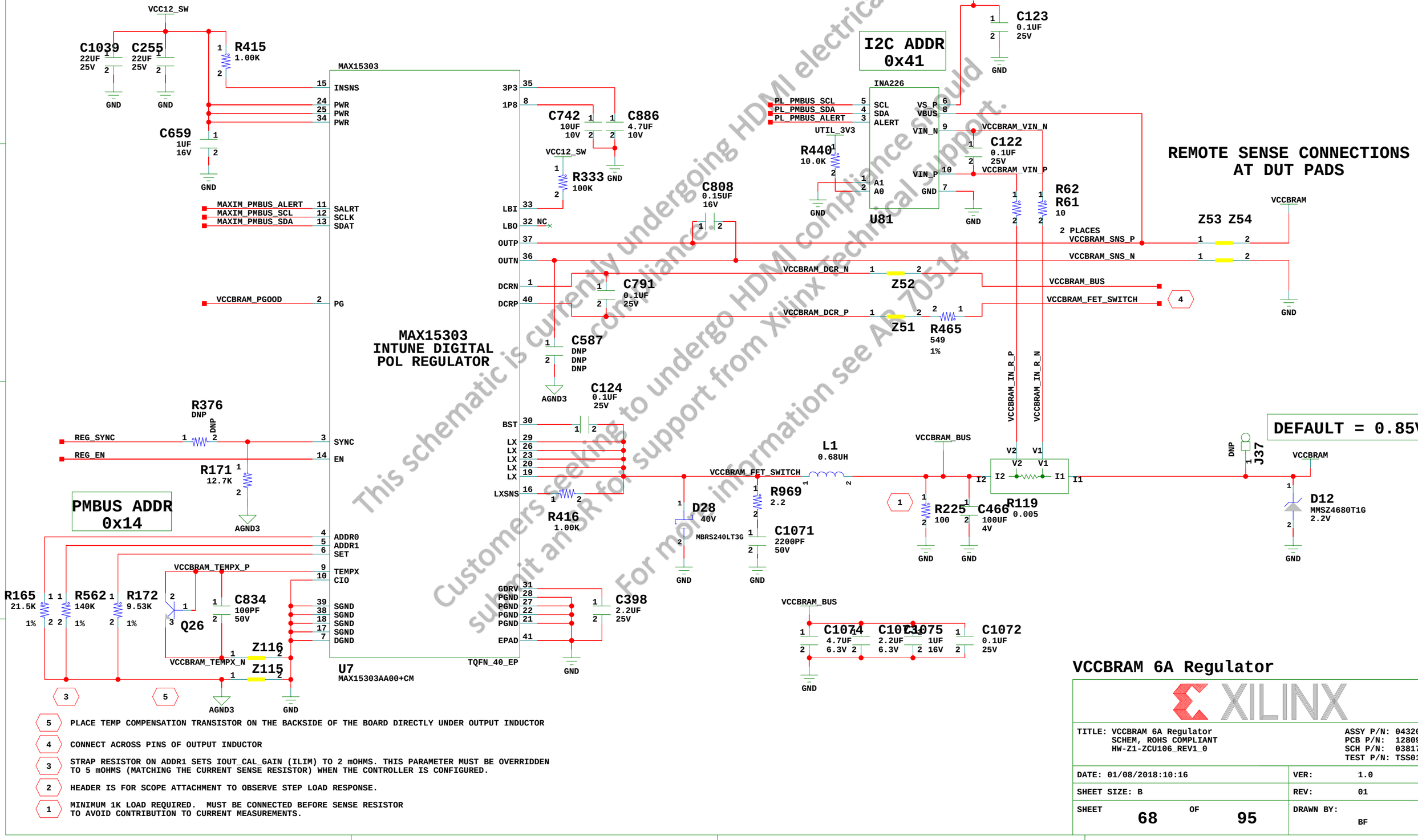
7 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER INDUCTOR L36

6 CONNECT ACROSS PINS OF INDUCTOR L36

VCCINT 40A Regulator

XILINX

TITLE: VCCINT 40A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 67 OF 95	DRAWN BY: BF


MAX15303
INTUNE DIGITAL
POL REGULATOR
U7
MAX15303AA00+CM
TQFN_40_EP
I2C ADDR
0x41
PMBUS ADDR
0x14
REMOTE SENSE CONNECTIONS
AT DUT PADS
DEFAULT = 0.85V
VCCBRAM 6A Regulator
INA226
U81
MAXIM_PMBUS_ALERT
MAXIM_PMBUS_SCL
MAXIM_PMBUS_SDA
VCCBRAM_PGOOD
REG_SYNC
REG_EN
VCCBRAM_TEMPX_P
VCCBRAM_TEMPX_N
VCCBRAM_DCR_N
VCCBRAM_DCR_P
VCCBRAM_SNS_P
VCCBRAM_SNS_N
VCCBRAM_BUS
VCCBRAM_FET_SWITCH
VCCBRAM_IN_R_P
VCCBRAM_IN_R_N
PL_PMBUS_SCL
PL_PMBUS_SDA
PL_PMBUS_ALERT
UTIL_3V3
VCC12_SW
VCCBRAM_VIN_N
VCCBRAM_VIN_P
2 PLACES
5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
CONNECT ACROSS PINS OF OUTPUT INDUCTOR
STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.
HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.
MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.
TITLE: VCCBRAM 6A Regulator
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0
ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186
DATE: 01/08/2018:10:16
VER: 1.0
SHEET SIZE: B
REV: 01
SHEET 68 OF 95
DRAWN BY: BF
This schematic is currently undergoing HDMI electrical compliance
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

UTIL_3V3

C1116
0.1UF
25V

GND

I2C ADDR
0x4A

INA226

PL_PMBUS_SCL
PL_PMBUS_SDA
PL_PMBUS_ALERT

SCL
SDA
ALERT
A1
A0
VS_P
VBUS
VIN_N
VIN_P
GND

U159

VCCINT_VCU_VIN_N
VCCINT_VCU_VIN_P

C1115
0.1UF
25V

R1003
R1002
10
2 PLACES

VCCINT_VCU_VIN_R_P
VCCINT_VCU_VIN_R_N

R1000
1%
2W

VCCINT_VCU

DEFAULT = 0.90V

VCCAUX_BUS

C1100
100UF
4V

C1099
100UF
4V

GND

MAX8556

IN_1
IN_2
IN_3
IN_4
IN_5
IN_6
OUT_7
OUT_8
OUT_9
OUT_10
OUT_11
POK_12
FB_13
GND_14
EP_17
EN_16
NC_15

U153

MAX8556ETE_T

VCCINT_VCU_PGOOD
VCCINT_PGOOD

VCCINT_VCU_BUS

C1102
10UF
10V

C1101
10UF
10V

0.002

GND

VCCINT_VCU_SNS

Z129

R972
DNP

R1001
392

C1107
DNP
DNP
DNP

R999
DNP

R987
487

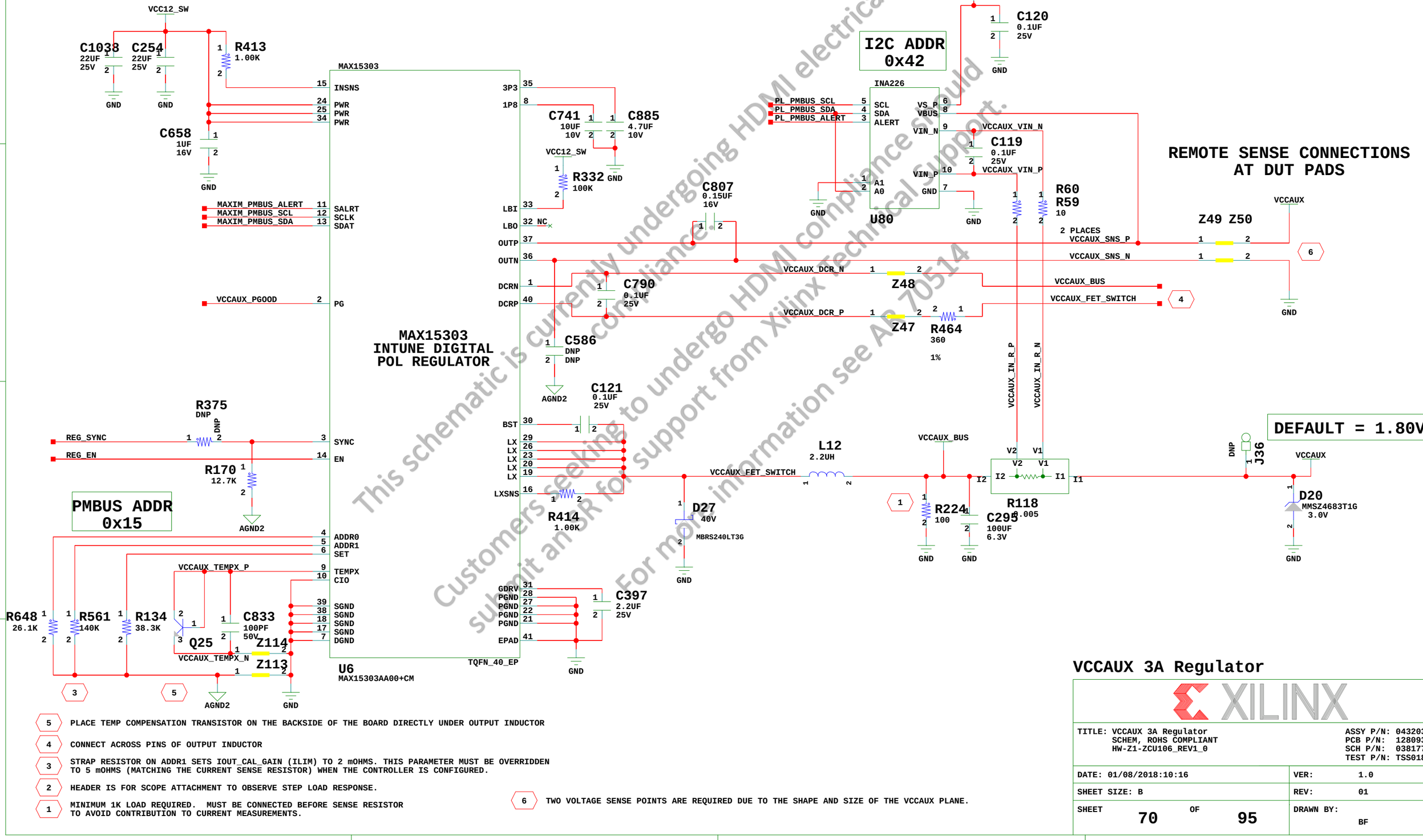
GND

Remote voltage sense trace must route dirctly to underneath U1

VCCINT_VCU 3A Regulator

XILINX

TITLE: VCCINT_VCU 3A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 69 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

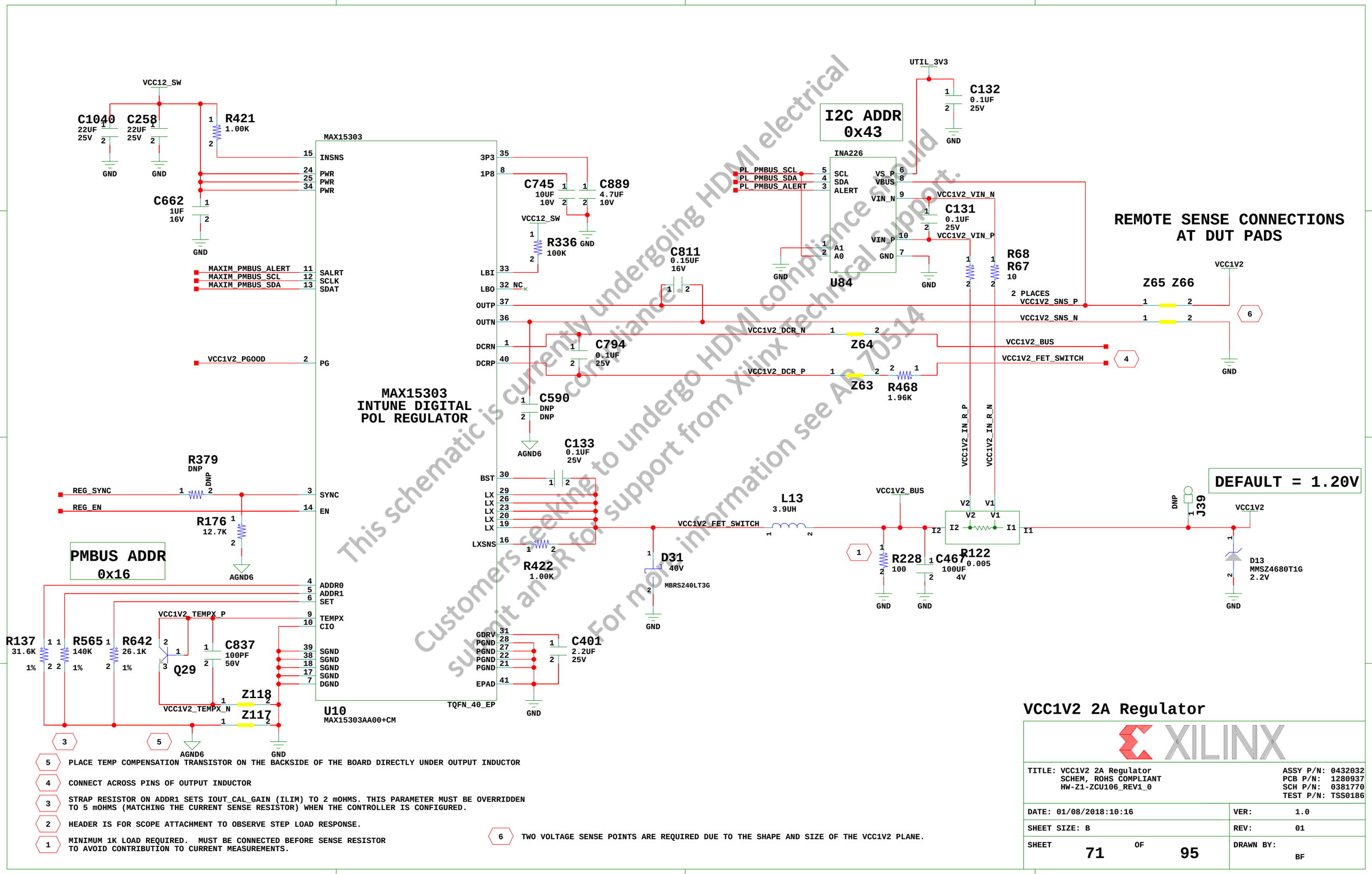
1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE VCCAUX PLANE.

VCCAUX 3A Regulator

XILINX

TITLE: VCCAUX 3A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 70 OF 95	DRAWN BY: BF


MAX15303
INTUNE DIGITAL
POL REGULATOR
U10
MAX15303AA00+CM
TQFN_40_EP
I2C ADDR
0x43
INA226
U84
PMBUS ADDR
0x16
REMOTE SENSE CONNECTIONS
AT DUT PADS
2 PLACES
DEFAULT = 1.20V
VCC1V2 2A Regulator
XILINX
TITLE: VCC1V2 2A Regulator
SCHEM, ROHS COMPLIANT
HW-Z1-ZCU106_REV1_0
ASSY P/N: 0432032
PCB P/N: 1280937
SCH P/N: 0381770
TEST P/N: TSS0186
DATE: 01/08/2018:10:16
VER: 1.0
SHEET SIZE: B
REV: 01
SHEET 71 OF 95
DRAWN BY: BF
5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR
3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.
2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.
1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.
6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE VCC1V2 PLANE.
This schematic is currently undergoing HDMI electrical compliance
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

MAX15303 INTUNE DIGITAL POL REGULATOR

I2C ADDR 0x44

PMBUS ADDR 0x17

DEFAULT = 3.3V

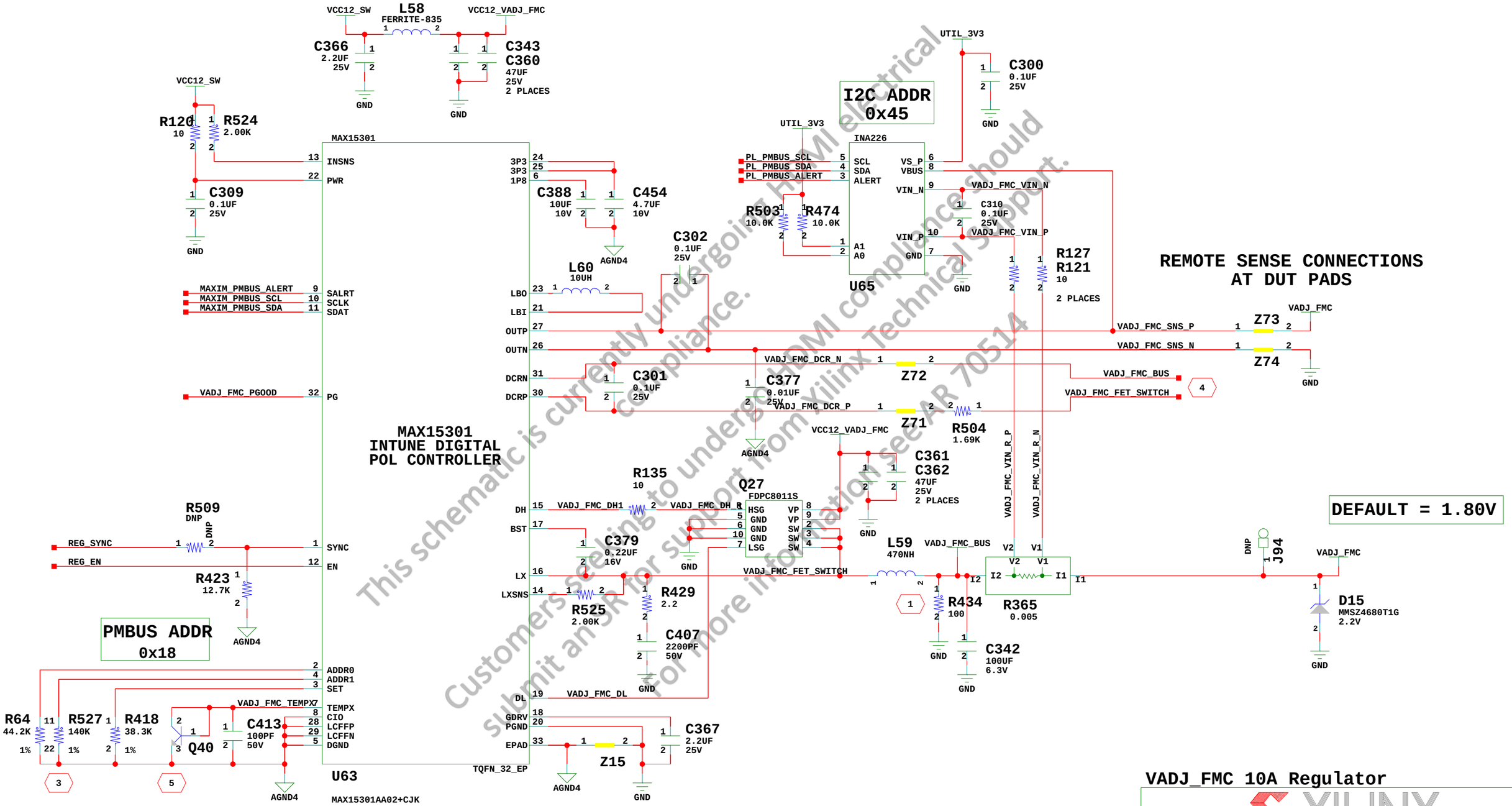
REMOTE SENSE CONNECTIONS AT DUT PADS

- 5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
- 4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR
- 3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.
- 2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.
- 1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.
- 6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE VCC3V3 PLANE.

VCC3V3 5A Regulator

XILINX

TITLE: VCC3V3 5A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 72 OF 95	DRAWN BY: BF



5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER INDUCTOR L3

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

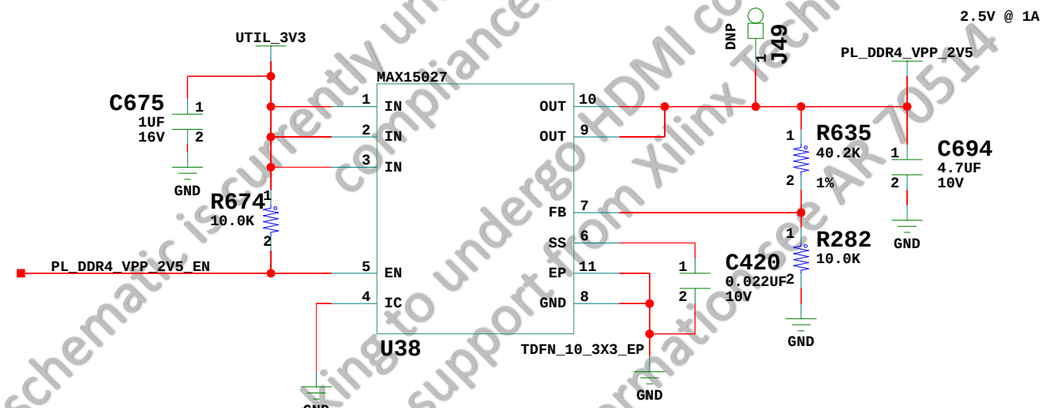
2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

VADJ_FMC 10A Regulator

XILINX

TITLE: VADJ_FMC 10A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 73 OF 95	DRAWN BY: BF

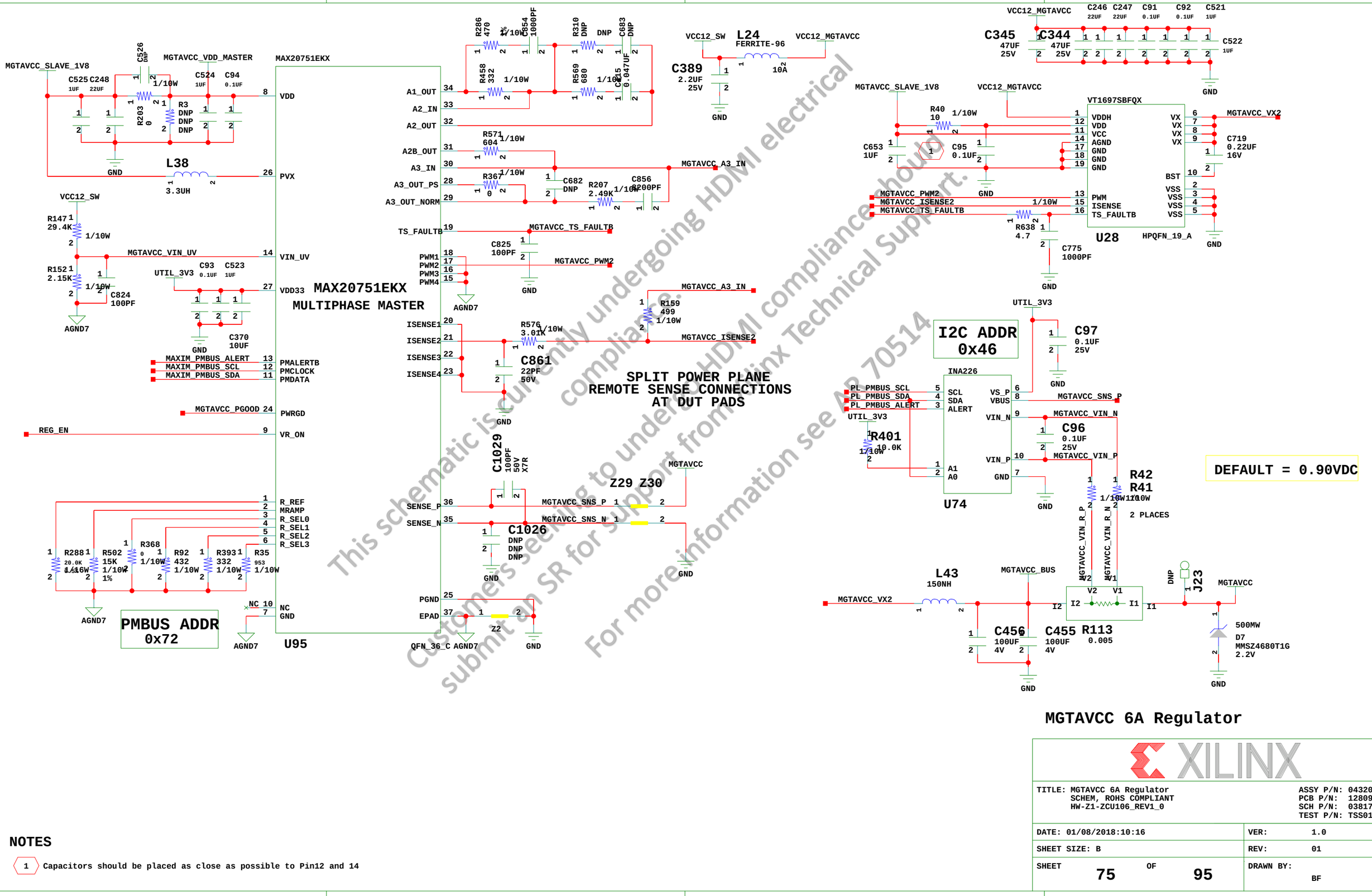


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PL_DDR4_VPP_2V5 1A Regulator

XILINX

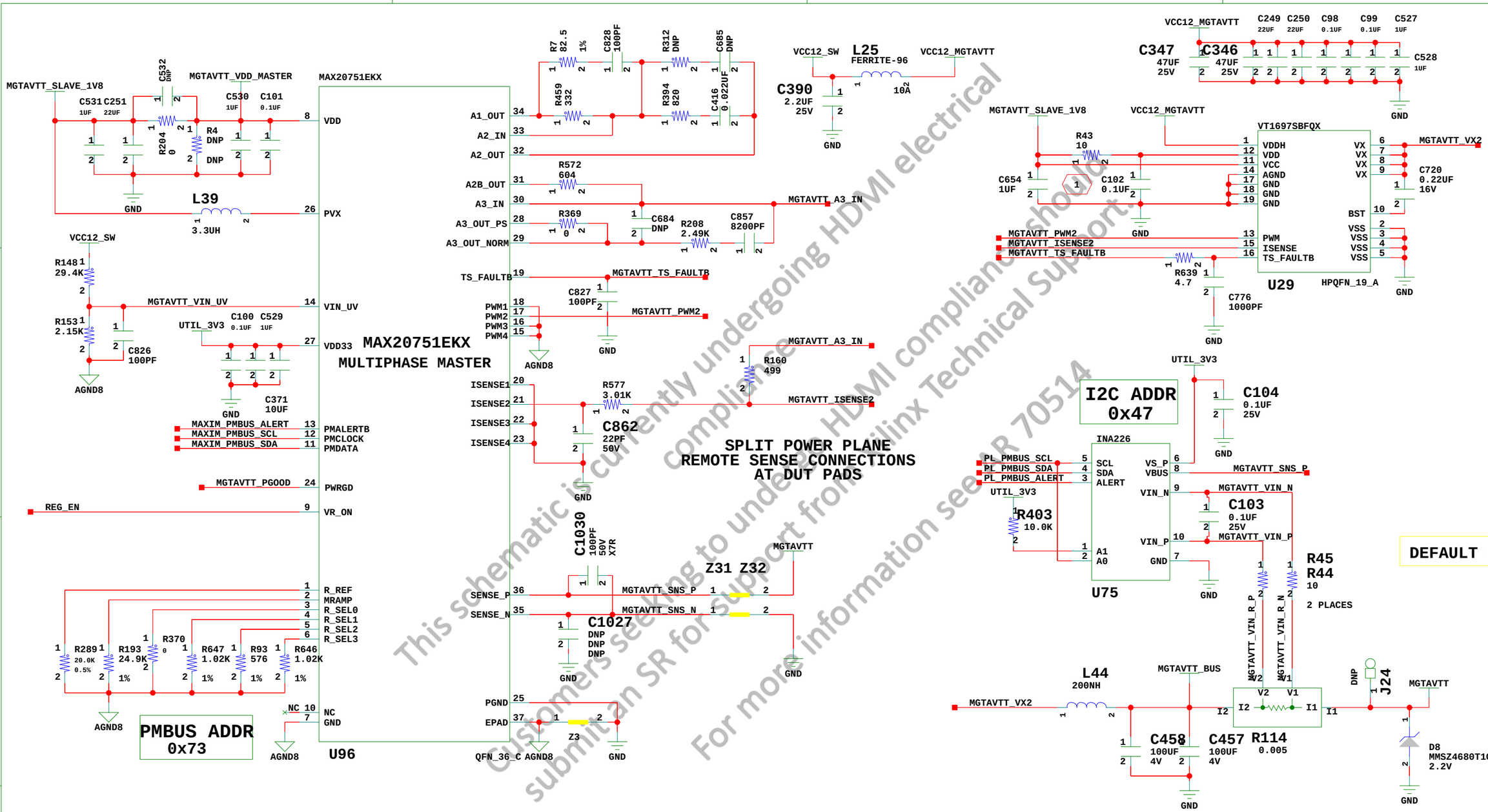
TITLE: PL_DDR4_VPP_2V5 1A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 74 OF 95	DRAWN BY: BF



NOTES

1. Capacitors should be placed as close as possible to Pin12 and 14

TITLE: MGTAVCC 6A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 75 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to use HDMI compliance should submit an SR for support. For more information see XTP 70514

NOTES

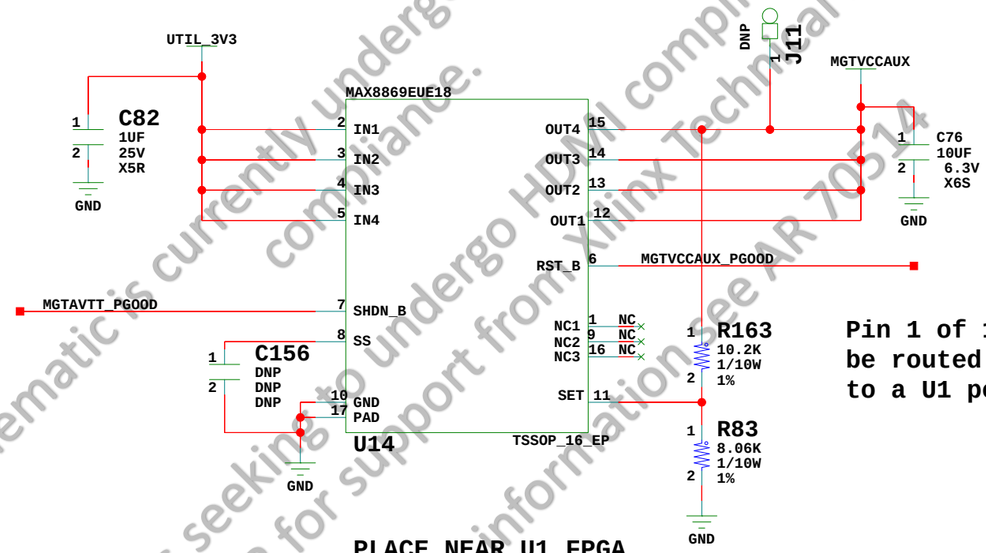
1. Capacitors should be placed as close as possible to Pin12 and 14

MGTAVTT 6A Regulator

XILINX

TITLE: MGTAVTT 6A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 76 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



Pin 1 of 10.2K resistor must be routed as a sense line directly to a U1 power pin

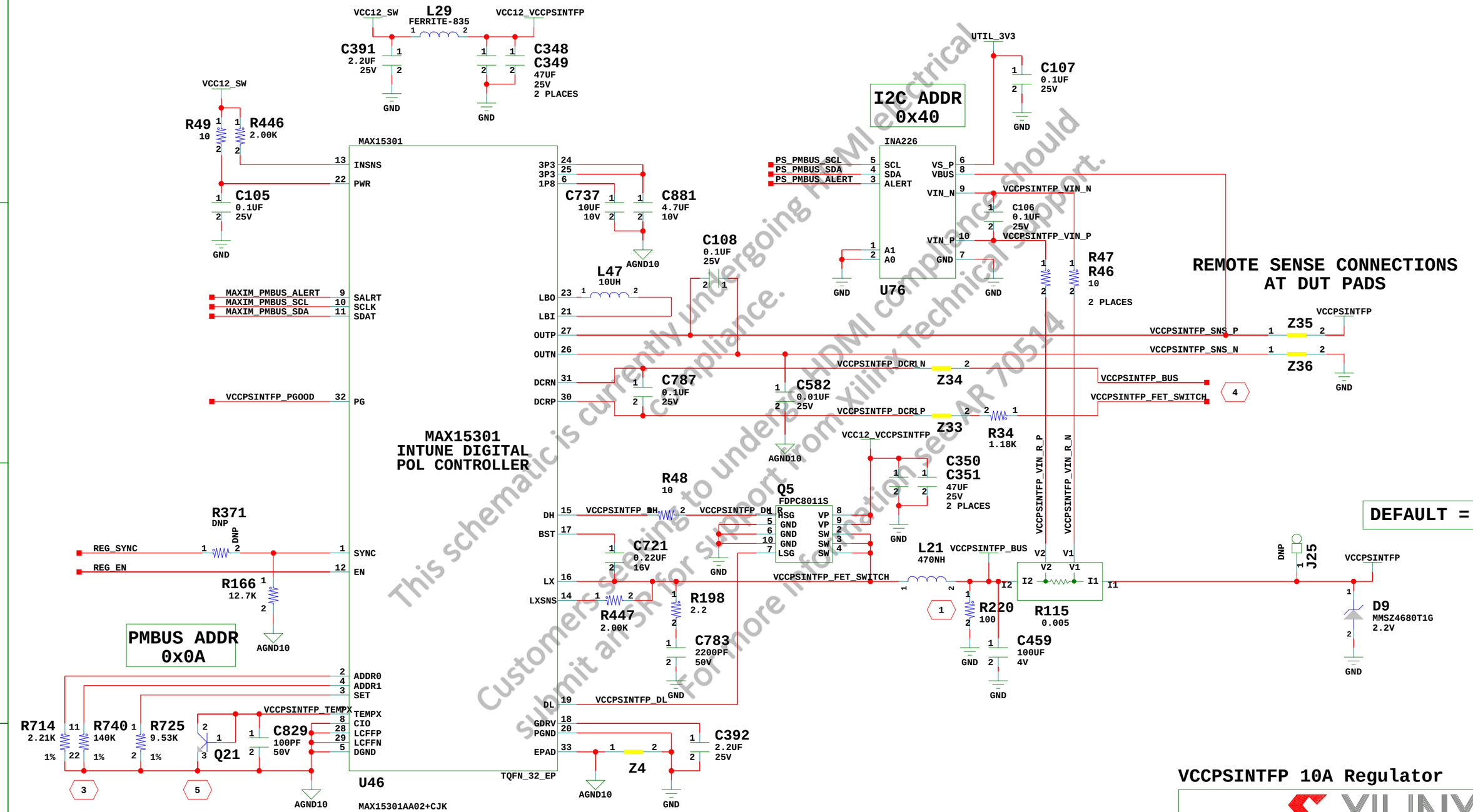
PLACE NEAR U1 FPGA

DEFAULT = 1.81VDC

MGTVCCAUX 1A Regulator

XILINX

TITLE: MGTVCCAUX 1A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 77 OF 95	DRAWN BY: BF



5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER INDUCTOR L3

4 CONNECT ACROSS PINS OF INDUCTOR L3

3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

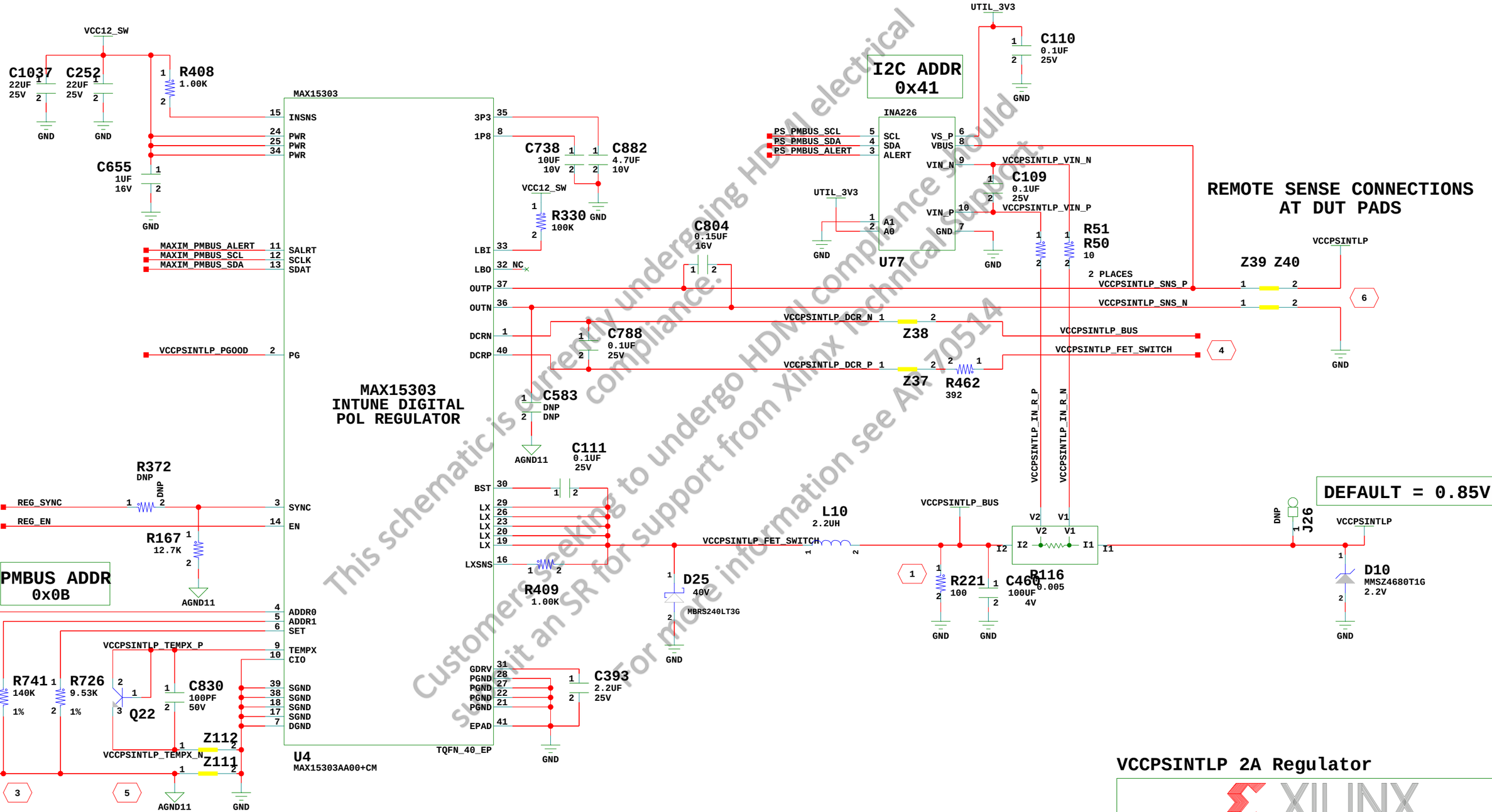
2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

VCCPSINTFP 10A Regulator

XILINX

TITLE: VCCPSINTFP 10A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432807 PCB P/N: 1280719 SCH P/N: 0381550 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: A.0
SHEET SIZE: B	REV: 01
SHEET 78 OF 95	DRAWN BY: BN



5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

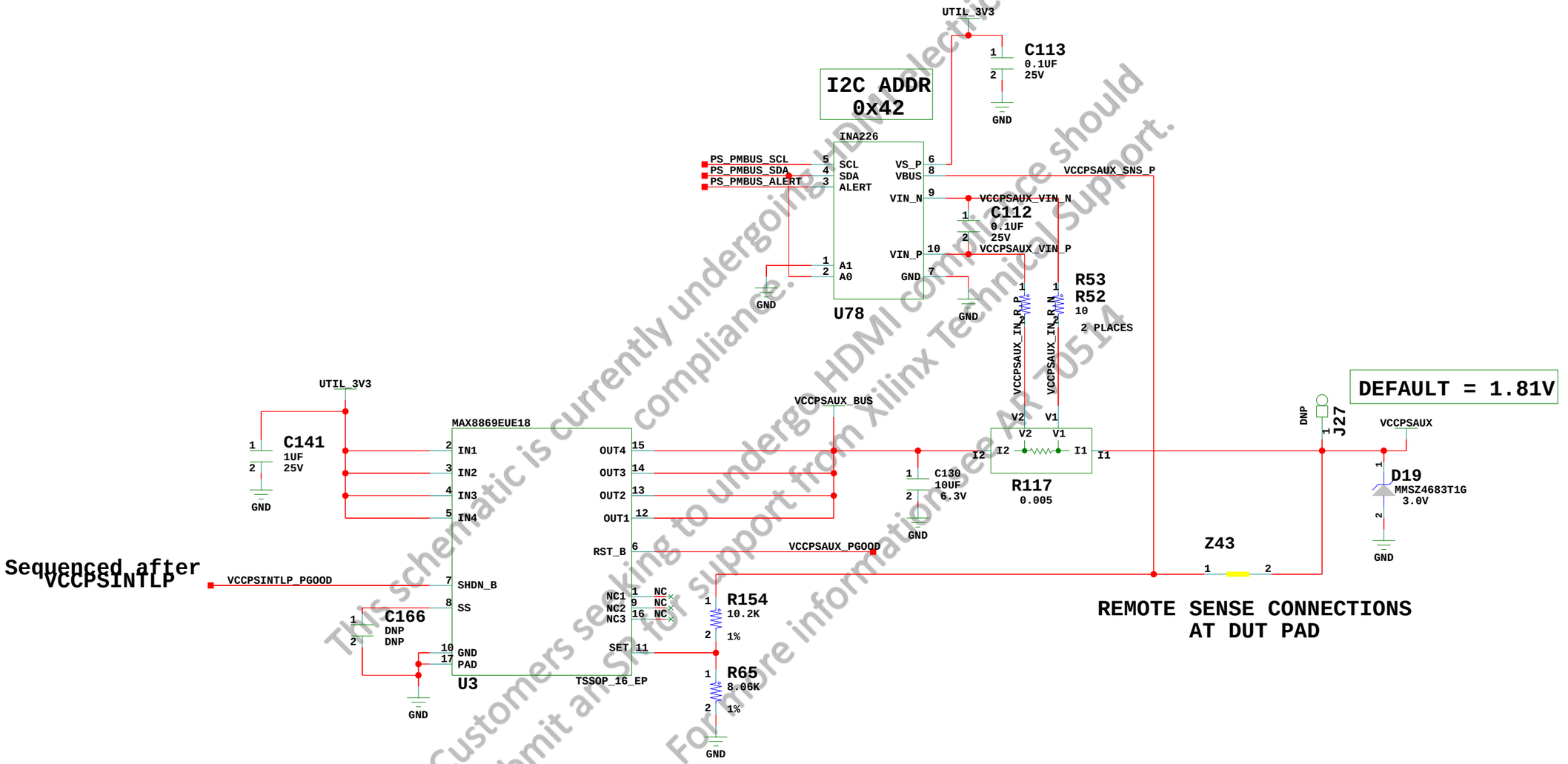
3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE VCCPSINTLP PLANE.

TITLE: VCCPSINTLP 2A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 79 OF 95	DRAWN BY: BF



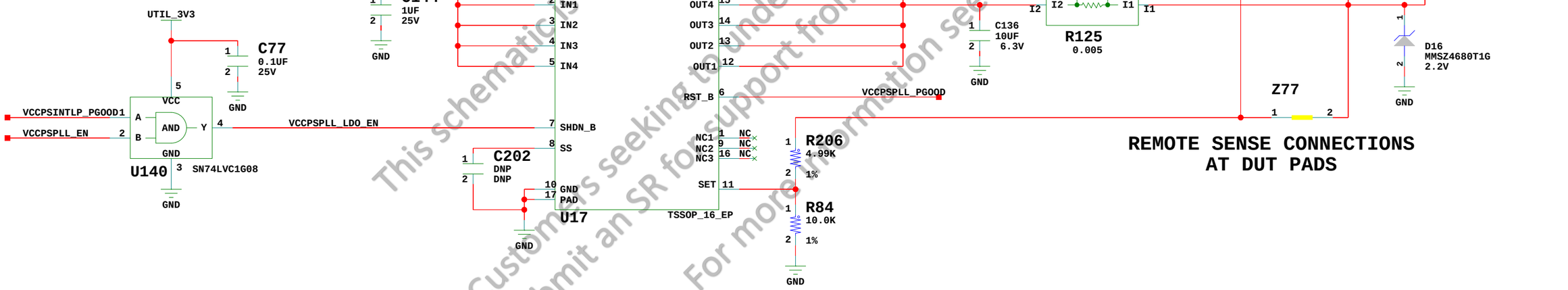
This schematic is currently undergoing HDMI electrical compliance. Customers seeking to submit an SR for support to undergo HDMI compliance should For more information support from Xilinx Technical Support. AR 70514

VCCPSAUX 500MA Regulator

XILINX

TITLE: VCCPSAUX 500MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 80 OF 95	DRAWN BY: BF

Sequenced after VCCPSINTLP, controlled by IO Expander



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

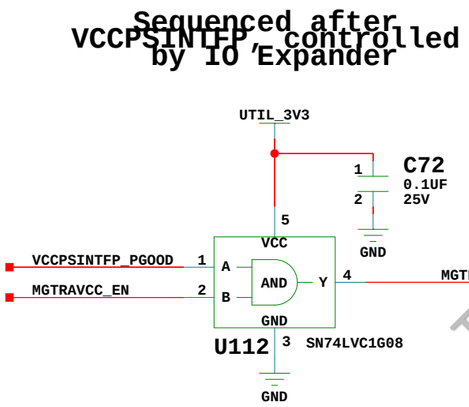
VCCPSPLL 200MA Regulator

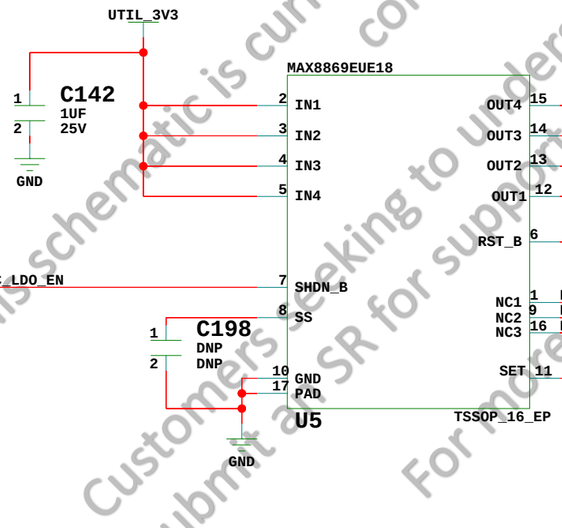
XILINX

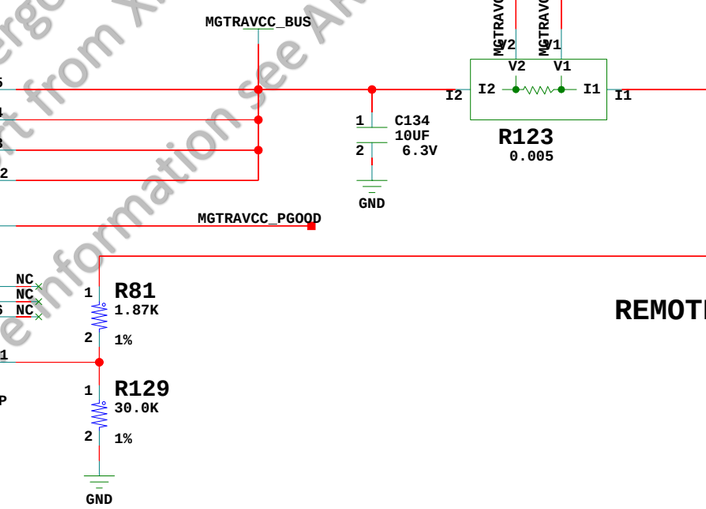
TITLE: VCCPSPLL 200MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 81 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit a SR for support from Xilinx Technical Support.
For more information see AR 70514

Sequenced after VCCPSINTFP, controlled by IO Expander







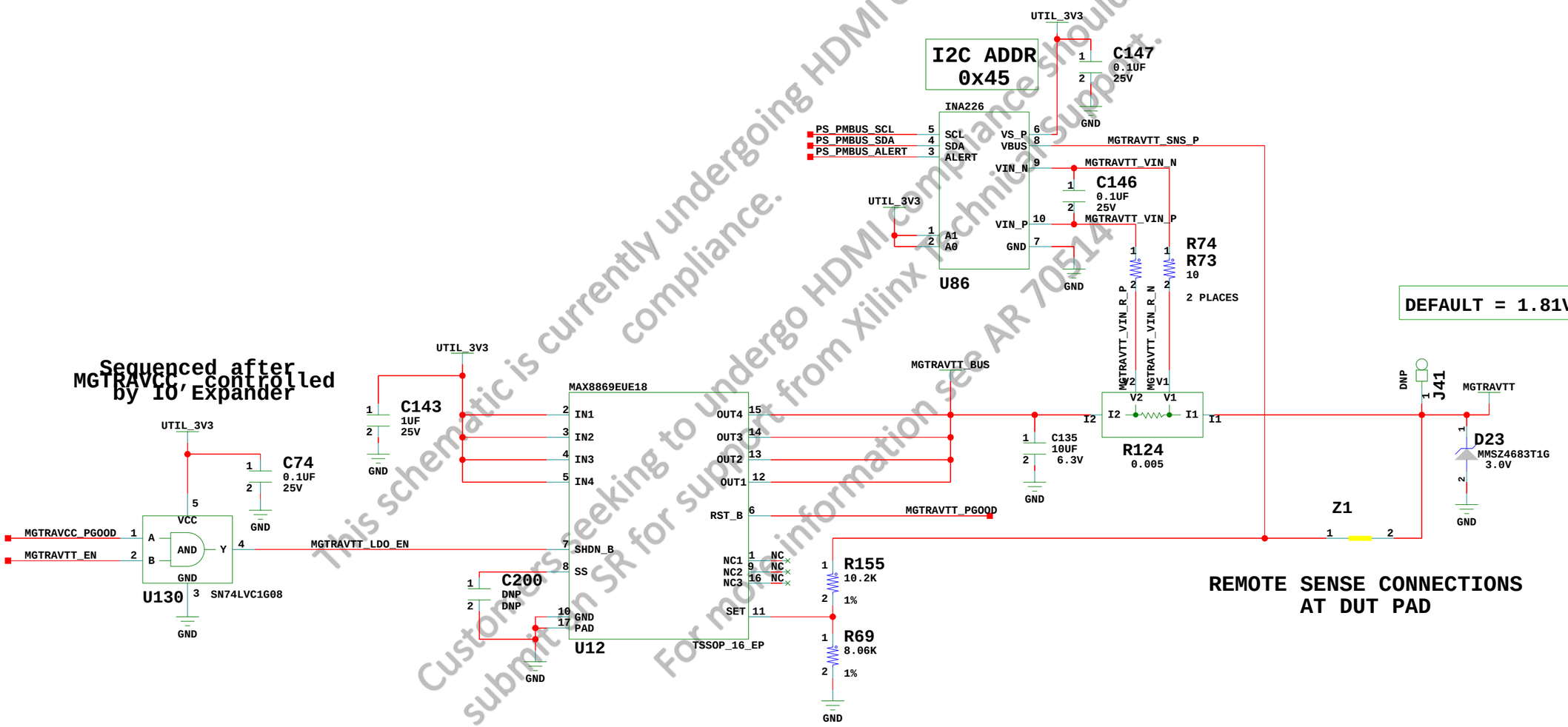
DEFAULT = 0.85VDC

REMOTE SENSE CONNECTIONS AT DUT PAD

MGTRAVCC 400MA Regulator

XILINX

TITLE: MGTRAVCC 400MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 82 OF 95	DRAWN BY: BF

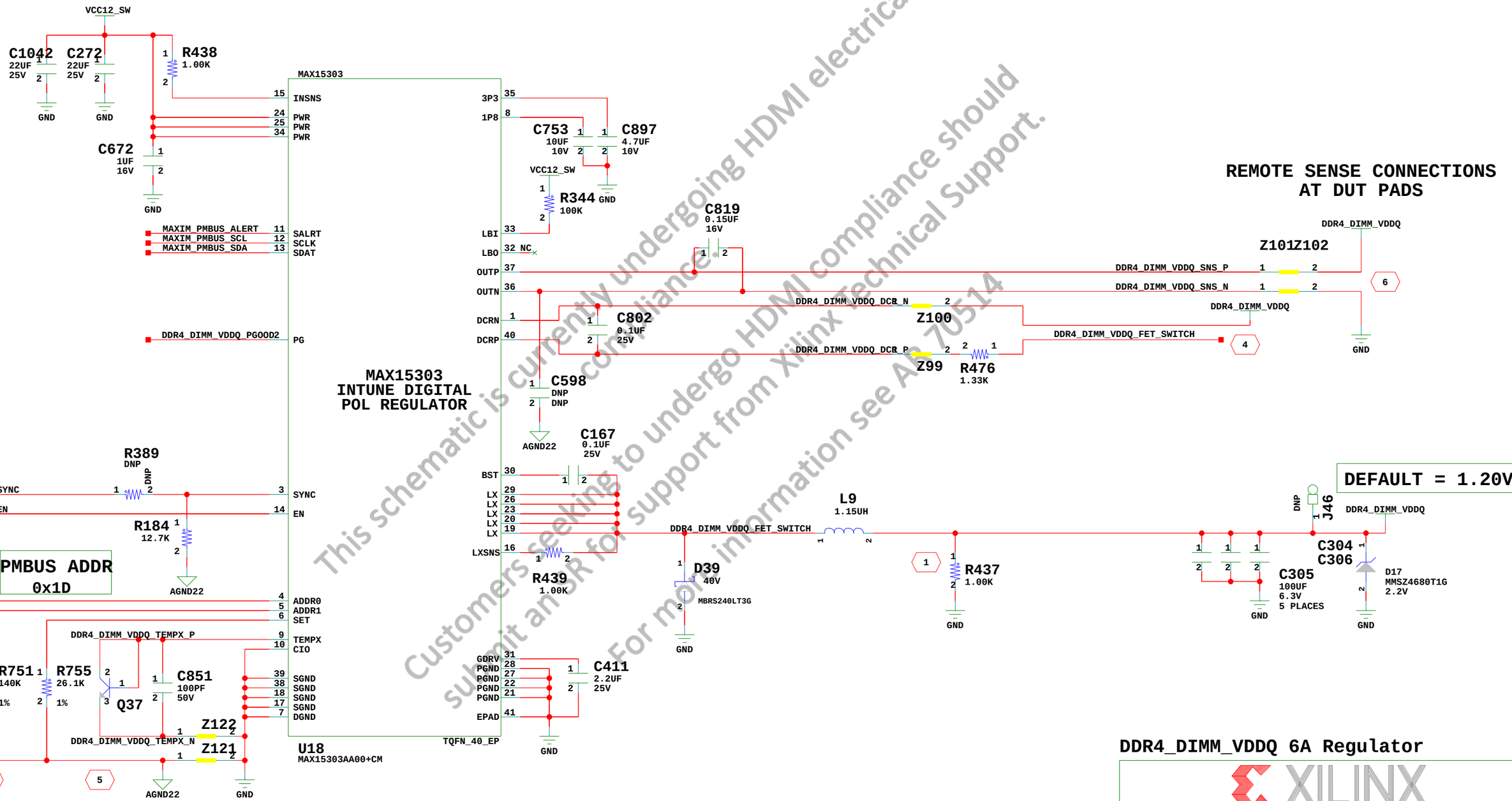


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

MGTRAVTT 100MA Regulator

XILINX

TITLE: MGTRAVTT 100MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 83 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

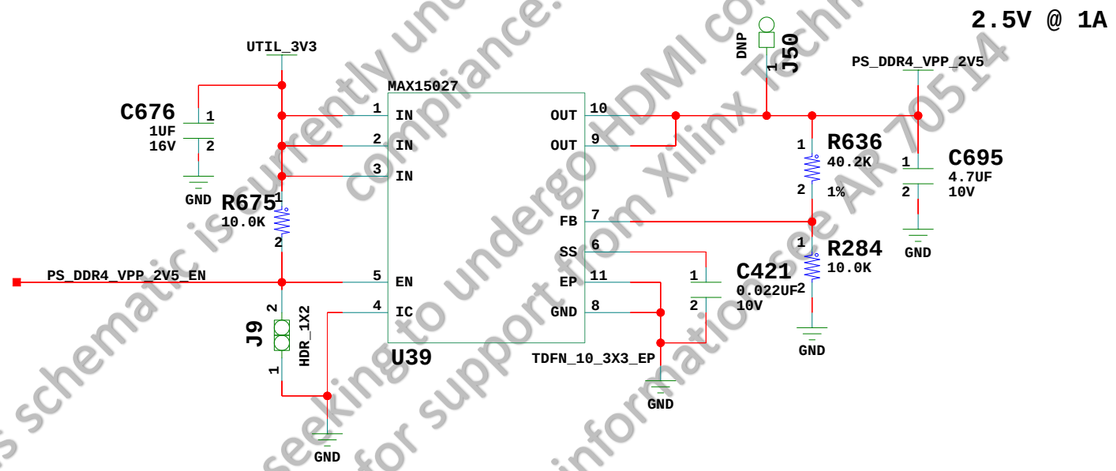
1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE DDR4_DIMM_VDDQ PLANE.

DDR4_DIMM_VDDQ 6A Regulator

XILINX

TITLE: DDR4_DIMM_VDDQ 6A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 84 OF 95	DRAWN BY: BF

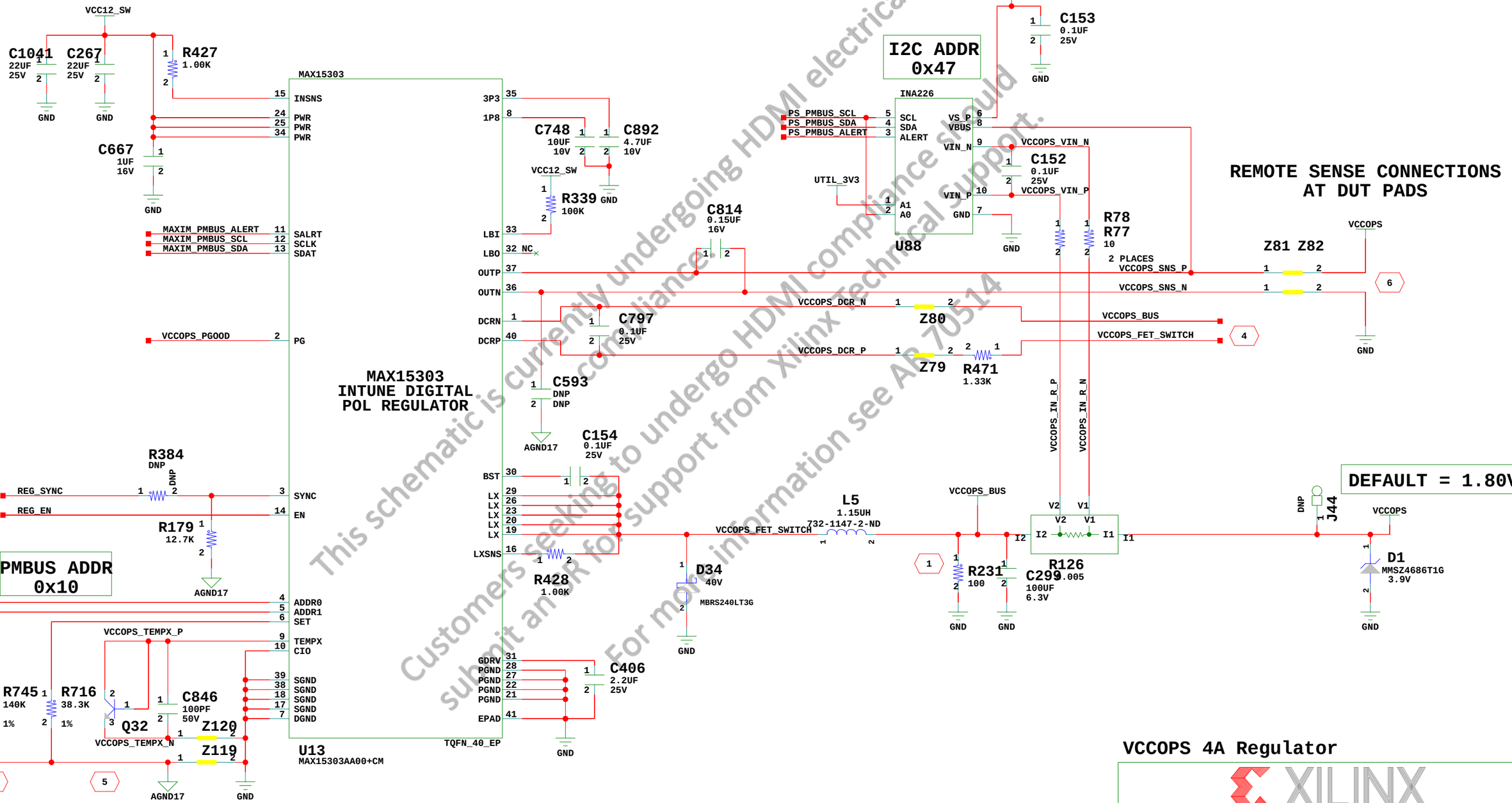


This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

PS_DDR4_VPP_2V5 1A Regulator

XILINX

TITLE: PS_DDR4_VPP_2V5 1A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 85 OF 95	DRAWN BY: BF



5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

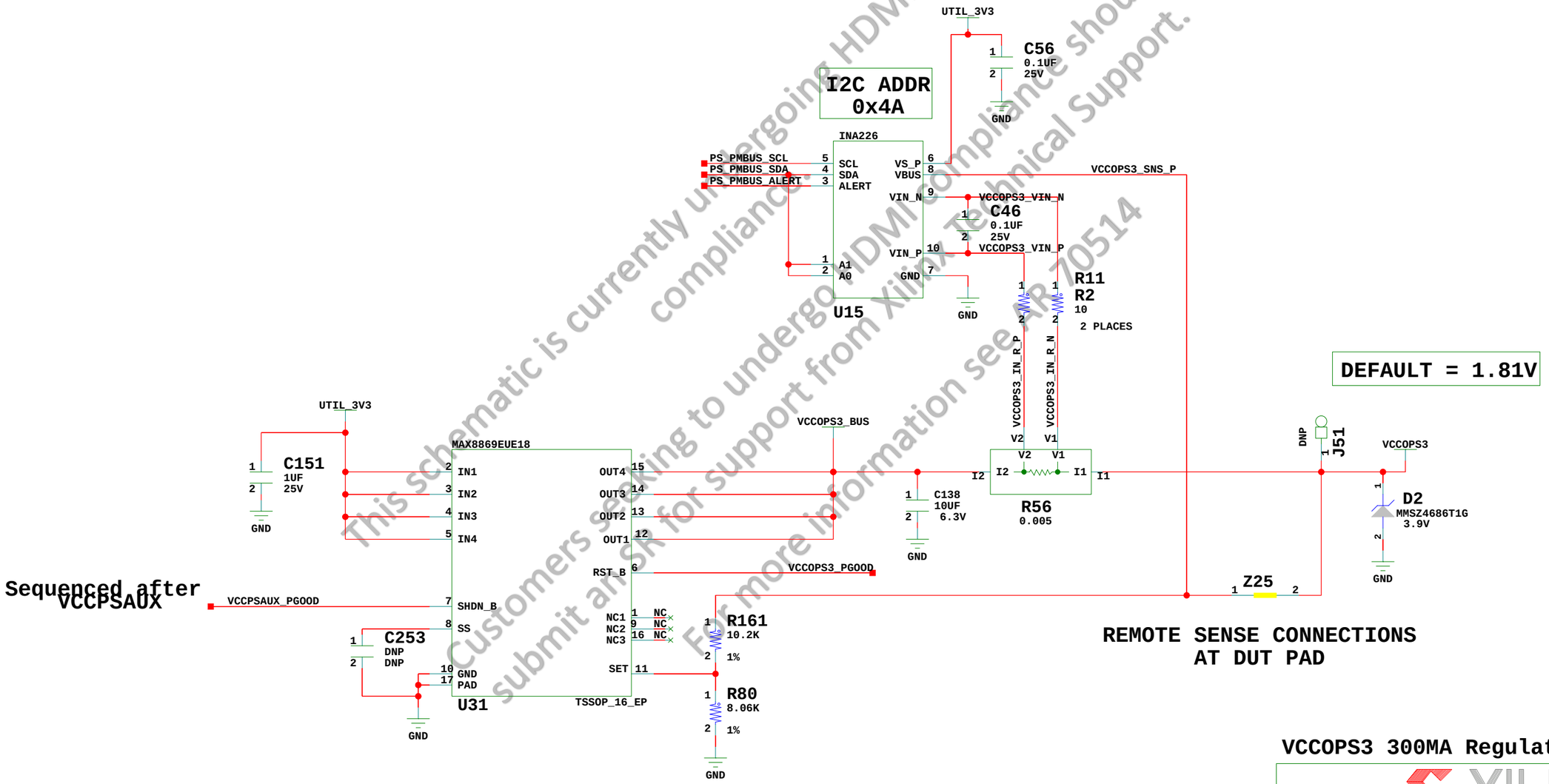
3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

6 TWO VOLTAGE SENSE POINTS ARE REQUIRED DUE TO THE SHAPE AND SIZE OF THE VCCOPS PLANE.

TITLE: VCCOPS 4A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 86 OF 95	DRAWN BY: BF

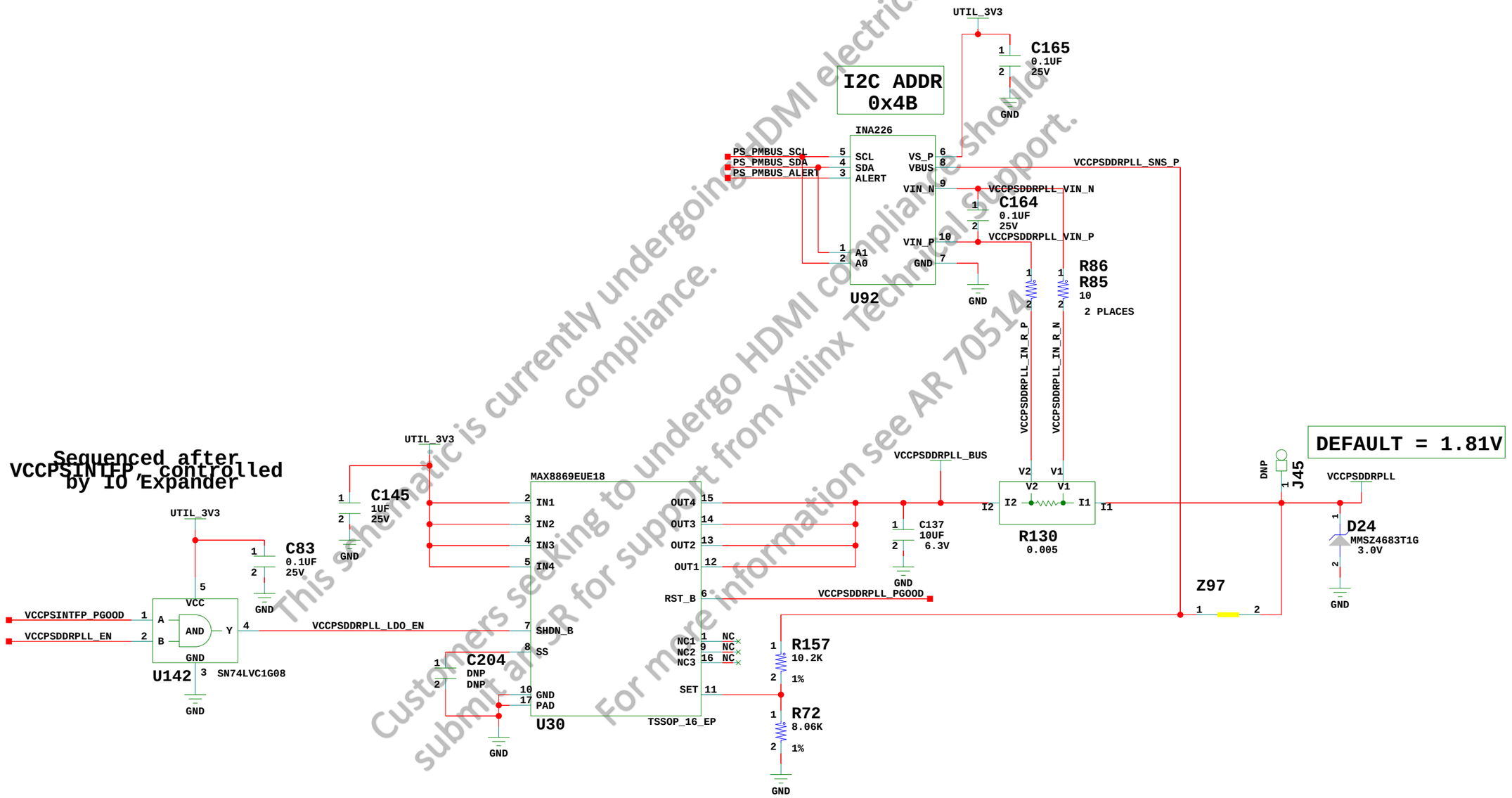


This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

VCCOPS3 300MA Regulator

XILINX

TITLE: VCCOPS3 300MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 87 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

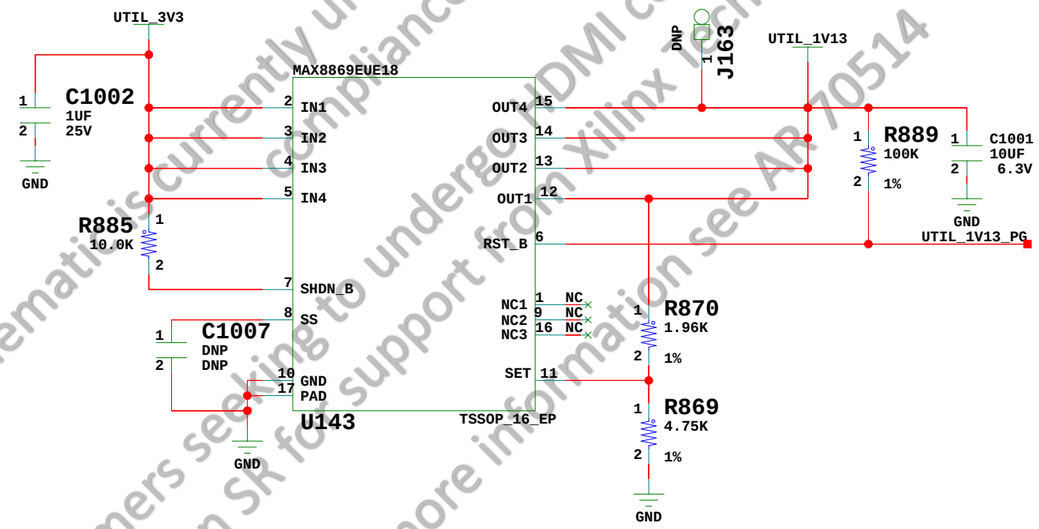
VCCPSDDRPLL 100MA Regulator

XILINX

TITLE: VCCPSDDRPLL 100MA Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 88 OF 95	DRAWN BY: BF

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

1.13V @ 1A



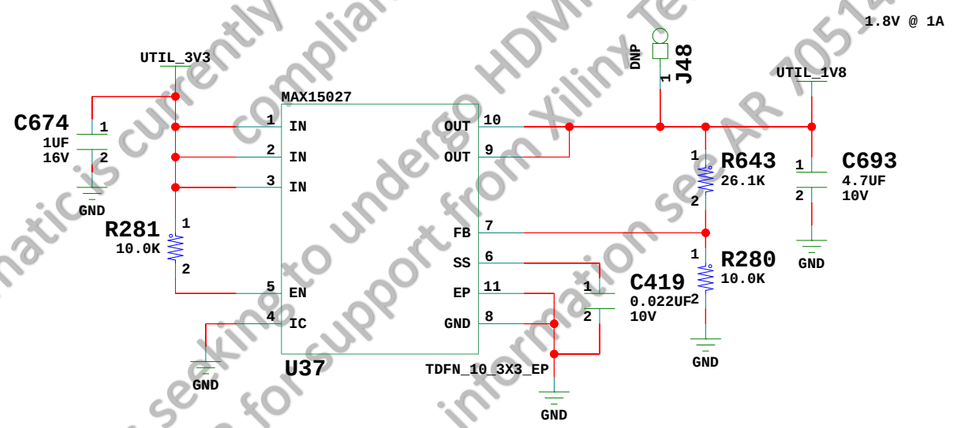
1.13V @ 1A satisfies voltage and total current requirements for TI ENET PHY, as well as HDMI TX and RX transcievers

UTIL_1V13 1A Regulator

XILINX

TITLE: UTIL_1V13 1A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 89 OF 95	DRAWN BY: BF

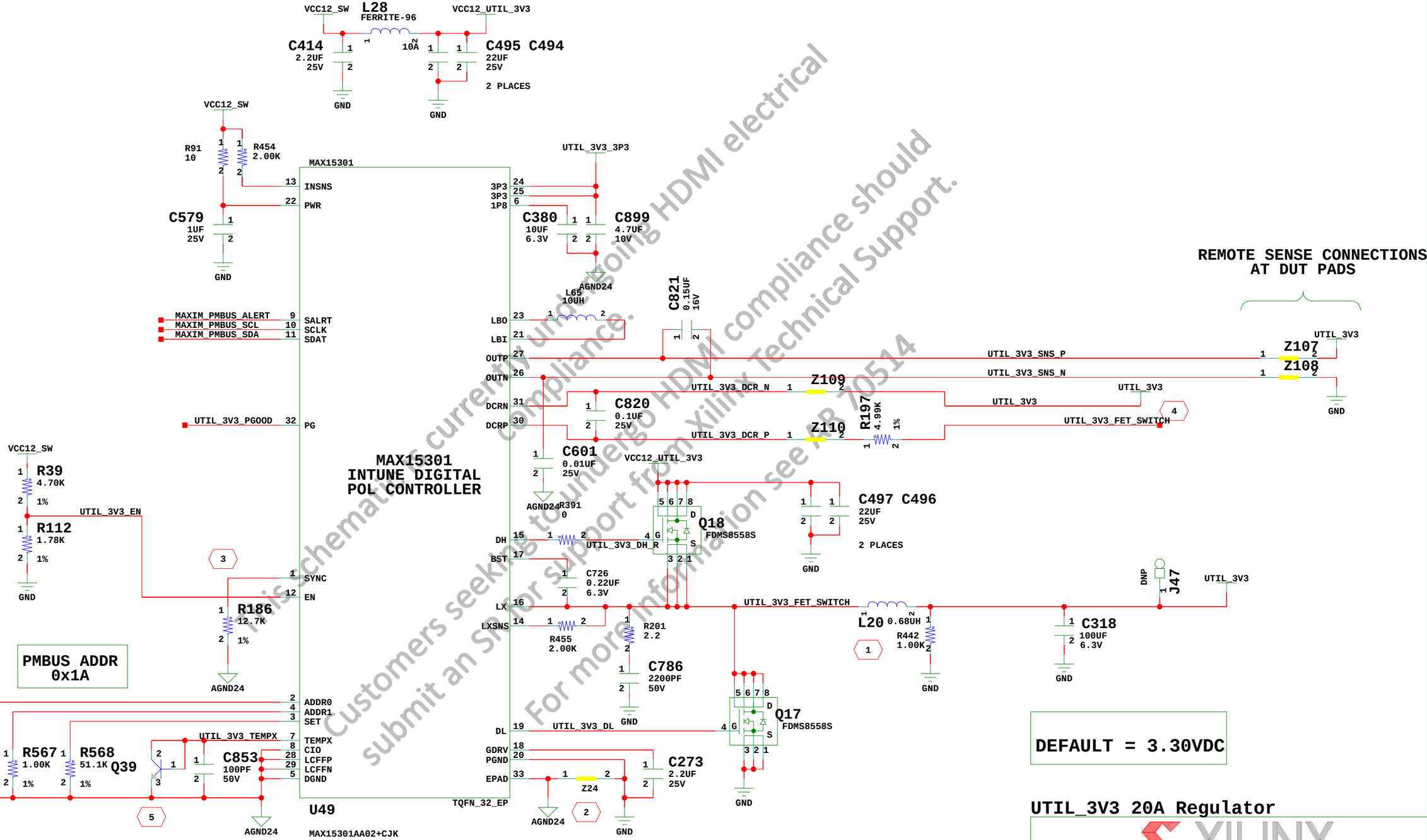
This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514



UTIL_1V8 1A Regulator

TITLE: UTIL_1V8 1A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 90 OF 95	DRAWN BY: BF

MAX15301 INTUNE DIGITAL POL CONTROLLER



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to build HDMI compliance should submit an SPR for support from Xilinx Technical Support.
For more information see AR 70514

3 UTIL_3V3 RAIL SWITCHING FREQUENCY IS 400kHz AND SHOULD NOT BE SYNCHRONIZED TO OTHER LOWER CURRENT RAILS OPERATING AT 600 kHZ.

2 CONNECT AGND TO GND AT OUTPUT CAPACITORS

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

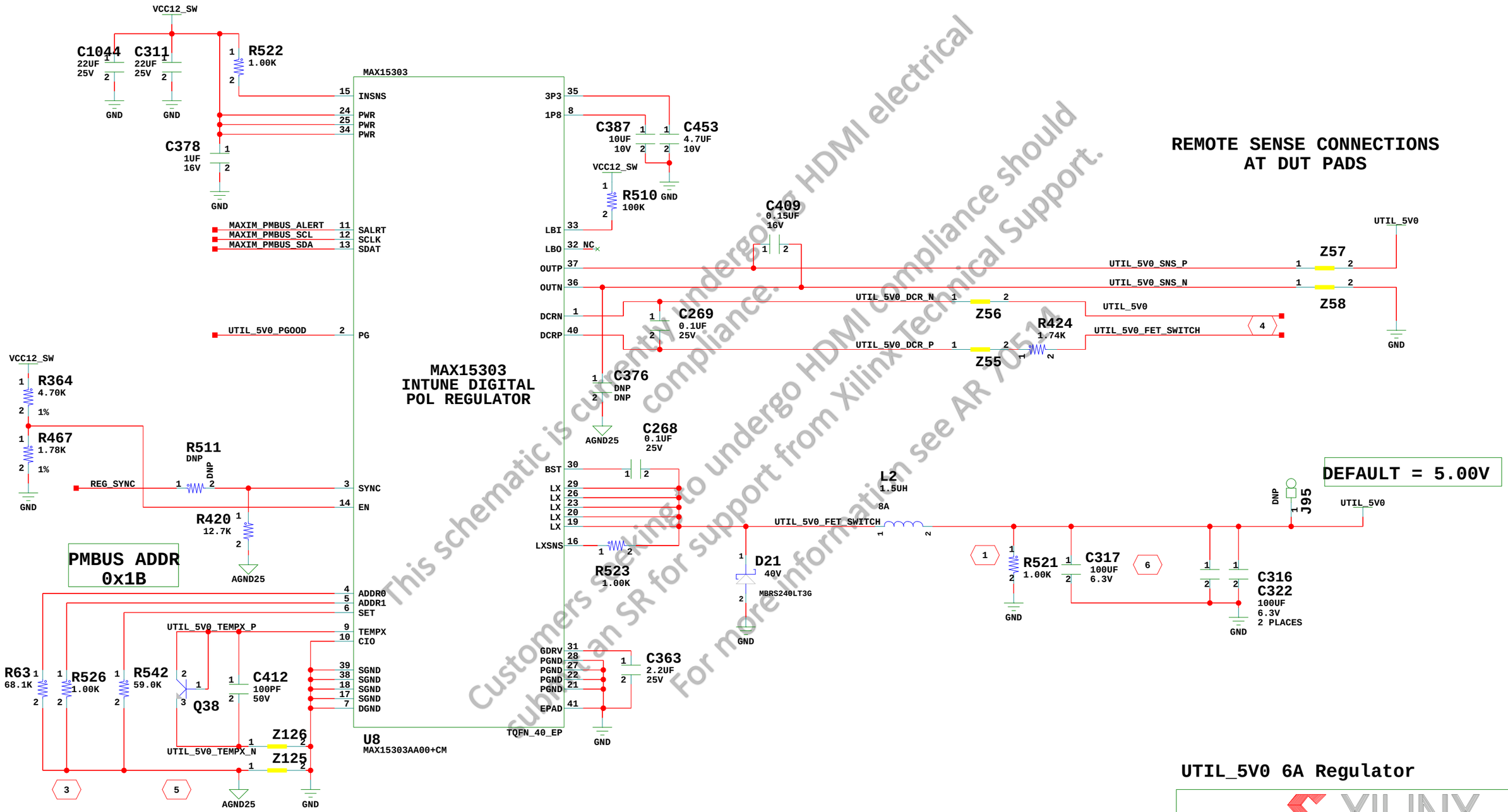
5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF OUTPUT INDUCTOR

UTIL_3V3 20A Regulator

XILINX

TITLE: UTIL_3V3 20A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 91 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

5 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR

4 CONNECT ACROSS PINS OF INDUCTOR

3 STRAP RESISTOR ON ADDR1 SETS IOUT_CAL_GAIN (ILIM) TO 2 mOHMS. THIS PARAMETER MUST BE OVERRIDDEN TO 5 mOHMS (MATCHING THE CURRENT SENSE RESISTOR) WHEN THE CONTROLLER IS CONFIGURED.

2 HEADER IS FOR SCOPE ATTACHMENT TO OBSERVE STEP LOAD RESPONSE.

1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

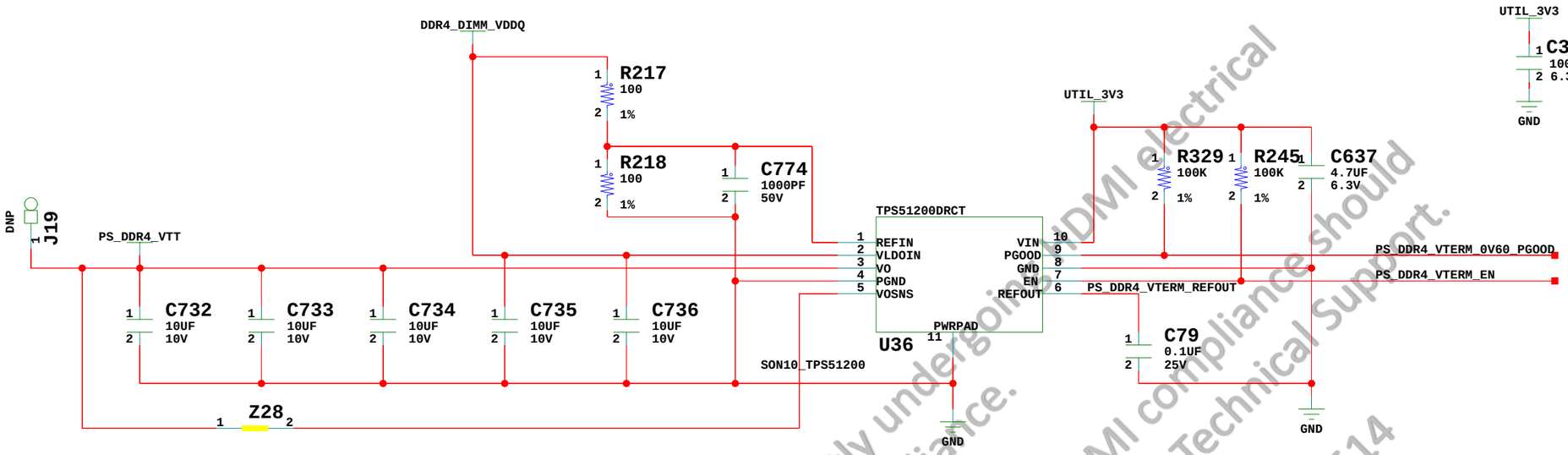
6 PLACE ONE 100uF CLOSE TO THE INDUCTOR, THE REST 100uF SHOULD BE PLACED CLOSE TO THE HEAVY LOAD SIDE

UTIL_5V0 6A Regulator

XILINX

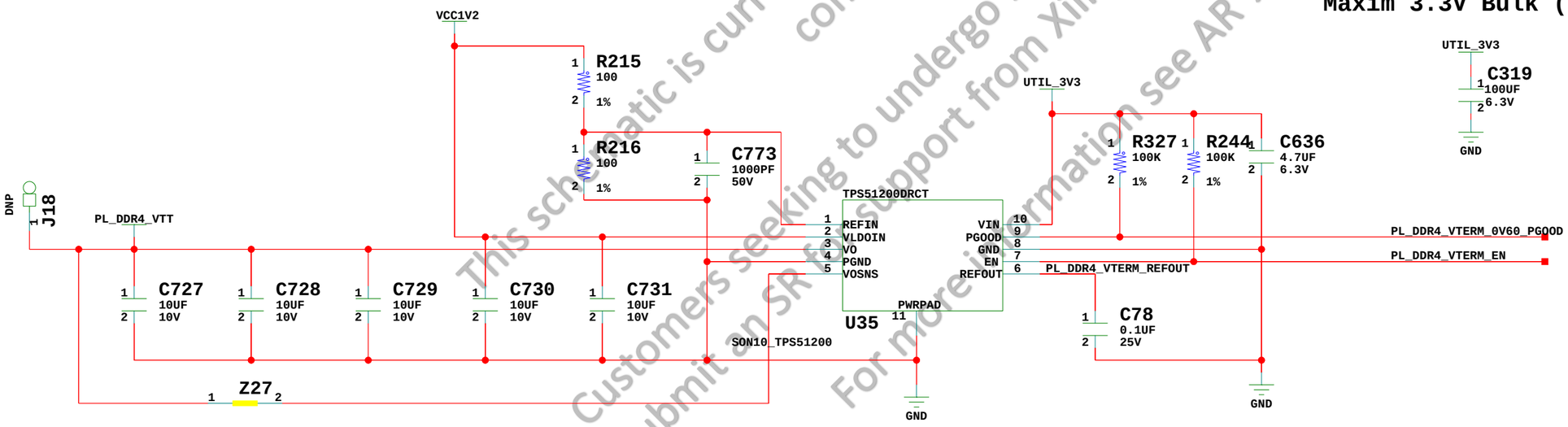
TITLE: UTIL_5V0 6A Regulator SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 92 OF 95	DRAWN BY: BF

Maxim 3.3V Bulk (near PS VTT)



Place sense near DDR4 DIMM

Maxim 3.3V Bulk (near PL VTT)



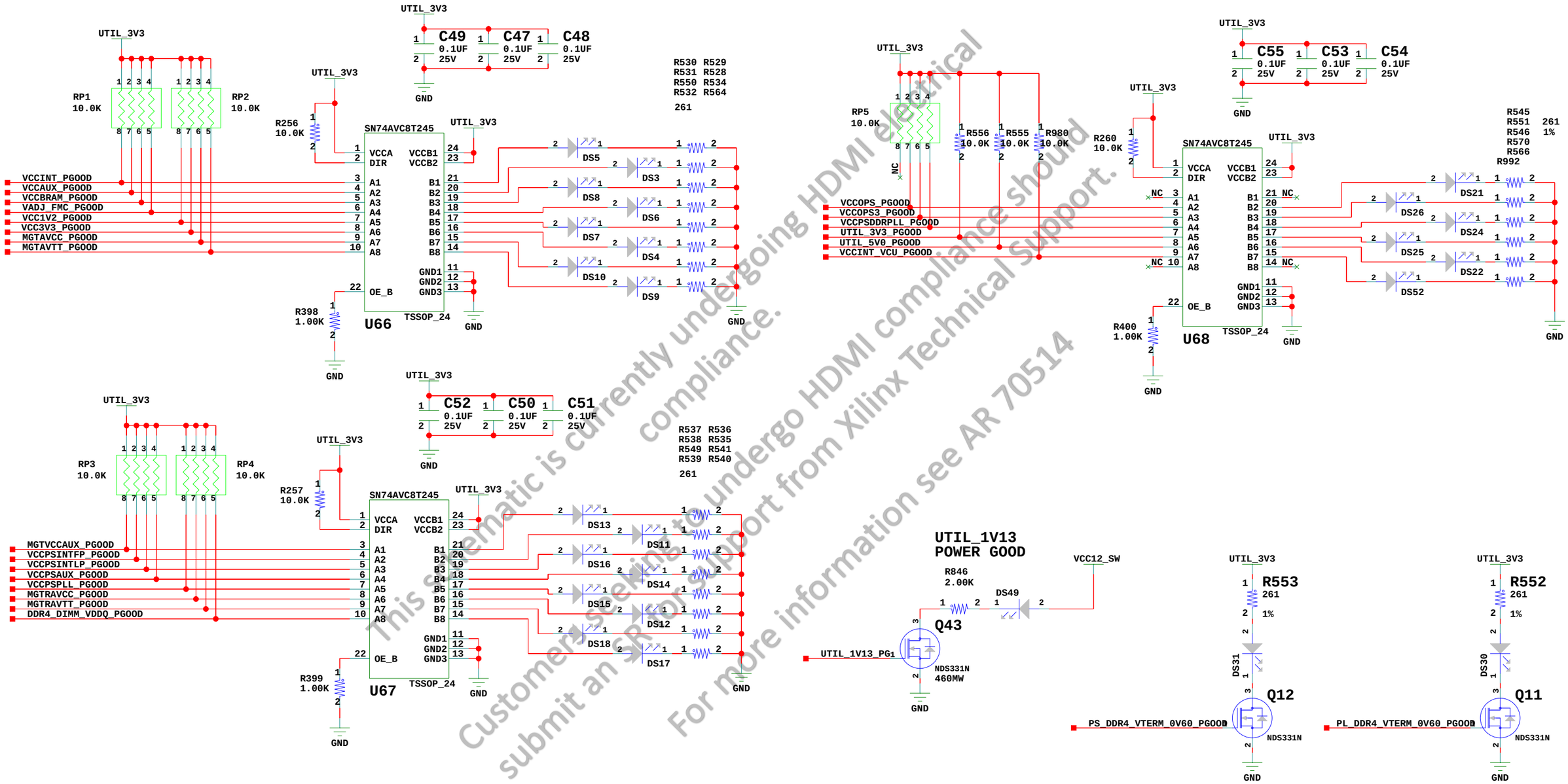
Place sense near DDR4 components

This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

DDR4 Termination Supply

XILINX

TITLE: DDR4 Termination Supply SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 93 OF 95	DRAWN BY: BF



This schematic is currently undergoing HDMI electrical compliance.
Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support.
For more information see AR 70514

Power Status LEDs

XILINX

TITLE: Power Status LEDs SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 94 OF 95	DRAWN BY: BF

PCIe Shorter Bracket Standoffs

HS1
12V Fan / Heatsink
12VDC

MMD1
DDR4_SODIMM

CBL1
NEMA 5-15P
IEC320-C13
110V 3 Prong
PC_POWER_CABLE
Power Cord

PB1
12v Power Brick
PWR_BRICK_12V

CBL2
USB A
USB Micro-B
USB Micro-B Cable
USB_MICRO_CABLE

MS01 STANDOFF_0P375
MW1 WASHER_FLAT_4
MRB1 RUBBER_BUMPER
MS1 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS2 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH1 GND

MS02 STANDOFF_0P250
MW2 WASHER_FLAT_4
MRB2 RUBBER_BUMPER
MS3 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS4 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH2 GND

MS03 STANDOFF_0P250
MW3 WASHER_FLAT_4
MRB3 RUBBER_BUMPER
MS5 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS6 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH3 GND

MS04 STANDOFF_0P375
MW4 WASHER_FLAT_4
MRB4 RUBBER_BUMPER
MS7 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 3/16"
MS8 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 3/16"
MH_156_300 MH4 GND

MS05 STANDOFF_0P375
MW5 WASHER_FLAT_4
MRB5 RUBBER_BUMPER
MS9 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS10 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 3/16"
MH_156_300 MH5 GND

MS06 STANDOFF_0P375
MW6 WASHER_FLAT_4
MRB6 RUBBER_BUMPER
MS11 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS12 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH6 GND

MS07 STANDOFF_0P375
MW7 WASHER_FLAT_4
MRB7 RUBBER_BUMPER
MS13 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS14 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH7 GND

MS08 STANDOFF_0P375
MW8 WASHER_FLAT_4
MRB8 RUBBER_BUMPER
MS15 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MS16 MACHINE_SCREW_4-40 #4-40 MACHINE SCREW 0.25"
MH_156_300 MH8 GND

PL1
KCU105_PCIE_PLATE

MJB1 JUMPER_BLOCK_2-PIN
MJB2 JUMPER_BLOCK_2-PIN
MJB3 JUMPER_BLOCK_2-PIN
MJB4 JUMPER_BLOCK_2-PIN
MJB5 JUMPER_BLOCK_2-PIN
MJB6 JUMPER_BLOCK_2-PIN
MJB7 JUMPER_BLOCK_2-PIN
MJB8 JUMPER_BLOCK_2-PIN
MJB9 JUMPER_BLOCK_2-PIN
MJB10 JUMPER_BLOCK_2-PIN
MJB11 JUMPER_BLOCK_2-PIN
MJB12 JUMPER_BLOCK_2-PIN
MJB13 JUMPER_BLOCK_2-PIN
MJB14 JUMPER_BLOCK_2-PIN

This schematic is currently undergoing HDMI electrical compliance testing. Customers seeking to undergo HDMI compliance should submit an SR for support from Xilinx Technical Support. For more information see AR 70514

Mechanical Components

XILINX

TITLE: Mechanical Components SCHEM, ROHS COMPLIANT HW-Z1-ZCU106_REV1_0	ASSY P/N: 0432032 PCB P/N: 1280937 SCH P/N: 0381770 TEST P/N: TSS0186
DATE: 01/08/2018:10:16	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 95 OF 95	DRAWN BY: BF